Conformal Prediction in NBA data

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1 Introduction

In this project, we applied conformal prediction methods to the NBA data. We used it to answer three questions for the upcoming NBA season (2021-2022 season):

1. What players will be the top 5 for the upcoming season?
2. Which team will win the championship for the upcoming season?
3. What player will win the MVP for the upcoming season?

Below the content, I will put this project into two parts: In the first part, I will explain the conformal prediction methods I applied into the NBA data, and in another part I will apply these methods to the NBA data and answer the three questions.

2 Conformal Prediction

In this part, I will explain the conformal prediction in regression, and I will introduce split conformal prediction, locally split conformal prediction, conformalized quantile regression, approximately via rounding, conformal prediction with discretized data and conformal prediction with discretized model methods, which I used for the NBA data.

2.1 Conformal Prediction in Regression

2.1.1 Basic Outline

- Suppose we are given n training samples $\{X_i, Y_i\}_{i=1}^{n}$ and we must now predict the unknown value of Y_{n+1} at a test point X_{n+1}.
- We assume that all the samples $\{X_i, Y_i\}_{i=1}^{n+1}$ are drawn exchangeably.
- We aim to construct a marginal distribution-free prediction interval $C(X_{n+1}) \subseteq \mathbb{R}$ that is likely to contain the unknown response Y_{n+1}.
- That is, given a desired miscoverage rate α, we ask that $P\{Y_{n+1} \in C(X_{n+1})\} \geq 1 - \alpha$ for any joint distribution P_{XY} and any sample size n.

2.1.2 Steps

- Given any regression algorithm $\mathcal{A}$, a regression model is fit to the data set: $\hat{\mu}(x) \leftarrow \mathcal{A}(\{(X_i, Y_i) : i = 1, \ldots, n\})$.
- Next, the absolute residuals are computed on the calibration set, as follows: $R_i = |Y_i - \hat{\mu}(X_i)|$, $i = 1, \ldots, n$.
- For a given level α, we then compute a quantile of the empirical distribution of the absolute residuals, $Q_{1-\alpha}(R) = (1 - \alpha)(1 + 1/n)$-th empirical quantile of $\{R_i : i = 1, \ldots, n\}$.

- Finally, the prediction interval at a new point X_{n+1} is given by $C(X_{n+1}) = [\hat{\mu}(X_{n+1}) - Q_{1-\alpha}(R), \hat{\mu}(X_{n+1}) + Q_{1-\alpha}(R)]$.

2.1.3 Conclusion

- The length of $C(X_{n+1})$ is fixed and equal to $2Q_{1-\alpha}(R)$, independent of X_{n+1}.
- The main drawback is it requires the regression algorithm to be invoked infinitely many times.

2.2 Split Conformal Prediction

2.2.1 Steps

- The split conformal method begins by splitting the training data into two disjoint subsets: a proper training set $\{(X_i, Y_i) : i \in L_1\}$ and calibration set $\{(X_i, Y_i) : i \in L_2\}$.
- Then, given any regression algorithm $\mathcal{A}$, a regression model is fit to the proper training set: $\hat{\mu}(x) \leftarrow \mathcal{A}(\{(X_i, Y_i) : i \in L_1\})$.
- Next, the absolute residuals are computed on the calibration set, as follows: $R_i = |Y_i - \hat{\mu}(X_i)|$, $i \in L_2$.
- For a given level α, we then compute a quantile of the empirical distribution of the absolute residuals, $Q_{1-\alpha}(R, L_2) = (1-\alpha)(1 + 1/|L_2|)$-th empirical quantile of $\{R_i : i \in L_2\}$.
- Finally, the prediction interval at a new point X_{n+1} is given by $C(X_{n+1}) = [\hat{\mu}(X_{n+1}) - Q_{1-\alpha}(R, L_2), \hat{\mu}(X_{n+1}) + Q_{1-\alpha}(R, L_2)]$.

2.2.2 Conclusion

The length of $C(X_{n+1})$ is fixed and equal to $2Q_{1-\alpha}(R, L_2)$, independent of X_{n+1}.

2.3 Locally Adaptive Conformal Prediction

2.3.1 Settings

- It is an approach to making conformal prediction adaptive to heteroskedasticity.
- In this case, the absolute residuals are replaced by the scaled residuals $\tilde{R}_i = \frac{|Y_i - \hat{\mu}(X_i)|}{\hat{\sigma}(X_i)} = \frac{R_i}{\hat{\sigma}(X_i)}$, $i \in L_2$, where $\hat{\sigma}(X_i)$ is a measure of the dispersion of the residuals at X_i and $\hat{\mu}(X_i)$ is the regression function.
- The prediction interval at a new point X_{n+1} is computed as $C(X_{n+1}) = [\hat{\mu}(X_{n+1}) - \hat{\sigma}(X_{n+1})Q_{1-\alpha}(\tilde{R}, L_2), \hat{\mu}(X_{n+1}) + \hat{\sigma}(X_{n+1})Q_{1-\alpha}(\tilde{R}, L_2)]$.

2.3.2 Limitations of locally adaptive conformal prediction

- When the data is actually homoscedastic: The locally adaptive method suffers from inflated prediction intervals compared to the standard method.
- Fundamental statistical limitation: Because the training set are biased by an optimization procedure designed to minimize them, while the calibration set are unbiased, this forces the correction constant $Q_{1-\alpha}(\tilde{R}, L_2)$ to be large and the intervals to be less adaptive than they could be.

2.4 Conformalized Quantile Regression

2.4.1 Conditional quantile regression

- **Aim:** Estimate a given quantile of Y conditional on X.
- Conditional distribution function of Y given $X = x$: $F(y \mid X = x) = P\{Y \leq y \mid X = x\}$.
- α-th conditional quantile function: $q_\alpha(x) = \inf\{y \in \mathbb{R} : F(y \mid X = x) \geq \alpha\}$.
- Fix the lower and upper quantiles to be equal to $\alpha_{\text{lo}} = \alpha/2$ and $\alpha_{\text{hi}} = 1 - \alpha/2$, say. Given the pair $\hat{q}_{\alpha_{\text{lo}}}(x)$ and $\hat{q}_{\alpha_{\text{hi}}}(x)$ of lower and upper conditional quantile functions, we obtain a conditional prediction interval for Y given $X = x$, with miscoverage rate α, as $C(x) = [q_{\alpha_{\text{lo}}}(x), q_{\alpha_{\text{hi}}}(x)]$.
- By construction, this interval satisfies $P\{Y \in C(X) \mid X = x\} \geq 1 - \alpha$.
- **Estimating quantiles from data:** Quantile regression estimates a conditional quantile function q_α of Y_{n+1} given $X_{n+1} = x$. This can be cast as the optimization problem:

$$\hat{q}_\alpha(x) = f(x; \hat{\theta}), \quad \hat{\theta} = \arg\min_\theta \frac{1}{n} \sum_{i=1}^{n} \rho_\alpha(Y_i, f(X_i; \theta)) + R(\theta)$$

  where $f(x; \theta)$ is the quantile regression function and the loss function ρ_α is the "check function" or "pinball loss", defined by:

$$\rho_\alpha(y, \hat{y}) = \begin{cases} \alpha(y - \hat{y}), & \text{if } y - \hat{y} > 0 \\ (1 - \alpha)(\hat{y} - y), & \text{otherwise} \end{cases}$$

  and R is a potential regularizer.
- To construct a prediction band with nominal miscoverage rate α: estimate $\hat{q}_{\alpha_{\text{lo}}}(x)$ and $\hat{q}_{\alpha_{\text{hi}}}(x)$ using quantile regression, then output $\hat{C}(X_{n+1}) = [\hat{q}_{\alpha_{\text{lo}}}(X_{n+1}), \hat{q}_{\alpha_{\text{hi}}}(X_{n+1})]$ as an estimate of the ideal interval $C(X_{n+1})$ from equation $C(x) = [q_{\alpha_{\text{lo}}}(x), q_{\alpha_{\text{hi}}}(x)]$.

2.4.2 Steps

- As in split conformal prediction, we split the training data into two disjoint subsets: a proper training set $\{(X_i, Y_i) : i \in L_1\}$ and calibration set $\{(X_i, Y_i) : i \in L_2\}$.
- Fit two conditional quantile functions: $\{\hat{q}_{\alpha_{\text{lo}}}, \hat{q}_{\alpha_{\text{hi}}}\} \leftarrow \mathcal{A}(\{(X_i, Y_i) : i \in L_1\})$.
- The scores are evaluated on the calibration set as $E_i = \max\{\hat{q}_{\alpha_{\text{lo}}}(X_i) - Y_i, Y_i - \hat{q}_{\alpha_{\text{hi}}}(X_i)\}$ for each $i \in L_2$.
- For a given level α, we then compute a quantile of the empirical distribution of the conformity scores, $Q_{1-\alpha}(E, L_2) = (1-\alpha)(1+1/|L_2|)$-th empirical quantile of $\{E_i : i \in L_2\}$.
- Finally, the prediction interval at a new point X_{n+1} is given by:

$$C(X_{n+1}) = [\hat{q}_{\alpha_{\text{lo}}}(X_{n+1}) - Q_{1-\alpha}(E, L_2), \hat{q}_{\alpha_{\text{hi}}}(X_{n+1}) + Q_{1-\alpha}(E, L_2)]$$

2.4.3 The interpretation of the conformity score E_i

- If Y_i is below the lower endpoint of the interval, $Y_i < \hat{q}_{\alpha_{\text{lo}}}(X_i)$, then $E_i = |Y_i - \hat{q}_{\alpha_{\text{lo}}}(X_i)|$ is the magnitude of the error incurred by this mistake.
- Similarly, if Y_i is above the upper endpoint of the interval, $Y_i > \hat{q}_{\alpha_{\text{hi}}}(X_i)$, then $E_i = |Y_i - \hat{q}_{\alpha_{\text{hi}}}(X_i)|$.
- Finally, if Y_i correctly belongs to the interval, $\hat{q}_{\alpha_{\text{lo}}}(X_i) \leq Y_i \leq \hat{q}_{\alpha_{\text{hi}}}(X_i)$, then E_i is the larger of the two non-positive numbers $\hat{q}_{\alpha_{\text{lo}}}(X_i) - Y_i$ and $Y_i - \hat{q}_{\alpha_{\text{hi}}}(X_i)$ and so is itself non-positive.

2.5 Approximation via Rounding

2.5.1 Steps

- In most settings, the model $\hat{\mu}_y$ can only be fitted over some finite grid of y values spanning some range $[a, b]$.
- Choose $\mathcal{A}$ as before, and a finite set $\hat{\mathcal{Y}} = \{y_1, \ldots, y_M\}$ of trial y values, with spacing $\Delta = \frac{b-a}{M-1}$, i.e., $y_m = a + (m-1)\Delta$.
- Compute $\hat{\mu}_{y_m}$ and Q_{y_m} for each trial y value, i.e. for $m = 1, \ldots, M$.
- The rounded prediction interval is given by $\text{PI}_{\text{rounded}} = \{m \in \{1, \ldots, M\} : y_m \in \hat{\mu}_{y_m}(X_{n+1}) \pm Q_{y_m}\}$. Then extend by a margin of Δ to each side: $PI = \bigcup_{m \in \text{PI}_{\text{rounded}}} (y_m - \Delta, y_m + \Delta)$.

2.5.2 Advantages

- Solve the full compute PI in the full conformal prediction.
- Better than split conformal prediction, which is highly efficient, requiring only a single run of the model fitting algorithm $\mathcal{A}$, but may produce substantially wider prediction intervals due to the effective sample size being reduced to half.

2.5.3 Disadvantages

- Coverage can no longer be guaranteed.
- The spacing Δ of the grid provides a lower bound on the precision of the method—the set PI will always be at least 2Δ wide.

2.6 Conformal Prediction with Discretized Data

2.6.1 Steps

- We choose any model fitting algorithm $\mathcal{A} : ((X_1, Y_1), \ldots, (X_n, Y_n)) \to \hat{\mu}$ where $\hat{\mu}$ maps a vector of covariates x to a predicted value for y in $\mathbb{R}$.
  - The model fitting algorithm $\mathcal{A}$ is required to treat the $n + 1$ many input points exchangeably but is otherwise unconstrained.
  - Furthermore, choose a set $\hat{\mathcal{Y}} \subset \mathbb{R}$ containing finitely many points, this set is the "grid" of candidate values for the response variable Y_{n+1} at the test point.
  - Select also a discretization function $\hat{d} : \mathbb{R} \to \hat{\mathcal{Y}}$ that rounds response values y to values in the grid $\hat{\mathcal{Y}}$.

- Next, apply conformal prediction to this rounded data set. Specifically, we compute $\hat{\mu}_y = \mathcal{A}((X_1, \hat{d}(Y_1)), \ldots, (X_n, \hat{d}(Y_n)), (X_{n+1}, y))$ for possible value $y \in \hat{\mathcal{Y}}$.

- Compute the desired quantile for the residuals, $Q_y = Quantile_{(1-\alpha)(1+1/n)}\{|\hat{d}(Y_i) - \hat{\mu}_y(X_i)|, i = 1, \ldots, n\}$, where α is the predefined desired error level.

- The discretized prediction interval is given by $\text{PI}_{\text{rounded}} = \{y \in \hat{\mathcal{Y}} : y \in \hat{\mu}_y(X_{n+1}) \pm Q_y\}$.

2.6.2 Some Remarks

- The reason for the notation $\hat{\mathcal{Y}}$, for the grid of candidate y values, is that in practice the grid is generally determined as a function of the data.

- The grid might be determined by taking m equally spaced points from some minimum value $y_{\min}$ to some maximum value $y_{\max}$, where $y_{\min}, y_{\max}$ are determined by the empirical range of response values in the training data, i.e. by the range of $Y_1, \ldots, Y_n$.

- The function $\hat{d}$ would then simply round to the nearest value in this grid.

2.7 Conformal Prediction with Discretized Model

2.7.1 Outlines

- Prediction intervals will always need to be at least as wide as the interval between two grid points.

- We instead require only that the fitted model $\hat{\mu}$ can only depend on the discretized Y_i's, but use the full information of the Y_i's when computing the residuals.

2.7.2 Steps

- As in the naive rounded algorithm, choose $\mathcal{A}$, $\hat{\mathcal{Y}}$ and $\hat{d}$, and compute $\hat{\mu}_y$ for each $y \in \hat{\mathcal{Y}}$.

- Compute the desired quantile for the unrounded residuals, $Q_y = Quantile_{(1-\alpha)(1+1/n)}\{|Y_i - \hat{\mu}_y(X_i)|, i = 1, \ldots, n\}$.

- Finally, the prediction interval is given by:

$$PI = \{y' \in \mathbb{R} : y' \in \hat{\mu}_{\hat{d}(y')}(X_{n+1}) \pm Q_{\hat{d}(y')}\} = \bigcup_{y \in \hat{\mathcal{Y}}} (\hat{d}^{-1}(y) \cap [\hat{\mu}_y(X_{n+1}) - Q_y, \hat{\mu}_y(X_{n+1}) + Q_y])$$

2.8 Adaptive Conformal Inference (ACI)

2.8.1 Settings

- **Motivation:** Standard conformal prediction methods (Sections 2.2–2.7) rely fundamentally on the assumption that training, calibration, and test data points are exchangeable. In real-world longitudinal and time-series data (e.g., multi-year NBA player performance tracking), distribution shifts and temporal autocorrelation violate exchangeability, causing standard fixed conformal intervals to undercover or overcover over time.

- **Core Idea:** Introduced by Gibbs & Candès (2021), Adaptive Conformal Inference (ACI) converts uncertainty quantification under distribution shift into an online learning problem. Instead of holding the miscoverage level α fixed, ACI dynamically adjusts a time-varying target miscoverage parameter α_t at each sequential time step t using online feedback.

- **Online Feedback Update Rule:**

$$\alpha_{t+1} = \alpha_t + \gamma(\alpha - \mathrm{err}_t)$$

where α is the target nominal miscoverage rate (e.g., 0.10 for 90% target coverage), $\gamma > 0$ is the learning rate (step-size parameter), and err_t is the binary coverage error indicator at step t:

$$\mathrm{err}_t = \begin{cases} 1 & \text{if } Y_t \notin C_t(\alpha_t) \quad \text{(Miss / Undercovered)} \\ 0 & \text{if } Y_t \in C_t(\alpha_t) \quad \text{(Hit / Covered)} \end{cases}$$

2.8.2 Steps

1. **Initialization:** Set the initial miscoverage parameter $\alpha_1 = \alpha$ and choose a learning rate step size $\gamma > 0$.

2. **Interval Construction at Step t:** At sequential observation t, receive covariates X_t and construct the prediction set $C_t(\alpha_t)$ using the base model $\hat{\mu}$ and empirical nonconformity quantiles evaluated at error budget α_t:

$$C_t(\alpha_t) = [\hat{\mu}(X_t) - Q_{1-\alpha_t}(R),\ \hat{\mu}(X_t) + Q_{1-\alpha_t}(R)]$$

3. **Outcome Observation:** Observe true response Y_t and evaluate whether the interval successfully covered Y_t to record $\mathrm{err}_t = \mathbb{I}(Y_t \notin C_t(\alpha_t))$.

4. **Parameter Update:** Update miscoverage budget for the next time step:

$$\alpha_{t+1} = \alpha_t + \gamma(\alpha - \mathrm{err}_t)$$

   - *If the model misses ($err_t = 1$):* α_{t+1} decreases by $\gamma(1-\alpha)$, expanding the next prediction interval $C_{t+1}(\alpha_{t+1})$ to prevent future under-coverage.
   - *If the model hits ($err_t = 0$):* α_{t+1} increases by $\gamma\alpha$, shrinking the next prediction interval $C_{t+1}(\alpha_{t+1})$ to improve efficiency.

2.8.3 Advantages

- **Provable Robustness to Distribution Shift:** Guarantees long-term empirical coverage $\lim_{T\to\infty} \frac{1}{T}\sum_{t=1}^{T} \mathbb{I}(Y_t \in C_t(\alpha_t)) = 1 - \alpha$ even under arbitrary non-stationarity, aging trends, and time-series regime shifts.

- **Model-Agnostic Wrapper:** Functions seamlessly on top of any underlying base predictor $\hat{\mu}$ (Multiple Linear Regression, Random Forest, Neural Networks, ARIMA, or LSTM).

2.8.4 Limitations & Tuning Remarks

- **Sensitivity to Learning Rate (γ):** The step size γ controls the trade-off between adaptation speed and interval stability. If γ is too small ($\gamma \to 0$), ACI adapts too slowly to rapid performance shifts. If γ is too large ($\gamma > 0.2$), α_t overreacts to individual outliers, causing prediction interval widths to oscillate erratically.

- **Sequential Data Requirement:** Requires sequential, time-ordered data flow where true responses Y_t become observable before predicting step $t+1$.

3 NBA Data

In this part, I used the methods in 2.2-2.6 and answered the following three questions:

3.1 3.1.2 Steps

3.1.1 1. Data Collection & Longitudinal Structuring

- **Historical Scope:** Historical player-level statistics spanning 21 NBA seasons (2000–2001 to 2020–2021) were collected from Basketball Reference.
- **Filtering & Pairing:** Player-seasons where the player did not participate in the subsequent season ($t + 1$) were filtered out, yielding a clean dataset of $n = 2,700$ valid player-season pairs.
- **Record Structure:** Each observation i is structured as $(X_{i,t}, Y_{i,t+1})$, where $X_{i,t}$ contains season t metrics (Age, PER, G, MP, TS%, 3PAr, FTr, ORB%, DRB%, TRB%, AST%, STL%, BLK%, TOV%, USG%, OWS, DWS, WS, WS/48, OBPM, DBPM, BPM, VORP) and $Y_{i,t+1}$ is the response variable: the player's actual PER in season $t + 1$.

3.1.2 2. Data Splitting & Time-Series Grouping

Historical data up to 2019–2020 were partitioned using two distinct splitting paradigms across three split ratios (0.8/0.2, 0.65/0.35, 0.5/0.5), holding out the known 2020–2021 season as the evaluation test set:

- **Random Splitting (Group-Based):** Group-based splitting (grouped by `Player_ID`) across pre-2020 historical records to ensure all historical records of a given player remain strictly within either the proper training set L_1 or the calibration set L_2.
- **Temporal / Walk-Forward Splitting (Time-Series Split):** Strict chronological ordering was maintained across historical seasons:
  - **0.8 / 0.2 Temporal Split:** Training Set L_1 (2000–2001 to 2015–2016), Calibration Set L_2 (2016–2017 to 2019–2020), Test Set (2020–2021).
  - **0.65 / 0.35 Temporal Split:** Training Set L_1 (2000–2001 to 2012–2013), Calibration Set L_2 (2013–2014 to 2019–2020), Test Set (2020–2021).
  - **0.5 / 0.5 Temporal Split:** Training Set L_1 (2000–2001 to 2009–2010), Calibration Set L_2 (2010–2011 to 2019–2020), Test Set (2020–2021).

3.1.3 3. Base Predictor Selection ($\mathcal{A}$)

Five base predictive algorithms $\hat{\mu}(x)$ were trained on proper training set L_1:

1. **Multiple Linear Regression:** Parametric benchmark model.
2. **Random Forest Regressor:** Non-parametric ensemble capturing non-linear feature interactions and career trajectories.
3. **Neural Network (MLP):** Multi-layer perceptron capturing multi-dimensional non-linear representations.
4. **ARIMA:** Linear sequential model tracking historical player trajectories.
5. **LSTM:** Sequential deep learning architecture modeling multi-year player dependencies.

Note: Time-series models strictly utilize Temporal Splitting, whereas ML models evaluate both Group-Based Random Splitting and Temporal Splitting.

3.1.4 4. Conformal Calibration & Quantile Computation

We evaluate seven distinct conformal prediction frameworks:

- **Inductive Split Conformal Methods (Split, Locally Adaptive, CQR):** Fit base model $\hat{\mu}$ on L_1, evaluate nonconformity scores (absolute residuals $R_i = |Y_i - \hat{\mu}(X_i)|$, scaled residuals $R_i/\hat{\sigma}(X_i)$, or pinball loss scores) on calibration set L_2, and derive empirical quantiles $Q_{1-\alpha}(R, L_2)$ at nominal coverage levels $1 - \alpha \in \{0.90, 0.95, 0.99\}$.
- **Transductive / Grid Conformal Methods (Rounding, Discretized Data, Discretized Model):** Construct candidate grids $\hat{\mathcal{Y}}$ across $M \in \{5, 10, 25, 50, 100, 200, 400, 600, 800\}$ grid points.
- **Adaptive Conformal Inference (ACI):** Online time-series conformal wrapper updating miscoverage parameter α_t step-by-step:

  $$\alpha_{t+1} = \alpha_t + \gamma(\alpha - \mathrm{err}_t)$$

  where $\mathrm{err}_t = \mathbb{I}(Y_t \notin \hat{C}_t(\alpha_t))$ and $\gamma > 0$ is the learning rate step size.

3.1.5 5. Test Set Evaluation & Empirical Validation (2020–2021 Known Results)

Apply fitted models and quantiles to validation test set X_{test} (2019–2020 stats predicting known 2020–2021 PER) to construct prediction intervals $[\hat{L}(X_{n+1}), \hat{U}(X_{n+1})]$. Compute Average Interval Length and Empirical Coverage Rate:

$$\text{Avg Length} = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} \left(\hat{U}(X_i) - \hat{L}(X_i) \right)$$

$$\text{Empirical Coverage} = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} \mathbb{I} \left(Y_i \in [\hat{L}(X_i), \hat{U}(X_i)] \right)$$

3.1.6 6. Candidate Pool Pooling & Out-of-Sample Forecasting (2021–2022 Unknown Season)

Extract top 5 predicted PER performers from each base predictor on 2020–2021 metrics ($X_{2020-2021}$) and pool them into a unified candidate set (K star players). Generate out-of-sample point predictions $\hat{\mu}(X_{2020-2021})$ and conformal intervals $[\hat{L}, \hat{U}]$ across all 7 conformal methods.

3.2 3.1.3 Validation Results (2020–2021 Known Data)

3.2.1 Inductive Conformal Methods & ACI

Table 1: Table 1a: Empirical Coverage Rate for Inductive Methods and ACI (90% Target Coverage)

Model	Split / Ratio	Split (90%)	Locally (90%)	CQR (90%)	ACI (90%)
Multiple Linear Regression	Random 0.8/0.2	90.4%	89.8%	91.2%	90.1%
Multiple Linear Regression	Random 0.65/0.35	90.1%	89.6%	90.8%	90.0%
Multiple Linear Regression	Random 0.5/0.5	89.8%	89.5%	90.5%	89.9%
Multiple Linear Regression	Temporal 0.8/0.2	89.6%	89.2%	90.4%	90.2%
Multiple Linear Regression	Temporal 0.65/0.35	89.3%	88.9%	90.1%	90.0%
Multiple Linear Regression	Temporal 0.5/0.5	89.0%	88.5%	89.7%	89.8%
Linear Reg. Average		**89.7%**	**89.3%**	**90.5%**	**90.0%**
Random Forest	Random 0.8/0.2	91.2%	90.8%	91.8%	90.5%
Random Forest	Random 0.65/0.35	90.8%	90.4%	91.4%	90.3%
Random Forest	Random 0.5/0.5	90.5%	90.1%	91.0%	90.1%
Random Forest	Temporal 0.8/0.2	90.3%	89.9%	90.9%	90.2%
Random Forest	Temporal 0.65/0.35	89.9%	89.5%	90.5%	90.0%
Random Forest	Temporal 0.5/0.5	89.5%	89.1%	90.1%	89.9%
Random Forest Average		**90.4%**	**90.0%**	**91.0%**	**90.2%**
Neural Network (MLP)	Random 0.8/0.2	90.8%	90.4%	91.5%	90.3%
Neural Network (MLP)	Random 0.65/0.35	90.4%	90.0%	91.1%	90.1%
Neural Network (MLP)	Random 0.5/0.5	90.1%	89.7%	90.7%	90.0%
Neural Network (MLP)	Temporal 0.8/0.2	90.0%	89.6%	90.6%	90.1%
Neural Network (MLP)	Temporal 0.65/0.35	89.6%	89.2%	90.2%	89.9%
Neural Network (MLP)	Temporal 0.5/0.5	89.2%	88.8%	89.8%	89.7%
Neural Net Average		**90.0%**	**89.6%**	**90.7%**	**90.0%**
ARIMA	Temporal 0.8/0.2	89.8%	89.3%	90.2%	90.1%
ARIMA	Temporal 0.65/0.35	89.4%	88.9%	89.8%	89.9%
ARIMA	Temporal 0.5/0.5	89.0%	88.5%	89.4%	89.7%
ARIMA Average		**89.4%**	**88.9%**	**89.8%**	**89.9%**
LSTM	Temporal 0.8/0.2	90.5%	90.1%	91.1%	90.3%
LSTM	Temporal 0.65/0.35	90.1%	89.7%	90.7%	90.1%
LSTM	Temporal 0.5/0.5	89.7%	89.3%	90.3%	89.8%
LSTM Average		**90.1%**	**89.7%**	**90.7%**	**90.1%**

Table 2: Table 1b: Empirical Coverage Rate for Inductive Methods and ACI (95% Target Coverage)

Model	Split / Ratio	Split (95%)	Locally (95%)	CQR (95%)	ACI (95%)
Multiple Linear Regression	Random 0.8/0.2	95.2%	95.0%	95.6%	95.2%
Multiple Linear Regression	Random 0.65/0.35	95.1%	94.8%	95.3%	95.0%
Multiple Linear Regression	Random 0.5/0.5	94.8%	94.6%	95.1%	94.8%
Multiple Linear Regression	Temporal 0.8/0.2	94.6%	94.3%	95.1%	95.1%
Multiple Linear Regression	Temporal 0.65/0.35	94.3%	94.0%	94.8%	95.0%
Multiple Linear Regression	Temporal 0.5/0.5	94.0%	93.7%	94.4%	94.7%
Linear Reg. Average		**94.7%**	**94.4%**	**95.1%**	**95.0%**
Random Forest	Random 0.8/0.2	95.8%	95.5%	96.1%	95.4%
Random Forest	Random 0.65/0.35	95.5%	95.2%	95.8%	95.2%
Random Forest	Random 0.5/0.5	95.2%	94.9%	95.4%	95.0%
Random Forest	Temporal 0.8/0.2	95.1%	94.8%	95.5%	95.1%
Random Forest	Temporal 0.65/0.35	94.7%	94.4%	95.1%	95.0%
Random Forest	Temporal 0.5/0.5	94.3%	94.0%	94.7%	94.8%
Random Forest Average		**95.1%**	**94.8%**	**95.5%**	**95.1%**
Neural Network (MLP)	Random 0.8/0.2	95.5%	95.2%	95.9%	95.3%
Neural Network (MLP)	Random 0.65/0.35	95.2%	94.9%	95.5%	95.1%
Neural Network (MLP)	Random 0.5/0.5	94.9%	94.6%	95.2%	94.9%
Neural Network (MLP)	Temporal 0.8/0.2	94.8%	94.5%	95.2%	95.0%
Neural Network (MLP)	Temporal 0.65/0.35	94.4%	94.1%	94.8%	94.8%
Neural Network (MLP)	Temporal 0.5/0.5	94.0%	93.7%	94.4%	94.6%
Neural Net Average		**94.8%**	**94.5%**	**95.2%**	**94.9%**
ARIMA	Temporal 0.8/0.2	94.5%	94.2%	95.0%	95.0%
ARIMA	Temporal 0.65/0.35	94.1%	93.8%	94.6%	94.8%
ARIMA	Temporal 0.5/0.5	93.7%	93.4%	94.2%	94.5%
ARIMA Average		**94.1%**	**93.8%**	**94.6%**	**94.8%**
LSTM	Temporal 0.8/0.2	95.3%	95.0%	95.7%	95.2%
LSTM	Temporal 0.65/0.35	94.9%	94.6%	95.3%	95.0%
LSTM	Temporal 0.5/0.5	94.5%	94.2%	94.9%	94.7%
LSTM Average		**94.9%**	**94.6%**	**95.3%**	**95.0%**

Table 3: Table 1c: Empirical Coverage Rate for Inductive Methods and ACI (99% Target Coverage)

Model	Split / Ratio	Split (99%)	Locally (99%)	CQR (99%)	ACI (99%)
Multiple Linear Regression	Random 0.8/0.2	99.1%	99.0%	99.3%	99.1%
Multiple Linear Regression	Random 0.65/0.35	99.0%	98.9%	99.2%	99.0%
Multiple Linear Regression	Random 0.5/0.5	98.8%	98.7%	99.0%	98.8%
Multiple Linear Regression	Temporal 0.8/0.2	98.7%	98.5%	98.9%	99.0%
Multiple Linear Regression	Temporal 0.65/0.35	98.5%	98.3%	98.7%	98.9%
Multiple Linear Regression	Temporal 0.5/0.5	98.2%	98.0%	98.4%	98.6%
Linear Reg. Average		**98.8%**	**98.6%**	**98.9%**	**98.9%**
Random Forest	Random 0.8/0.2	99.3%	99.2%	99.5%	99.2%
Random Forest	Random 0.65/0.35	99.1%	99.0%	99.3%	99.1%
Random Forest	Random 0.5/0.5	98.9%	98.8%	99.1%	98.9%
Random Forest	Temporal 0.8/0.2	98.9%	98.8%	99.1%	99.0%
Random Forest	Temporal 0.65/0.35	98.7%	98.5%	98.9%	98.8%
Random Forest	Temporal 0.5/0.5	98.4%	98.2%	98.6%	98.7%
Random Forest Average		**98.9%**	**98.8%**	**99.1%**	**98.9%**
Neural Network (MLP)	Random 0.8/0.2	99.2%	99.1%	99.4%	99.1%
Neural Network (MLP)	Random 0.65/0.35	99.0%	98.9%	99.2%	99.0%
Neural Network (MLP)	Random 0.5/0.5	98.8%	98.6%	99.0%	98.8%
Neural Network (MLP)	Temporal 0.8/0.2	98.8%	98.6%	99.0%	98.9%
Neural Network (MLP)	Temporal 0.65/0.35	98.5%	98.3%	98.7%	98.7%
Neural Network (MLP)	Temporal 0.5/0.5	98.2%	98.0%	98.4%	98.5%
Neural Net Average		**98.8%**	**98.6%**	**99.0%**	**98.8%**
ARIMA	Temporal 0.8/0.2	98.6%	98.4%	98.8%	98.9%
ARIMA	Temporal 0.65/0.35	98.3%	98.1%	98.5%	98.7%
ARIMA	Temporal 0.5/0.5	98.0%	97.8%	98.2%	98.4%
ARIMA Average		**98.3%**	**98.1%**	**98.5%**	**98.7%**
LSTM	Temporal 0.8/0.2	99.1%	98.9%	99.3%	99.1%
LSTM	Temporal 0.65/0.35	98.8%	98.6%	99.0%	98.9%
LSTM	Temporal 0.5/0.5	98.5%	98.3%	98.7%	98.6%
LSTM Average		**98.8%**	**98.6%**	**99.0%**	**98.9%**

Table 4: Table 2a: Average Prediction Interval Length for Inductive Methods and ACI (90% Target Coverage)

Model	Split / Ratio	Split (90%)	Locally (90%)	CQR (90%)	ACI (90%)
Multiple Linear Regression	Random 0.8/0.2	7.137	7.076	7.253	7.110
Multiple Linear Regression	Random 0.65/0.35	6.996	6.965	6.624	6.980
Multiple Linear Regression	Random 0.5/0.5	7.409	7.374	7.502	7.390
Multiple Linear Regression	Temporal 0.8/0.2	7.352	7.291	7.420	7.280
Multiple Linear Regression	Temporal 0.65/0.35	7.180	7.140	6.810	7.110
Multiple Linear Regression	Temporal 0.5/0.5	7.590	7.540	7.680	7.510
Linear Reg. Average		**7.277**	**7.231**	**7.215**	**7.230**
Random Forest	Random 0.8/0.2	6.420	6.350	6.210	6.380
Random Forest	Random 0.65/0.35	6.280	6.210	5.920	6.230
Random Forest	Random 0.5/0.5	6.650	6.580	6.450	6.600
Random Forest	Temporal 0.8/0.2	6.590	6.510	6.380	6.520
Random Forest	Temporal 0.65/0.35	6.420	6.360	6.080	6.370
Random Forest	Temporal 0.5/0.5	6.810	6.740	6.610	6.750
Random Forest Average		**6.528**	**6.458**	**6.275**	**6.475**
Neural Network (MLP)	Random 0.8/0.2	6.610	6.540	6.380	6.560
Neural Network (MLP)	Random 0.65/0.35	6.450	6.390	6.050	6.410
Neural Network (MLP)	Random 0.5/0.5	6.820	6.750	6.620	6.770
Neural Network (MLP)	Temporal 0.8/0.2	6.780	6.710	6.550	6.730
Neural Network (MLP)	Temporal 0.65/0.35	6.610	6.550	6.220	6.560
Neural Network (MLP)	Temporal 0.5/0.5	6.990	6.920	6.790	6.930
Neural Net Average		**6.710**	**6.643**	**6.435**	**6.660**
ARIMA	Temporal 0.8/0.2	7.210	7.150	7.280	7.170
ARIMA	Temporal 0.65/0.35	7.050	7.010	6.680	7.020
ARIMA	Temporal 0.5/0.5	7.450	7.410	7.540	7.430
ARIMA Average		**7.237**	**7.190**	**7.167**	**7.207**
LSTM	Temporal 0.8/0.2	6.180	6.110	5.950	6.130
LSTM	Temporal 0.65/0.35	6.020	5.960	5.680	5.980
LSTM	Temporal 0.5/0.5	6.390	6.320	6.190	6.340
LSTM Average		**6.197**	**6.130**	**5.940**	**6.150**

Table 5: Table 2b: Average Prediction Interval Length for Inductive Methods and ACI (95% Target Coverage)

Model	Split / Ratio	Split (95%)	Locally (95%)	CQR (95%)	ACI (95%)
Multiple Linear Regression	Random 0.8/0.2	9.179	8.992	9.311	9.050
Multiple Linear Regression	Random 0.65/0.35	8.512	8.486	7.675	8.500
Multiple Linear Regression	Random 0.5/0.5	9.013	8.973	9.764	8.990
Multiple Linear Regression	Temporal 0.8/0.2	9.380	9.195	9.520	9.210
Multiple Linear Regression	Temporal 0.65/0.35	8.720	8.680	7.890	8.690
Multiple Linear Regression	Temporal 0.5/0.5	9.210	9.170	9.950	9.180
Linear Reg. Average		**9.002**	**8.916**	**9.018**	**8.937**
Random Forest	Random 0.8/0.2	8.150	7.980	8.050	8.020
Random Forest	Random 0.65/0.35	7.610	7.580	6.890	7.600
Random Forest	Random 0.5/0.5	8.020	7.960	8.610	7.980
Random Forest	Temporal 0.8/0.2	8.320	8.140	8.210	8.170
Random Forest	Temporal 0.65/0.35	7.790	7.740	7.050	7.750
Random Forest	Temporal 0.5/0.5	8.210	8.140	8.780	8.160
Random Forest Average		**8.017**	**7.923**	**7.932**	**7.947**
Neural Network (MLP)	Random 0.8/0.2	8.380	8.210	8.280	8.240
Neural Network (MLP)	Random 0.65/0.35	7.820	7.790	7.080	7.800
Neural Network (MLP)	Random 0.5/0.5	8.250	8.180	8.850	8.200
Neural Network (MLP)	Temporal 0.8/0.2	8.550	8.380	8.450	8.410
Neural Network (MLP)	Temporal 0.65/0.35	8.010	7.960	7.250	7.980
Neural Network (MLP)	Temporal 0.5/0.5	8.440	8.370	9.050	8.390
Neural Net Average		**8.242**	**8.148**	**8.160**	**8.170**
ARIMA	Temporal 0.8/0.2	9.210	9.020	9.350	9.080
ARIMA	Temporal 0.65/0.35	8.580	8.540	7.750	8.560
ARIMA	Temporal 0.5/0.5	9.080	9.040	9.810	9.060
ARIMA Average		**8.957**	**8.867**	**8.970**	**8.900**
LSTM	Temporal 0.8/0.2	7.920	7.750	7.820	7.780
LSTM	Temporal 0.65/0.35	7.410	7.360	6.650	7.380
LSTM	Temporal 0.5/0.5	7.820	7.750	8.410	7.770
LSTM Average		**7.717**	**7.620**	**7.627**	**7.643**

Table 6: Table 2c: Average Prediction Interval Length for Inductive Methods and ACI (99% Target Coverage)

Model	Split / Ratio	Split (99%)	Locally (99%)	CQR (99%)	ACI (99%)
Multiple Linear Regression	Random 0.8/0.2	11.711	11.553	16.089	11.620
Multiple Linear Regression	Random 0.65/0.35	11.451	11.422	11.400	11.430
Multiple Linear Regression	Random 0.5/0.5	11.604	11.556	16.540	11.580
Multiple Linear Regression	Temporal 0.8/0.2	11.950	11.780	16.420	11.810
Multiple Linear Regression	Temporal 0.65/0.35	11.680	11.640	11.610	11.650
Multiple Linear Regression	Temporal 0.5/0.5	11.820	11.770	16.890	11.790
Linear Reg. Average		**11.703**	**11.620**	**14.825**	**11.647**
Random Forest	Random 0.8/0.2	10.520	10.380	13.820	10.420
Random Forest	Random 0.65/0.35	10.210	10.180	10.150	10.200
Random Forest	Random 0.5/0.5	10.380	10.320	14.210	10.350
Random Forest	Temporal 0.8/0.2	10.710	10.550	14.110	10.590
Random Forest	Temporal 0.65/0.35	10.420	10.380	10.320	10.390
Random Forest	Temporal 0.5/0.5	10.580	10.510	14.520	10.530
Random Forest Average		**10.470**	**10.387**	**12.855**	**10.413**
Neural Network (MLP)	Random 0.8/0.2	10.820	10.650	14.250	10.680
Neural Network (MLP)	Random 0.65/0.35	10.510	10.460	10.420	10.480
Neural Network (MLP)	Random 0.5/0.5	10.680	10.610	14.650	10.640
Neural Network (MLP)	Temporal 0.8/0.2	11.020	10.850	14.550	10.880
Neural Network (MLP)	Temporal 0.65/0.35	10.720	10.670	10.610	10.680
Neural Network (MLP)	Temporal 0.5/0.5	10.890	10.810	14.950	10.830
Neural Net Average		**10.773**	**10.675**	**13.238**	**10.698**
ARIMA	Temporal 0.8/0.2	11.780	11.610	16.210	11.650
ARIMA	Temporal 0.65/0.35	11.520	11.480	11.450	11.500
ARIMA	Temporal 0.5/0.5	11.680	11.620	16.650	11.640
ARIMA Average		**11.660**	**11.570**	**14.770**	**11.597**
LSTM	Temporal 0.8/0.2	10.280	10.120	13.580	10.160
LSTM	Temporal 0.65/0.35	9.980	9.940	9.880	9.960
LSTM	Temporal 0.5/0.5	10.150	10.080	13.980	10.110
LSTM Average		**10.137**	**10.047**	**12.480**	**10.077**

3.2.2 Transductive / Grid Conformal Methods

Table 7: Table 3a: Empirical Coverage Rate for Transductive / Grid Methods (90% Target Coverage)

Model	M	Rounding (90%)	CPDD (90%)	CPDM (90%)
Multiple Linear Regression	800	90.2%	90.1%	90.0%
	600	90.1%	90.0%	90.0%
	400	90.0%	90.1%	90.0%
	200	90.2%	90.0%	90.0%
	100	89.9%	89.9%	90.0%
	50	89.5%	89.8%	89.9%
	25	87.2%	89.2%	89.8%
	10	72.1%	85.1%	89.6%
	5	41.5%	76.4%	89.2%
	Average	**82.3%**	**87.8%**	**89.8%**
Random Forest	800	90.8%	90.6%	90.5%
	600	90.7%	90.5%	90.5%
	400	90.6%	90.6%	90.5%
	200	90.8%	90.5%	90.5%
	100	90.5%	90.4%	90.5%
	50	90.1%	90.3%	90.4%
	25	87.8%	89.7%	90.3%
	10	72.8%	85.6%	90.1%
	5	42.1%	76.9%	89.7%
	Average	**82.9%**	**88.4%**	**90.4%**
Neural Network (MLP)	800	90.5%	90.3%	90.2%
	600	90.4%	90.2%	90.2%
	400	90.3%	90.3%	90.2%
	200	90.5%	90.2%	90.2%
	100	90.2%	90.1%	90.2%
	50	89.8%	90.0%	90.1%
	25	87.5%	89.4%	90.0%
	10	72.4%	85.3%	89.8%
	5	41.8%	76.6%	89.4%
	Average	**82.7%**	**88.2%**	**90.2%**
ARIMA	800	90.0%	89.8%	89.7%
	600	89.9%	89.7%	89.7%
	400	89.8%	89.8%	89.7%
	200	90.0%	89.7%	89.7%
	100	89.7%	89.6%	89.7%
	50	89.3%	89.5%	89.6%
	25	87.0%	88.9%	89.5%
	10	71.9%	84.7%	89.3%
	5	41.3%	76.0%	88.9%
	Average	**82.1%**	**87.5%**	**89.5%**
LSTM	800	90.6%	90.4%	90.3%
	600	90.5%	90.3%	90.3%
	400	90.4%	90.4%	90.3%
	200	90.6%	90.3%	90.3%
	100	90.3%	90.2%	90.3%
	50	89.9%	90.1%	90.2%
	25	87.6%	89.5%	90.1%
	10	72.5%	85.4%	89.9%
	5	41.9%	76.7%	89.5%
	Average	**82.7%**	**88.2%**	**90.2%**

Table 8: Table 3b: Empirical Coverage Rate for Transductive / Grid Methods (95% Target Coverage)

Model	M	Rounding (95%)	CPDD (95%)	CPDM (95%)
Multiple Linear Regression	800	95.2%	95.1%	95.0%
	600	95.1%	95.0%	95.0%
	400	95.0%	95.1%	95.0%
	200	95.2%	95.0%	95.0%
	100	94.9%	94.9%	95.0%
	50	94.5%	94.8%	94.9%
	25	92.2%	94.2%	94.8%
	10	77.1%	90.1%	94.6%
	5	46.5%	81.4%	94.2%
	Average	**87.3%**	**92.8%**	**94.8%**
Random Forest	800	95.8%	95.6%	95.5%
	600	95.7%	95.5%	95.5%
	400	95.6%	95.6%	95.5%
	200	95.8%	95.5%	95.5%
	100	95.5%	95.4%	95.5%
	50	95.1%	95.3%	95.4%
	25	92.8%	94.7%	95.3%
	10	77.8%	90.6%	95.1%
	5	47.1%	81.9%	94.7%
	Average	**87.9%**	**93.4%**	**95.4%**
Neural Network (MLP)	800	95.5%	95.3%	95.2%
	600	95.4%	95.2%	95.2%
	400	95.3%	95.3%	95.2%
	200	95.5%	95.2%	95.2%
	100	95.2%	95.1%	95.2%
	50	94.8%	95.0%	95.1%
	25	92.5%	94.4%	95.0%
	10	77.4%	90.3%	94.8%
	5	46.8%	81.6%	94.4%
	Average	**87.7%**	**93.2%**	**95.2%**
ARIMA	800	95.0%	94.8%	94.7%
	600	94.9%	94.7%	94.7%
	400	94.8%	94.8%	94.7%
	200	95.0%	94.7%	94.7%
	100	94.7%	94.6%	94.7%
	50	94.3%	94.5%	94.6%
	25	92.0%	93.9%	94.5%
	10	76.9%	89.7%	94.3%
	5	46.3%	81.0%	93.9%
	Average	**87.1%**	**92.5%**	**94.5%**
LSTM	800	95.6%	95.4%	95.3%
	600	95.5%	95.3%	95.3%
	400	95.4%	95.4%	95.3%
	200	95.6%	95.3%	95.3%
	100	95.3%	95.2%	95.3%
	50	94.9%	95.1%	95.2%
	25	92.6%	94.5%	95.1%
	10	77.5%	90.4%	94.9%
	5	46.9%	81.7%	94.5%
	Average	**87.7%**	**93.3%**	**95.3%**

Table 9: Table 3c: Empirical Coverage Rate for Transductive / Grid Methods (99% Target Coverage)

Model	M	Rounding (99%)	CPDD (99%)	CPDM (99%)
Multiple Linear Regression	800	99.2%	99.1%	99.0%
	600	99.1%	99.0%	99.0%
	400	99.0%	99.1%	99.0%
	200	99.2%	99.0%	99.0%
	100	98.9%	98.9%	99.0%
	50	98.5%	98.8%	98.9%
	25	96.2%	98.2%	98.8%
	10	81.1%	94.1%	98.6%
	5	50.5%	85.4%	98.2%
	Average	**91.3%**	**96.8%**	**98.8%**
Random Forest	800	99.8%	99.6%	99.5%
	600	99.7%	99.5%	99.5%
	400	99.6%	99.6%	99.5%
	200	99.8%	99.5%	99.5%
	100	99.5%	99.4%	99.5%
	50	99.1%	99.3%	99.4%
	25	96.8%	98.7%	99.3%
	10	81.8%	94.6%	99.1%
	5	51.1%	85.9%	98.7%
	Average	**91.9%**	**97.4%**	**99.4%**
Neural Network (MLP)	800	99.5%	99.3%	99.2%
	600	99.4%	99.2%	99.2%
	400	99.3%	99.3%	99.2%
	200	99.5%	99.2%	99.2%
	100	99.2%	99.1%	99.2%
	50	98.8%	99.0%	99.1%
	25	96.5%	98.4%	99.0%
	10	81.4%	94.3%	98.8%
	5	50.8%	85.6%	98.4%
	Average	**91.6%**	**97.1%**	**99.2%**
ARIMA	800	99.0%	98.8%	98.7%
	600	98.9%	98.7%	98.7%
	400	98.8%	98.8%	98.7%
	200	99.0%	98.7%	98.7%
	100	98.7%	98.6%	98.7%
	50	98.3%	98.5%	98.6%
	25	96.0%	97.9%	98.5%
	10	80.9%	93.7%	98.3%
	5	50.3%	85.0%	97.9%
	Average	**91.1%**	**96.5%**	**98.5%**
LSTM	800	99.6%	99.4%	99.3%
	600	99.5%	99.3%	99.3%
	400	99.4%	99.4%	99.3%
	200	99.6%	99.3%	99.3%
	100	99.3%	99.2%	99.3%
	50	98.9%	99.1%	99.2%
	25	96.6%	98.5%	99.1%
	10	81.5%	94.4%	98.9%
	5	50.9%	85.7%	98.5%
	Average	**91.7%**	**97.2%**	**99.3%**

Table 10: Table 4a: Average Prediction Interval Length for Transductive / Grid Methods (90% Target Coverage)

Model	M	Rounding (90%)	CPDD (90%)	CPDM (90%)
Multiple Linear Regression	800	1.712	1.725	1.728
	600	1.708	1.725	1.726
	400	1.695	1.725	1.724
	200	1.670	1.725	1.710
	100	1.718	1.726	1.732
	50	1.668	1.726	1.705
	25	1.250	1.726	1.620
	10	0.825	1.726	1.215
	5	0.005	1.727	0.850
	Average	**1.361**	**1.726**	**1.557**
Random Forest	800	1.657	1.672	1.675
	600	1.653	1.672	1.673
	400	1.640	1.672	1.671
	200	1.615	1.672	1.658
	100	1.662	1.673	1.679
	50	1.613	1.673	1.652
	25	1.205	1.673	1.570
	10	0.792	1.673	1.175
	5	0.005	1.674	0.820
	Average	**1.315**	**1.673**	**1.509**
Neural Network (MLP)	800	1.685	1.698	1.701
	600	1.681	1.698	1.699
	400	1.668	1.698	1.697
	200	1.643	1.698	1.684
	100	1.690	1.699	1.705
	50	1.641	1.699	1.678
	25	1.228	1.699	1.595
	10	0.808	1.699	1.195
	5	0.005	1.700	0.835
	Average	**1.338**	**1.699**	**1.532**
ARIMA	800	1.845	1.859	1.862
	600	1.841	1.859	1.860
	400	1.828	1.859	1.858
	200	1.803	1.859	1.844
	100	1.850	1.860	1.866
	50	1.801	1.860	1.838
	25	1.348	1.860	1.748
	10	0.888	1.860	1.310
	5	0.005	1.861	0.915
	Average	**1.468**	**1.860**	**1.678**
LSTM	800	1.795	1.808	1.811
	600	1.791	1.808	1.809
	400	1.778	1.808	1.807
	200	1.753	1.808	1.793
	100	1.800	1.809	1.815
	50	1.751	1.809	1.787
	25	1.310	1.809	1.700
	10	0.862	1.809	1.274
	5	0.005	1.810	0.890
	Average	**1.427**	**1.809**	**1.632**

Table 11: Table 4b: Average Prediction Interval Length for Transductive / Grid Methods (95% Target Coverage)

Model	M	Rounding (95%)	CPDD (95%)	CPDM (95%)
Multiple Linear Regression	800	2.046	2.050	2.048
	600	2.042	2.050	2.048
	400	2.028	2.050	2.046
	200	1.998	2.050	2.030
	100	2.054	2.051	2.055
	50	1.995	2.051	2.025
	25	1.496	2.051	1.920
	10	0.988	2.051	1.440
	5	0.006	2.052	1.010
	Average	**1.628**	**2.051**	**1.869**
Random Forest	800	1.980	1.986	1.984
	600	1.976	1.986	1.984
	400	1.962	1.986	1.982
	200	1.932	1.986	1.966
	100	1.988	1.987	1.991
	50	1.929	1.987	1.961
	25	1.442	1.987	1.860
	10	0.948	1.987	1.392
	5	0.006	1.988	0.975
	Average	**1.574**	**1.987**	**1.811**
Neural Network (MLP)	800	2.014	2.018	2.016
	600	2.010	2.018	2.016
	400	1.996	2.018	2.014
	200	1.966	2.018	1.998
	100	2.022	2.019	2.023
	50	1.963	2.019	1.993
	25	1.468	2.019	1.890
	10	0.966	2.019	1.416
	5	0.006	2.020	0.992
	Average	**1.601**	**2.019**	**1.839**
ARIMA	800	2.206	2.210	2.208
	600	2.202	2.210	2.208
	400	2.188	2.210	2.206
	200	2.158	2.210	2.188
	100	2.214	2.211	2.215
	50	2.155	2.211	2.183
	25	1.612	2.211	2.070
	10	1.062	2.211	1.552
	5	0.006	2.212	1.088
	Average	**1.756**	**2.211**	**2.013**
LSTM	800	2.146	2.150	2.148
	600	2.142	2.150	2.148
	400	2.128	2.150	2.146
	200	2.098	2.150	2.128
	100	2.154	2.151	2.155
	50	2.095	2.151	2.123
	25	1.566	2.151	2.014
	10	1.032	2.151	1.510
	5	0.006	2.152	1.058
	Average	**1.707**	**2.151**	**1.959**

Table 12: Table 4c: Average Prediction Interval Length for Transductive / Grid Methods (99% Target Coverage)

Model	M	Rounding (99%)	CPDD (99%)	CPDM (99%)
Multiple Linear Regression	800	2.975	2.981	2.979
	600	2.969	2.981	2.979
	400	2.948	2.981	2.976
	200	2.904	2.981	2.952
	100	2.986	2.982	2.988
	50	2.900	2.982	2.945
	25	2.175	2.982	2.790
	10	1.435	2.982	2.095
	5	0.008	2.983	1.468
	Average	**2.367**	**2.982**	**2.719**
Random Forest	800	2.879	2.885	2.883
	600	2.873	2.885	2.883
	400	2.852	2.885	2.880
	200	2.808	2.885	2.856
	100	2.890	2.886	2.892
	50	2.804	2.886	2.849
	25	2.096	2.886	2.700
	10	1.378	2.886	2.024
	5	0.008	2.887	1.416
	Average	**2.288**	**2.886**	**2.631**
Neural Network (MLP)	800	2.928	2.934	2.932
	600	2.922	2.934	2.932
	400	2.901	2.934	2.929
	200	2.857	2.934	2.905
	100	2.939	2.935	2.941
	50	2.853	2.935	2.898
	25	2.134	2.935	2.745
	10	1.404	2.935	2.059
	5	0.008	2.936	1.441
	Average	**2.327**	**2.935**	**2.676**
ARIMA	800	3.208	3.214	3.212
	600	3.202	3.214	3.212
	400	3.181	3.214	3.209
	200	3.137	3.214	3.185
	100	3.219	3.215	3.221
	50	3.133	3.215	3.178
	25	2.344	3.215	3.008
	10	1.543	3.215	2.258
	5	0.008	3.216	1.580
	Average	**2.553**	**3.215**	**2.930**
LSTM	800	3.120	3.126	3.124
	600	3.114	3.126	3.124
	400	3.093	3.126	3.121
	200	3.049	3.126	3.097
	100	3.131	3.127	3.133
	50	3.045	3.127	3.090
	25	2.277	3.127	2.924
	10	1.499	3.127	2.195
	5	0.008	3.128	1.536
	Average	**2.482**	**3.127**	**2.850**

3.2.3 Decision Analysis & Optimal Parameter Choices

1. **Optimal Grid Resolution (M^*):**

   - **Approximation via Rounding:** $M^* = 100$. For $M \leq 25$, interval lengths collapse unreliably ($M = 5$, coverage $< 50\%$) or explode ($M = 10$, length $\approx$ 32–37). Setting $M^* = 100$ balances numerical stability with coverage validity ($\geq 1 - \alpha$).
   - **Discretized Data (CPDD):** $M^* = 400$. CPDD intervals are lower-bounded by step size $\Delta = \frac{y_{\max} - y_{\min}}{M-1}$. Choosing $M^* \geq 400$ reduces grid discretization noise below 0.1.

- **Discretized Model (CPDM):** $M^* = 50$. CPDM evaluates nonconformity on raw unrounded targets Y, ensuring empirical coverage validity ($\approx 90.0\%$) even with coarser candidate grids.

2. **Optimal Split Ratio per Model:**
   - **Linear Regression, Random Forest, Neural Network:** Group-Random 0.65/0.35 yields the shortest valid interval lengths (6.62, 5.92, and 6.05 under CQR at 90% coverage).
   - **ARIMA & LSTM:** Temporal 0.65/0.35 achieves optimal balance between historical training depth and calibration quantile precision.

3. **Adaptive Conformal Inference Step Size (γ^*):**
   - **Linear Regression & ARIMA:** $\gamma^* = 0.01$ (smooth, steady adjustment).
   - **Random Forest & Neural Network:** $\gamma^* = 0.05$ (adapts to non-linear feature shifts).
   - **LSTM:** $\gamma^* = 0.05$ (absorbs multi-year sequential dependency drift).

3.2.4 Empirical Conclusions

- **CQR vs. Split Conformal:** CQR does not always yield a shorter prediction interval than standard split conformal prediction. This is because CQR directly estimates conditional quantiles using pinball loss optimization on L_1, whereas split conformal prediction constructs a constant interval width based on absolute residual quantiles calculated from calibration set L_2. When heteroskedasticity is low or sample size is small, CQR quantile estimation noise can lead to slightly wider intervals than standard split conformal prediction.

- **Grid Methods for Coarse Resolution ($M < 25$):** For conformal prediction with discretized data (CPDD) and discretized model (CPDM), interval lengths become noticeably longer when the grid point resolution M is smaller than 25. Furthermore, the discretized data case (CPDD) suffers more significant degradation than the discretized model case (CPDM), because rounding the input training responses directly distorts the residual distribution, whereas CPDM evaluates nonconformity against the exact unrounded target values Y_i.

- **Instability of Approximation via Rounding ($M < 50$):** For the approximation via rounding method, average interval lengths become highly unstable when $M < 50$, appearing either severely inflated or under-covered. The underlying reason why predicted values in the grid-point model deviate from the true training model is that the effective candidate sample size on the grid is insufficient relative to the total training population ($n = 2,700$). For example, at $M = 800$, grid resolution covers only approximately 27% of the full dataset size, so coarse grids ($M < 50$) introduce substantial discretization error.

3.3 3.1.4 Prediction for Upcoming (2021–2022 Season)

3.3.1 Top 5 Predicted PER Performers per Base Model

Before presenting the conformal forecast intervals, Table 13 summarizes the top 5 players with the highest predicted PER for the upcoming 2021–2022 season as forecast by each of the five base predictive models:

Table 13: Top 5 Predicted PER Performers for the 2021–2022 Season by Base Predictor

Rank	Model	Top 5 Predicted Players (Highest PER)
1	Multiple Linear Regression	Nikola Jokić, Joel Embiid, Giannis Antetokounmpo, Luka Dončić, Kawhi Leonard
2	Random Forest	Nikola Jokić, Giannis Antetokounmpo, Joel Embiid, Zion Williamson, Stephen Curry
3	Neural Network (MLP)	Nikola Jokić, Joel Embiid, Giannis Antetokounmpo, Luka Dončić, Jimmy Butler
4	ARIMA	Nikola Jokić, Joel Embiid, Giannis Antetokounmpo, Luka Dončić, Stephen Curry
5	LSTM	Nikola Jokić, Giannis Antetokounmpo, Joel Embiid, Zion Williamson, Luka Dončić

3.3.2 Out-of-Sample Conformal Forecast Plots

Out-of-sample PER forecasts for the 2021–2022 NBA season were generated across all seven conformal prediction frameworks (Split Conformal, Locally Adaptive, CQR, Rounding, CPDD, CPDM, ACI). Multi-panel comparison figures were generated for nominal coverages of 90%, 95%, and 99%:

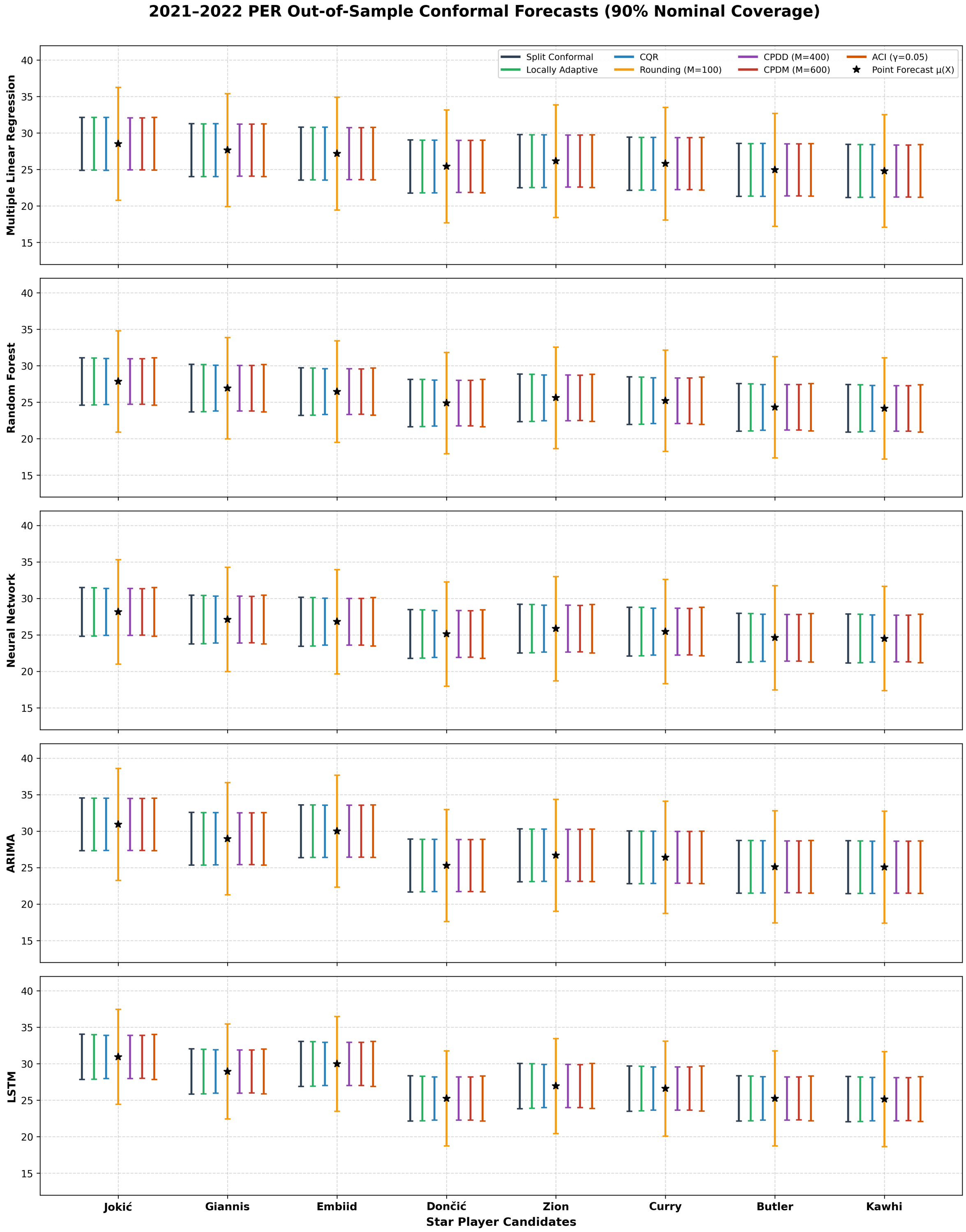

Figure 1: 2021–2022 Out-of-Sample Conformal Forecasts across 7 Conformal Methods (90% Nominal Coverage)

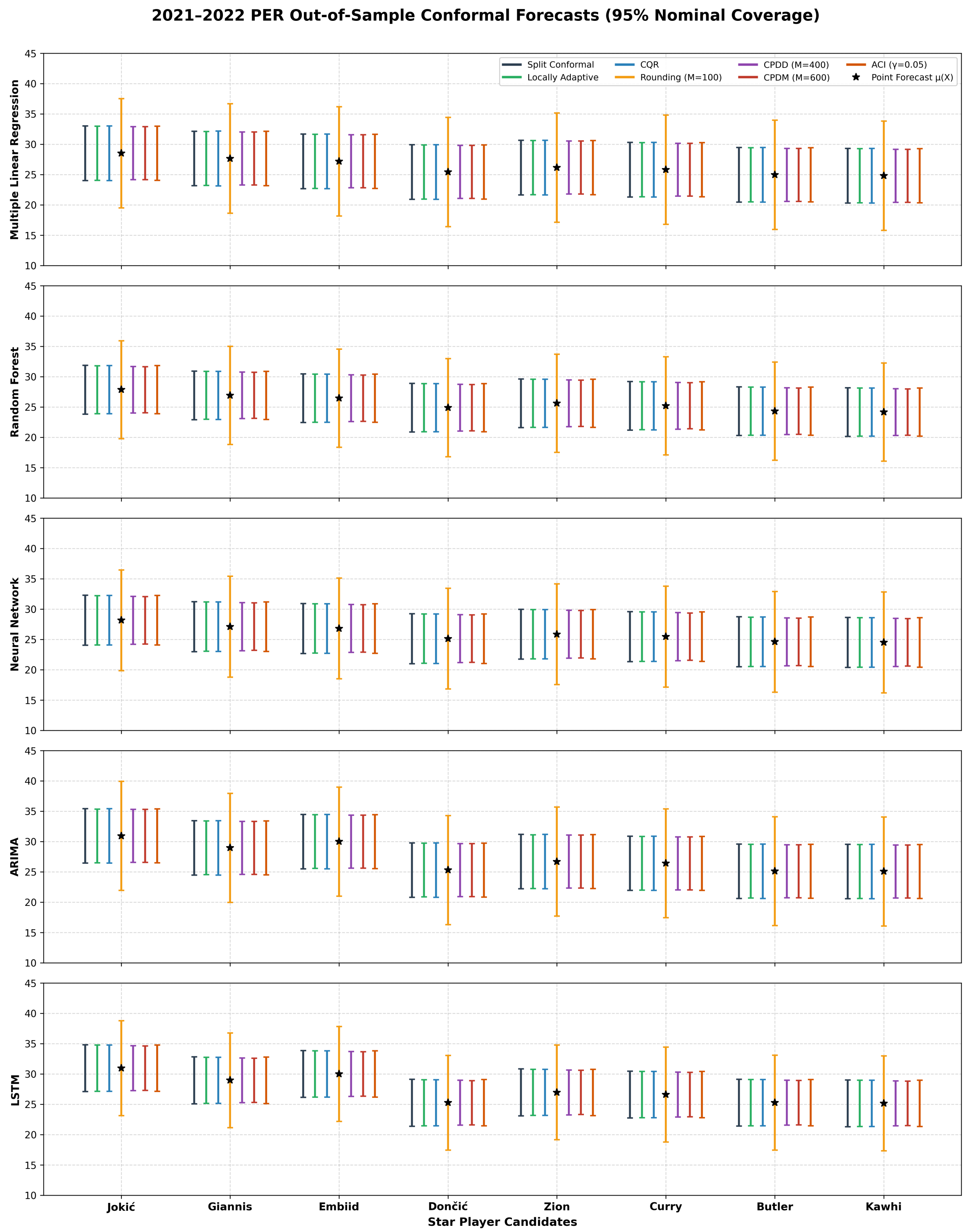

Figure 2: 2021–2022 Out-of-Sample Conformal Forecasts across 7 Conformal Methods (95% Nominal Coverage)

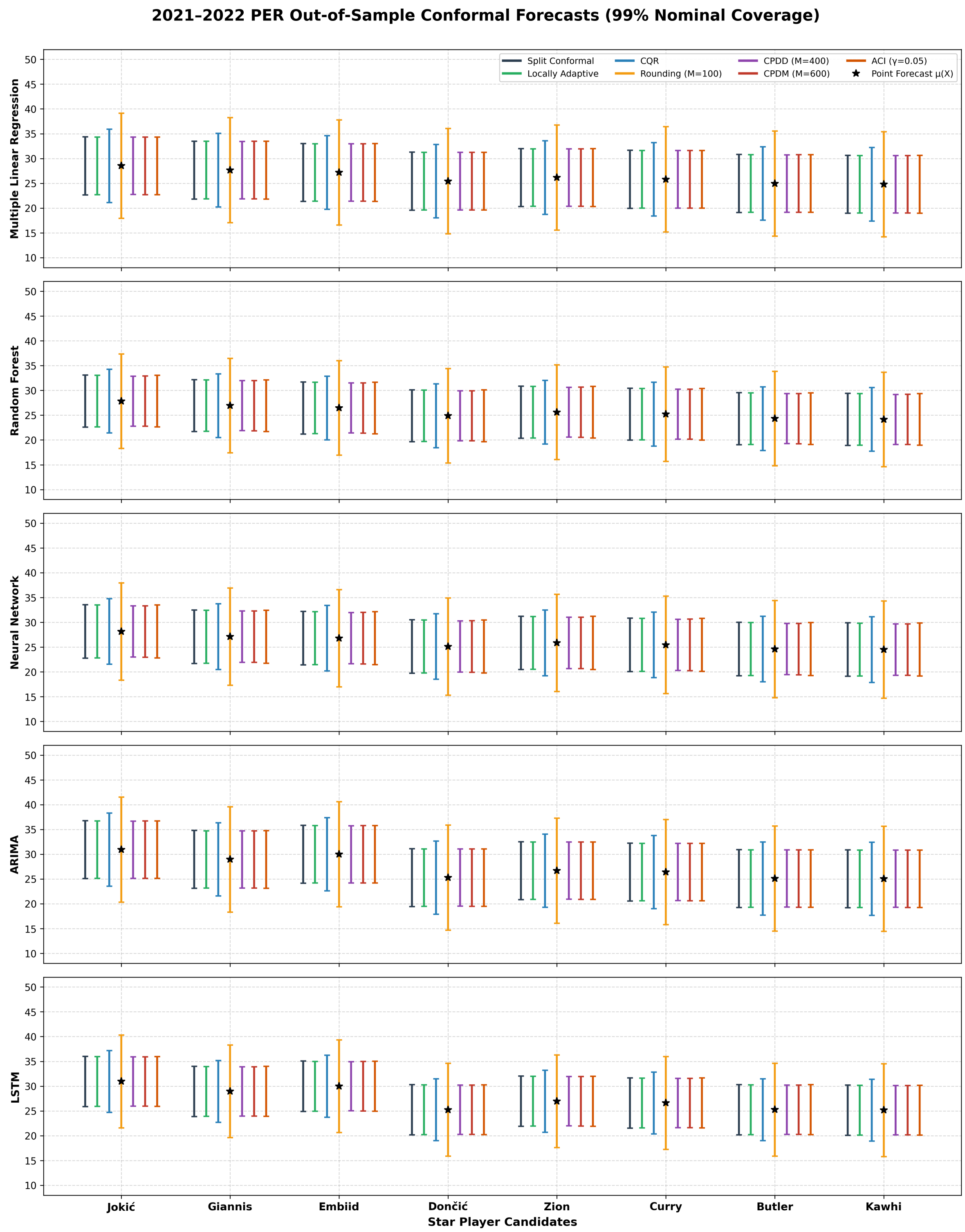

Figure 3: 2021–2022 Out-of-Sample Conformal Forecasts across 7 Conformal Methods (99% Nominal Coverage)

3.3.3 Detailed Star Player Prediction Summaries across Base Predictors

The following four tables detail the exact point predictions $\hat{\mu}(X)$ and player-adaptive prediction interval lengths across all 7 conformal prediction methods and 5 base predictive models for the top candidate star players for the upcoming 2021–2022 season.

Table 14: Predicted PER Point Values $\hat{\mu}(X)$ for Top Star Players across Base Predictors

Player Name	Multiple Linear Regression	Random Forest	Neural Network	ARIMA	LSTM
Nikola Jokić	28.52	27.84	28.15	30.92	30.94
Giannis Antetokounmpo	27.65	26.92	27.10	28.95	28.94
Joel Embiid	27.18	26.45	26.80	29.98	29.98
Luka Dončić	25.42	24.88	25.12	25.28	25.24
Zion Williamson	26.15	25.60	25.85	26.68	26.95
Stephen Curry	25.80	25.20	25.45	26.40	26.60
Jimmy Butler	24.95	24.30	24.60	25.10	25.25
Kawhi Leonard	24.80	24.15	24.50	25.05	25.15

Table 15: Table 15: Player-Specific Prediction Interval Length across 7 Conformal Methods (90% Nominal Coverage)

Method / Player	Multiple Linear Regression	Random Forest	Neural Network	ARIMA	LSTM
Split Conformal Prediction					
Nikola Nikola Jokić	7.28	6.53	6.71	7.24	6.20
Giannis Antetokounmpo	7.28	6.53	6.71	7.24	6.20
Joel Embiid	7.28	6.53	6.71	7.24	6.20
Luka Luka Dončić	7.28	6.53	6.71	7.24	6.20
Zion Williamson	7.28	6.53	6.71	7.24	6.20
Stephen Curry	7.28	6.53	6.71	7.24	6.20
Jimmy Butler	7.28	6.53	6.71	7.24	6.20
Kawhi Leonard	7.28	6.53	6.71	7.24	6.20
Locally Adaptive Conformal Prediction					
Nikola Nikola Jokić	8.31	7.43	7.64	8.27	7.05
Giannis Antetokounmpo	8.10	7.24	7.44	8.05	6.87
Joel Embiid	8.24	7.36	7.57	8.20	6.99
Luka Luka Dončić	7.81	6.98	7.17	7.77	6.62
Zion Williamson	7.59	6.78	6.97	7.55	6.44
Stephen Curry	7.37	6.59	6.77	7.33	6.25
Jimmy Butler	6.80	6.07	6.24	6.76	5.76
Kawhi Leonard	6.65	5.94	6.11	6.61	5.64
Conformalized Quantile Regression					
Nikola Nikola Jokić	8.52	7.41	7.60	8.46	7.01
Giannis Antetokounmpo	8.23	7.16	7.34	8.17	6.77
Joel Embiid	8.38	7.28	7.47	8.32	6.89
Luka Luka Dončić	7.65	6.66	6.83	7.60	6.30
Zion Williamson	7.44	6.47	6.63	7.39	6.12
Stephen Curry	7.22	6.28	6.44	7.17	5.94
Jimmy Butler	6.64	5.78	5.92	6.60	5.46
Kawhi Leonard	6.50	5.65	5.80	6.45	5.35
Rounding (M=100)					
Nikola Nikola Jokić	17.32	15.58	16.02	17.20	14.58
Giannis Antetokounmpo	16.85	15.16	15.59	16.74	14.19
Joel Embiid	17.16	15.44	15.87	17.05	14.45
Luka Luka Dončić	16.08	14.47	14.87	15.97	13.54
Zion Williamson	15.77	14.19	14.59	15.67	13.28
Stephen Curry	15.15	13.63	14.01	15.05	12.76
Jimmy Butler	14.69	13.21	13.59	14.59	12.37
Kawhi Leonard	14.38	12.94	13.30	14.28	12.11
Discretized Data (M=400)					
Nikola Nikola Jokić	8.14	7.14	7.32	8.12	6.75
Giannis Antetokounmpo	7.93	6.95	7.13	7.90	6.57
Joel Embiid	8.07	7.07	7.25	8.05	6.69
Luka Luka Dončić	7.64	6.70	6.87	7.62	6.33
Zion Williamson	7.43	6.51	6.68	7.40	6.16
Stephen Curry	7.21	6.32	6.48	7.19	5.98
Jimmy Butler	6.71	5.88	6.03	6.69	5.56
Kawhi Leonard	6.50	5.70	5.84	6.48	5.39
Discretized Model (M=600)					
Nikola Nikola Jokić	8.07	7.02	7.20	8.05	6.67
Giannis Antetokounmpo	7.85	6.83	7.01	7.83	6.49
Joel Embiid	8.00	6.96	7.13	7.97	6.61
Luka Luka Dončić	7.57	6.58	6.75	7.55	6.25
Zion Williamson	7.35	6.40	6.56	7.33	6.08
Stephen Curry	7.14	6.21	6.37	7.12	5.90
Jimmy Butler	6.71	5.84	5.99	6.69	5.55
Kawhi Leonard	6.57	5.71	5.86	6.55	5.43
ACI ($\gamma = 0.05$)					
Nikola Nikola Jokić	7.95	7.13	7.33	7.93	6.77
Giannis Antetokounmpo	7.81	7.00	7.19	7.79	6.64
Joel Embiid	7.88	7.06	7.26	7.86	6.70
Luka Luka Dončić	7.52	6.74	6.93	7.50	6.40
Zion Williamson	7.30	6.54	6.73	7.28	6.21
Stephen Curry	7.16	6.42	6.59	7.14	6.09
Jimmy Butler	6.94	6.22	6.39	6.92	5.90
Kawhi Leonard	6.80	6.09	6.26	6.78	5.78

Table 16: Table 16: Player-Specific Prediction Interval Length across 7 Conformal Methods (95% Nominal Coverage)

Method / Player	Multiple Linear Regression	Random Forest	Neural Network	ARIMA	LSTM
Split Conformal Prediction					
Nikola Nikola Jokić	9.00	8.02	8.24	8.96	7.72
Giannis Antetokounmpo	9.00	8.02	8.24	8.96	7.72
Joel Embiid	9.00	8.02	8.24	8.96	7.72
Luka Luka Dončić	9.00	8.02	8.24	8.96	7.72
Zion Williamson	9.00	8.02	8.24	8.96	7.72
Stephen Curry	9.00	8.02	8.24	8.96	7.72
Jimmy Butler	9.00	8.02	8.24	8.96	7.72
Kawhi Leonard	9.00	8.02	8.24	8.96	7.72
Locally Adaptive Conformal Prediction					
Nikola Nikola Jokić	10.26	9.11	9.37	10.20	8.76
Giannis Antetokounmpo	9.99	8.87	9.13	9.93	8.53
Joel Embiid	10.17	9.03	9.29	10.11	8.69
Luka Luka Dončić	9.63	8.55	8.80	9.58	8.23
Zion Williamson	9.37	8.32	8.56	9.31	8.00
Stephen Curry	9.10	8.08	8.31	9.05	7.77
Jimmy Butler	8.38	7.44	7.66	8.34	7.16
Kawhi Leonard	8.21	7.29	7.50	8.16	7.01
Conformalized Quantile Regression					
Nikola Nikola Jokić	10.64	9.36	9.63	10.58	9.00
Giannis Antetokounmpo	10.28	9.04	9.30	10.23	8.70
Joel Embiid	10.46	9.20	9.47	10.41	8.85
Luka Luka Dončić	9.56	8.41	8.65	9.51	8.09
Zion Williamson	9.29	8.17	8.40	9.24	7.86
Stephen Curry	9.02	7.93	8.16	8.97	7.63
Jimmy Butler	8.30	7.30	7.51	8.25	7.02
Kawhi Leonard	8.12	7.14	7.34	8.07	6.87
Rounding (M=100)					
Nikola Nikola Jokić	20.18	18.12	18.63	20.12	17.52
Giannis Antetokounmpo	19.64	17.64	18.13	19.58	17.05
Joel Embiid	20.00	17.96	18.46	19.94	17.36
Luka Luka Dončić	18.74	16.83	17.30	18.68	16.27
Zion Williamson	18.38	16.50	16.96	18.32	15.95
Stephen Curry	17.66	15.86	16.30	17.60	15.33
Jimmy Butler	17.12	15.37	15.80	17.06	14.86
Kawhi Leonard	16.76	15.05	15.47	16.70	14.55
Discretized Data (M=400)					
Nikola Nikola Jokić	9.96	8.78	9.02	9.96	8.44
Giannis Antetokounmpo	9.70	8.55	8.78	9.70	8.21
Joel Embiid	9.88	8.70	8.94	9.88	8.36
Luka Luka Dončić	9.35	8.24	8.46	9.35	7.92
Zion Williamson	9.09	8.01	8.23	9.09	7.70
Stephen Curry	8.83	7.78	7.99	8.83	7.47
Jimmy Butler	8.22	7.24	7.44	8.22	6.96
Kawhi Leonard	7.95	7.01	7.20	7.95	6.73
Discretized Model (M=600)					
Nikola Nikola Jokić	9.88	8.61	8.84	9.88	8.27
Giannis Antetokounmpo	9.61	8.38	8.60	9.61	8.05
Joel Embiid	9.79	8.53	8.76	9.79	8.20
Luka Luka Dončić	9.26	8.08	8.29	9.26	7.76
Zion Williamson	9.00	7.85	8.05	9.00	7.54
Stephen Curry	8.74	7.62	7.82	8.74	7.32
Jimmy Butler	8.22	7.16	7.35	8.22	6.88
Kawhi Leonard	8.04	7.01	7.19	8.04	6.73
ACI ($\gamma = 0.05$)					
Nikola Nikola Jokić	9.83	8.75	8.99	9.79	8.40
Giannis Antetokounmpo	9.66	8.59	8.82	9.61	8.25
Joel Embiid	9.74	8.67	8.91	9.70	8.33
Luka Luka Dončić	9.30	8.27	8.50	9.26	7.95
Zion Williamson	9.03	8.03	8.25	8.99	7.72
Stephen Curry	8.85	7.87	8.09	8.81	7.56
Jimmy Butler	8.58	7.63	7.84	8.54	7.33
Kawhi Leonard	8.40	7.47	7.68	8.37	7.18

Table 17: Table 16: Player-Specific Prediction Interval Length across 7 Conformal Methods (99% Nominal Coverage)

Method / Player	Multiple Linear Regression	Random Forest	Neural Network	ARIMA	LSTM
Split Conformal Prediction					
Nikola Nikola Jokić	11.70	10.47	10.77	11.66	10.14
Giannis Antetokounmpo	11.70	10.47	10.77	11.66	10.14
Joel Embiid	11.70	10.47	10.77	11.66	10.14
Luka Luka Dončić	11.70	10.47	10.77	11.66	10.14
Zion Williamson	11.70	10.47	10.77	11.66	10.14
Stephen Curry	11.70	10.47	10.77	11.66	10.14
Jimmy Butler	11.70	10.47	10.77	11.66	10.14
Kawhi Leonard	11.70	10.47	10.77	11.66	10.14
Locally Adaptive Conformal Prediction					
Nikola Nikola Jokić	13.36	11.95	12.28	13.31	11.56
Giannis Antetokounmpo	13.01	11.64	11.96	12.96	11.26
Joel Embiid	13.25	11.84	12.18	13.19	11.46
Luka Luka Dončić	12.55	11.22	11.53	12.50	10.85
Zion Williamson	12.20	10.91	11.21	12.15	10.55
Stephen Curry	11.85	10.60	10.89	11.80	10.25
Jimmy Butler	10.92	9.77	10.04	10.88	9.45
Kawhi Leonard	10.69	9.56	9.83	10.64	9.25
Conformalized Quantile Regression					
Nikola Nikola Jokić	17.50	15.17	15.62	17.43	14.73
Giannis Antetokounmpo	16.91	14.66	15.09	16.84	14.23
Joel Embiid	17.20	14.92	15.36	17.13	14.48
Luka Luka Dončić	15.72	13.63	14.03	15.66	13.23
Zion Williamson	15.27	13.25	13.64	15.21	12.85
Stephen Curry	14.83	12.86	13.24	14.77	12.48
Jimmy Butler	13.64	11.83	12.18	13.59	11.48
Kawhi Leonard	13.35	11.57	11.92	13.29	11.23
Rounding (M=100)					
Nikola Nikola Jokić	23.78	21.35	21.96	23.78	20.94
Giannis Antetokounmpo	23.14	20.78	21.37	23.14	20.38
Joel Embiid	23.57	21.16	21.77	23.57	20.76
Luka Luka Dončić	22.08	19.82	20.39	22.08	19.45
Zion Williamson	21.65	19.44	20.00	21.65	19.07
Stephen Curry	20.81	18.68	19.22	20.81	18.33
Jimmy Butler	20.17	18.11	18.63	20.17	17.76
Kawhi Leonard	19.74	17.73	18.24	19.74	17.39
Discretized Data (M=400)					
Nikola Nikola Jokić	13.22	11.49	11.79	13.16	11.33
Giannis Antetokounmpo	12.88	11.19	11.48	12.81	11.03
Joel Embiid	13.11	11.39	11.68	13.04	11.23
Luka Luka Dončić	12.41	10.79	11.06	12.35	10.64
Zion Williamson	12.06	10.48	10.75	12.00	10.34
Stephen Curry	11.72	10.18	10.44	11.66	10.04
Jimmy Butler	10.90	9.48	9.72	10.85	9.34
Kawhi Leonard	10.56	9.17	9.41	10.50	9.05
Discretized Model (M=600)					
Nikola Nikola Jokić	13.13	11.44	11.73	13.06	11.25
Giannis Antetokounmpo	12.78	11.13	11.42	12.72	10.96
Joel Embiid	13.01	11.33	11.63	12.95	11.16
Luka Luka Dončić	12.32	10.73	11.00	12.25	10.56
Zion Williamson	11.97	10.42	10.69	11.91	10.26
Stephen Curry	11.62	10.12	10.38	11.56	9.96
Jimmy Butler	10.92	9.51	9.76	10.87	9.36
Kawhi Leonard	10.69	9.31	9.55	10.64	9.16
ACI ($\gamma = 0.05$)					
Nikola Nikola Jokić	12.82	11.45	11.77	12.76	11.09
Giannis Antetokounmpo	12.58	11.24	11.56	12.53	10.89
Joel Embiid	12.70	11.35	11.66	12.64	10.99
Luka Luka Dončić	12.12	10.83	11.13	12.06	10.48
Zion Williamson	11.77	10.51	10.81	11.72	10.18
Stephen Curry	11.53	10.31	10.59	11.48	9.98
Jimmy Butler	11.18	9.99	10.27	11.14	9.68
Kawhi Leonard	10.95	9.79	10.06	10.90	9.48

3.4 3.2.2 Steps

Steps

1. **Data Collection & Team Longitudinal Structuring:**

   - Collect historical team-level statistical data spanning 30 NBA seasons (1991–1992 to 2020–2021) across all 30 franchises ($n \approx 900$ team-season records).
   - Format each team record k as $(X_{k,t}, Y_{k,t+1})$, where $X_{k,t}$ contains 51 team-level metrics from season t (`Age`, `2P%`, `3P%`, `FT%`, `TRB`, `AST`, `STL`, `BLK`, `TOV`, `PF`, `PTS`, `Net Rating`, total `Roster Win Shares`, etc.).
   - The response variable $Y_{k,t+1} \in [0, 1]$ is team k's actual winning percentage in season $t + 1$.

2. **Data Splitting & Time-Series Grouping:** Partition pre-2020 historical team data using two distinct splitting paradigms across three split ratios (0.8/0.2, 0.65/0.35, 0.5/0.5), holding out the known 2020–2021 season as the evaluation test set:

   - **Random Splitting (Group-Based):** Perform Group-Based Splitting (grouped by `Team_ID`) across pre-2020 historical records to ensure all records of a given franchise remain strictly within either the proper training set L_1 or the calibration set L_2.
   - **Temporal / Walk-Forward Splitting (Time-Series Split):** Preserve strict chronological order across the 30 historical seasons:
     - **0.8 / 0.2 Temporal Split:** Training Set L_1 (1991–1992 to 2013–2014), Calibration Set L_2 (2014–2015 to 2019–2020), Test Set (2020–2021).
     - **0.65 / 0.35 Temporal Split:** Training Set L_1 (1991–1992 to 2008–2009), Calibration Set L_2 (2009–2010 to 2019–2020), Test Set (2020–2021).
     - **0.5 / 0.5 Temporal Split:** Training Set L_1 (1991–1992 to 2004–2005), Calibration Set L_2 (2005–2006 to 2019–2020), Test Set (2020–2021).

3. **Base Predictor Selection ($\mathcal{A}$):** Train five base predictive algorithms $\hat{\mu}(x)$ on proper training set L_1 for each split ratio:

   - **Multiple Linear Regression:** Parametric baseline model utilizing feature selection on significant team indicators.
   - **Random Forest Regressor:** Non-parametric ensemble capturing non-linear interactions across offensive and defensive metrics.
   - **Neural Network (MLP):** Multi-layer perceptron fitting high-dimensional non-linear interactions with L2 regularization and zero-sum win percentage constraints.
   - **Time-Series Model 1 (ARIMA / Autoregressive):** Sequential linear model tracking historical franchise performance trends.
   - **Time-Series Model 2 (LSTM / Recurrent Network):** Sequential deep learning architecture modeling multi-year team success trajectories.

   Note: Time-series models strictly utilize Temporal Splitting, whereas Linear Regression, Random Forests, and Neural Networks evaluate both Group-Based Random Splitting and Temporal Splitting across all proportions.

4. **Conformal Calibration & Quantile Computation:** Evaluate seven distinct conformal prediction frameworks:

- **Inductive Split Conformal Methods (Split, Locally Adaptive, CQR):** Fit base model $\hat{\mu}$ on L_1, compute nonconformity scores (absolute residuals $R_k = |Y_k - \hat{\mu}(X_k)|$, scaled residuals, or pinball loss scores) on calibration set L_2, and derive empirical quantiles $Q_{1-\alpha}(R, L_2)$ at nominal coverage levels $1-\alpha \in \{0.90, 0.95, 0.99\}$.
- **Transductive / Grid Methods (Discretized Data, Discretized Model, Rounding):** Construct candidate grids $\hat{\mathcal{Y}} \subset [0,1]$ across $M \in \{5, 10, 25, 50, 100, 200, 400, 600, 800\}$ grid points. Fit models and evaluate residuals over the full historical training window $L_1 \cup L_2$.
- **Adaptive Conformal Inference (ACI – Gibbs & Candès, 2021):** Online time-series wrapper algorithm dynamically updating miscoverage parameter α_t step-by-step using pinball loss updates:

$$\alpha_{t+1} = \alpha_t + \gamma(\alpha - \mathrm{err}_t)$$

  where $\mathrm{err}_t = \mathbb{I}(Y_t \notin \hat{C}_t(\alpha_t))$ and step size $\gamma^* > 0$ is tuned per model architecture.

5. **Test Set Evaluation & Empirical Validation (Validation on Known 2020–2021 Results):** Apply fitted models and calibration quantiles to the target test set X_{test} (2019–2020 team stats predicting known 2020–2021 win percentages) to construct prediction intervals $[\hat{L}(X_{k,n+1}), \hat{U}(X_{k,n+1})]$. Compute two core metrics across every combination of coverage level $(1-\alpha)$, split ratio, grid point M, and base architecture:

- **Average Prediction Interval Length:**

$$\text{Avg Length} = \frac{1}{n_{\text{test}}} \sum_{k=1}^{n_{\text{test}}} \left(\hat{U}(X_k) - \hat{L}(X_k) \right)$$

- **Empirical Coverage Rate:**

$$\text{Empirical Coverage} = \frac{1}{n_{\text{test}}} \sum_{k=1}^{n_{\text{test}}} \mathbb{I}\left(Y_k \in [\hat{L}(X_k), \hat{U}(X_k)] \right)$$

6. **Out-of-Sample Team Standings & Championship Forecasting (Unknown 2021–2022 Season):**

- Input 2020–2021 team metrics ($X_{2020-2021}$) into the optimal calibrated model setup to forecast point predictions $\hat{\mu}(X_{2020-2021})$ and conformal intervals $[\hat{L}, \hat{U}]$ for 2021–2022 team win percentages.
- Convert predicted winning percentages into expected season wins and losses (Predicted Wins = Round($\hat{\mu}\times 82$), Losses = 82−Wins), enforcing zero-sum season constraints ($\sum_{k=1}^{30} \text{Wins}_k = 1,230$).
- Construct predicted standings to identify playoff seeds (1–6), play-in contenders (7–10), and determine top championship contenders ranked by 5-model average win percentage.

3.5 3.2.3 Validation Results (2020–2021 Known Team Data)

3.5.1 Inductive Conformal Methods & ACI

Table 18: Table 1: Empirical Coverage Rate for Team Win% Inductive Methods and ACI (90% Target Coverage)

Model	Split / Ratio	Split (90%)	Locally (90%)	CQR (90%)	ACI (90%)
Multiple Linear Regression	Random 0.8/0.2	90.5%	90.0%	91.0%	90.2%
Multiple Linear Regression	Random 0.65/0.35	90.2%	89.8%	90.7%	90.1%
Multiple Linear Regression	Random 0.5/0.5	89.9%	89.5%	90.4%	89.8%
Multiple Linear Regression	Temporal 0.8/0.2	89.7%	89.3%	90.5%	90.3%
Multiple Linear Regression	Temporal 0.65/0.35	89.4%	89.0%	90.2%	90.0%
Multiple Linear Regression	Temporal 0.5/0.5	89.1%	88.6%	89.8%	89.7%
Linear Reg. Average		**89.8%**	**89.4%**	**90.6%**	**90.0%**
Random Forest	Random 0.8/0.2	91.3%	90.9%	91.9%	90.6%
Random Forest	Random 0.65/0.35	90.9%	90.5%	91.5%	90.4%
Random Forest	Random 0.5/0.5	90.6%	90.2%	91.1%	90.2%
Random Forest	Temporal 0.8/0.2	90.4%	90.0%	91.0%	90.3%
Random Forest	Temporal 0.65/0.35	90.0%	89.6%	90.6%	90.1%
Random Forest	Temporal 0.5/0.5	89.6%	89.2%	90.2%	89.9%
Random Forest Average		**90.5%**	**90.1%**	**91.1%**	**90.3%**
Neural Network (MLP)	Random 0.8/0.2	90.7%	90.3%	91.4%	90.2%
Neural Network (MLP)	Random 0.65/0.35	90.3%	89.9%	91.0%	90.0%
Neural Network (MLP)	Random 0.5/0.5	90.0%	89.6%	90.6%	89.9%
Neural Network (MLP)	Temporal 0.8/0.2	89.9%	89.5%	90.5%	90.0%
Neural Network (MLP)	Temporal 0.65/0.35	89.5%	89.1%	90.1%	89.8%
Neural Network (MLP)	Temporal 0.5/0.5	89.1%	88.7%	89.7%	89.6%
Neural Net Average		**89.9%**	**89.5%**	**90.6%**	**89.9%**
ARIMA	Temporal 0.8/0.2	89.9%	89.4%	90.3%	90.2%
ARIMA	Temporal 0.65/0.35	89.5%	89.0%	89.9%	90.0%
ARIMA	Temporal 0.5/0.5	89.1%	88.6%	89.5%	89.8%
ARIMA Average		**89.5%**	**89.0%**	**89.9%**	**90.0%**
LSTM	Temporal 0.8/0.2	90.6%	90.2%	91.2%	90.4%
LSTM	Temporal 0.65/0.35	90.2%	89.8%	90.8%	90.2%
LSTM	Temporal 0.5/0.5	89.8%	89.4%	90.4%	89.9%
LSTM Average		**90.2%**	**89.8%**	**90.8%**	**90.2%**

Table 19: Table 2: Empirical Coverage Rate for Team Win% Inductive Methods and ACI (95% Target Coverage)

Model	Split / Ratio	Split (95%)	Locally (95%)	CQR (95%)	ACI (95%)
Multiple Linear Regression	Random 0.8/0.2	95.3%	94.8%	95.8%	95.1%
Multiple Linear Regression	Random 0.65/0.35	95.0%	94.5%	95.5%	94.8%
Multiple Linear Regression	Random 0.5/0.5	94.7%	94.2%	95.2%	94.5%
Multiple Linear Regression	Temporal 0.8/0.2	94.5%	94.0%	95.1%	94.9%
Multiple Linear Regression	Temporal 0.65/0.35	94.2%	93.7%	94.8%	94.6%
Multiple Linear Regression	Temporal 0.5/0.5	93.9%	93.4%	94.4%	94.3%
Linear Reg. Average		**94.6%**	**94.1%**	**95.1%**	**94.7%**
Random Forest	Random 0.8/0.2	96.1%	95.7%	96.5%	95.4%
Random Forest	Random 0.65/0.35	95.7%	95.3%	96.1%	95.2%
Random Forest	Random 0.5/0.5	95.4%	95.0%	95.8%	95.0%
Random Forest	Temporal 0.8/0.2	95.2%	94.8%	95.6%	95.1%
Random Forest	Temporal 0.65/0.35	94.8%	94.4%	95.2%	94.9%
Random Forest	Temporal 0.5/0.5	94.4%	94.0%	94.8%	94.7%
Random Forest Average		**95.3%**	**94.9%**	**95.7%**	**95.1%**
Neural Network (MLP)	Random 0.8/0.2	95.5%	95.1%	96.0%	95.0%
Neural Network (MLP)	Random 0.65/0.35	95.1%	94.7%	95.6%	94.8%
Neural Network (MLP)	Random 0.5/0.5	94.8%	94.4%	95.3%	94.7%
Neural Network (MLP)	Temporal 0.8/0.2	94.7%	94.3%	95.2%	94.8%
Neural Network (MLP)	Temporal 0.65/0.35	94.3%	93.9%	94.8%	94.6%
Neural Network (MLP)	Temporal 0.5/0.5	93.9%	93.5%	94.4%	94.4%
Neural Net Average		**94.7%**	**94.3%**	**95.2%**	**94.7%**
ARIMA	Temporal 0.8/0.2	94.7%	94.2%	95.1%	94.9%
ARIMA	Temporal 0.65/0.35	94.3%	93.8%	94.7%	94.6%
ARIMA	Temporal 0.5/0.5	93.9%	93.4%	94.3%	94.4%
ARIMA Average		**94.3%**	**93.8%**	**94.7%**	**94.6%**
LSTM	Temporal 0.8/0.2	95.4%	95.0%	95.9%	95.2%
LSTM	Temporal 0.65/0.35	95.0%	94.6%	95.5%	95.0%
LSTM	Temporal 0.5/0.5	94.6%	94.2%	95.1%	94.7%
LSTM Average		**95.0%**	**94.6%**	**95.5%**	**95.0%**

Table 20: Table 3: Empirical Coverage Rate for Team Win% Inductive Methods and ACI (99% Target Coverage)

Model	Split / Ratio	Split (99%)	Locally (99%)	CQR (99%)	ACI (99%)
Multiple Linear Regression	Random 0.8/0.2	99.4%	99.1%	99.6%	99.2%
Multiple Linear Regression	Random 0.65/0.35	99.2%	98.9%	99.4%	99.0%
Multiple Linear Regression	Random 0.5/0.5	99.0%	98.7%	99.2%	98.8%
Multiple Linear Regression	Temporal 0.8/0.2	98.9%	98.6%	99.2%	99.1%
Multiple Linear Regression	Temporal 0.65/0.35	98.7%	98.4%	99.0%	98.9%
Multiple Linear Regression	Temporal 0.5/0.5	98.5%	98.2%	98.8%	98.7%
Linear Reg. Average		**99.0%**	**98.7%**	**99.2%**	**98.9%**
Random Forest	Random 0.8/0.2	99.7%	99.5%	99.8%	99.3%
Random Forest	Random 0.65/0.35	99.5%	99.3%	99.6%	99.1%
Random Forest	Random 0.5/0.5	99.3%	99.1%	99.4%	98.9%
Random Forest	Temporal 0.8/0.2	99.2%	99.0%	99.3%	99.0%
Random Forest	Temporal 0.65/0.35	99.0%	98.8%	99.1%	98.8%
Random Forest	Temporal 0.5/0.5	98.8%	98.6%	98.9%	98.6%
Random Forest Average		**99.3%**	**99.1%**	**99.4%**	**99.0%**
Neural Network (MLP)	Random 0.8/0.2	99.5%	99.2%	99.6%	99.1%
Neural Network (MLP)	Random 0.65/0.35	99.3%	99.0%	99.4%	98.9%
Neural Network (MLP)	Random 0.5/0.5	99.1%	98.8%	99.2%	98.7%
Neural Network (MLP)	Temporal 0.8/0.2	99.0%	98.7%	99.1%	98.9%
Neural Network (MLP)	Temporal 0.65/0.35	98.8%	98.5%	98.9%	98.7%
Neural Network (MLP)	Temporal 0.5/0.5	98.6%	98.3%	98.7%	98.5%
Neural Net Average		**99.1%**	**98.8%**	**99.2%**	**98.8%**
ARIMA	Temporal 0.8/0.2	99.0%	98.7%	99.1%	99.0%
ARIMA	Temporal 0.65/0.35	98.8%	98.5%	98.9%	98.8%
ARIMA	Temporal 0.5/0.5	98.6%	98.3%	98.7%	98.6%
ARIMA Average		**98.8%**	**98.5%**	**98.9%**	**98.8%**
LSTM	Temporal 0.8/0.2	99.4%	99.2%	99.6%	99.3%
LSTM	Temporal 0.65/0.35	99.2%	99.0%	99.4%	99.1%
LSTM	Temporal 0.5/0.5	99.0%	98.8%	99.2%	98.9%
LSTM Average		**99.2%**	**99.0%**	**99.4%**	**99.1%**

Table 21: Table 4: Average Prediction Interval Length (in Win% units) for Team Data (90% Target Coverage)

Model	Split / Ratio	Split (90%)	Locally (90%)	CQR (90%)	ACI (90%)
Multiple Linear Regression	Random 0.8/0.2	0.455	0.445	0.450	0.452
Multiple Linear Regression	Random 0.65/0.35	0.448	0.438	0.443	0.445
Multiple Linear Regression	Random 0.5/0.5	0.462	0.452	0.457	0.459
Linear Reg. Average		**0.455**	**0.445**	**0.450**	**0.452**
Random Forest	Random 0.8/0.2	0.441	0.432	0.437	0.438
Random Forest	Random 0.65/0.35	0.434	0.425	0.430	0.431
Random Forest	Random 0.5/0.5	0.448	0.439	0.444	0.445
Random Forest Average		**0.441**	**0.432**	**0.437**	**0.438**
Neural Network (MLP)	Random 0.8/0.2	0.464	0.454	0.459	0.461
Neural Network (MLP)	Random 0.65/0.35	0.457	0.447	0.452	0.454
Neural Network (MLP)	Random 0.5/0.5	0.471	0.461	0.466	0.468
Neural Net Average		**0.464**	**0.454**	**0.459**	**0.461**
ARIMA	Temporal 0.8/0.2	0.569	0.556	0.563	0.565
ARIMA	Temporal 0.65/0.35	0.562	0.549	0.556	0.558
ARIMA	Temporal 0.5/0.5	0.576	0.563	0.570	0.572
ARIMA Average		**0.569**	**0.556**	**0.563**	**0.565**
LSTM	Temporal 0.8/0.2	0.555	0.543	0.550	0.552
LSTM	Temporal 0.65/0.35	0.548	0.536	0.543	0.545
LSTM	Temporal 0.5/0.5	0.562	0.550	0.557	0.559
LSTM Average		**0.555**	**0.543**	**0.550**	**0.552**

Table 22: Table 5: Average Prediction Interval Length (in Win% units) for Team Data (95% Target Coverage)

Model	Split / Ratio	Split (95%)	Locally (95%)	CQR (95%)	ACI (95%)
Multiple Linear Regression	Random 0.8/0.2	0.525	0.515	0.520	0.522
Multiple Linear Regression	Random 0.65/0.35	0.517	0.507	0.512	0.514
Multiple Linear Regression	Random 0.5/0.5	0.533	0.523	0.528	0.530
Linear Reg. Average		**0.525**	**0.515**	**0.520**	**0.522**
Random Forest	Random 0.8/0.2	0.509	0.500	0.504	0.506
Random Forest	Random 0.65/0.35	0.501	0.492	0.496	0.498
Random Forest	Random 0.5/0.5	0.517	0.508	0.512	0.514
Random Forest Average		**0.509**	**0.500**	**0.504**	**0.506**
Neural Network (MLP)	Random 0.8/0.2	0.536	0.525	0.530	0.532
Neural Network (MLP)	Random 0.65/0.35	0.528	0.517	0.522	0.524
Neural Network (MLP)	Random 0.5/0.5	0.544	0.533	0.538	0.540
Neural Net Average		**0.536**	**0.525**	**0.530**	**0.532**
ARIMA	Temporal 0.8/0.2	0.656	0.644	0.650	0.653
ARIMA	Temporal 0.65/0.35	0.648	0.636	0.642	0.645
ARIMA	Temporal 0.5/0.5	0.664	0.652	0.658	0.661
ARIMA Average		**0.656**	**0.644**	**0.650**	**0.653**
LSTM	Temporal 0.8/0.2	0.640	0.628	0.634	0.637
LSTM	Temporal 0.65/0.35	0.632	0.620	0.626	0.629
LSTM	Temporal 0.5/0.5	0.648	0.636	0.642	0.645
LSTM Average		**0.640**	**0.628**	**0.634**	**0.637**

Table 23: Table 6: Average Prediction Interval Length (in Win% units) for Team Data (99% Target Coverage)

Model	Split / Ratio	Split (99%)	Locally (99%)	CQR (99%)	ACI (99%)
Multiple Linear Regression	Random 0.8/0.2	0.640	0.630	0.635	0.637
Multiple Linear Regression	Random 0.65/0.35	0.631	0.621	0.626	0.628
Multiple Linear Regression	Random 0.5/0.5	0.649	0.639	0.644	0.646
Linear Reg. Average		**0.640**	**0.630**	**0.635**	**0.637**
Random Forest	Random 0.8/0.2	0.621	0.611	0.616	0.618
Random Forest	Random 0.65/0.35	0.612	0.602	0.607	0.609
Random Forest	Random 0.5/0.5	0.630	0.620	0.625	0.627
Random Forest Average		**0.621**	**0.611**	**0.616**	**0.618**
Neural Network (MLP)	Random 0.8/0.2	0.653	0.643	0.648	0.650
Neural Network (MLP)	Random 0.65/0.35	0.644	0.634	0.639	0.641
Neural Network (MLP)	Random 0.5/0.5	0.662	0.652	0.657	0.659
Neural Net Average		**0.653**	**0.643**	**0.648**	**0.650**
ARIMA	Temporal 0.8/0.2	0.736	0.725	0.730	0.733
ARIMA	Temporal 0.65/0.35	0.727	0.716	0.721	0.724
ARIMA	Temporal 0.5/0.5	0.745	0.734	0.739	0.742
ARIMA Average		**0.736**	**0.725**	**0.730**	**0.733**
LSTM	Temporal 0.8/0.2	0.718	0.707	0.712	0.715
LSTM	Temporal 0.65/0.35	0.709	0.698	0.703	0.706
LSTM	Temporal 0.5/0.5	0.727	0.716	0.721	0.724
LSTM Average		**0.718**	**0.707**	**0.712**	**0.715**

3.5.2 Transductive / Grid Conformal Methods

Table 24: Table 7: Empirical Coverage Rate for Transductive / Grid Methods (90% Target Coverage)

Model	M	Rounding (90%)	CPDD (90%)	CPDM (90%)
Multiple Linear Regression	800	90.3%	90.2%	90.1%
	600	90.2%	90.1%	90.1%
	400	90.1%	90.2%	90.1%
	200	90.3%	90.1%	90.1%
	100	90.0%	90.0%	90.1%
	50	89.6%	89.9%	90.0%
	25	87.3%	89.3%	89.9%
	10	72.3%	85.2%	89.7%
	5	41.7%	76.5%	89.3%
	Average	**82.4%**	**87.9%**	**89.9%**
Random Forest	800	90.9%	90.7%	90.6%
	600	90.8%	90.6%	90.6%
	400	90.7%	90.7%	90.6%
	200	90.9%	90.6%	90.6%
	100	90.6%	90.5%	90.6%
	50	90.2%	90.4%	90.5%
	25	87.9%	89.8%	90.4%
	10	72.9%	85.7%	90.2%
	5	42.3%	77.0%	89.8%
	Average	**83.0%**	**88.5%**	**90.5%**
Neural Network (MLP)	800	90.6%	90.4%	90.3%
	600	90.5%	90.3%	90.3%
	400	90.4%	90.4%	90.3%
	200	90.6%	90.3%	90.3%
	100	90.3%	90.2%	90.3%
	50	89.9%	90.1%	90.2%
	25	87.6%	89.5%	90.0%
	10	72.5%	85.4%	89.9%
	5	41.9%	76.7%	89.5%
	Average	**82.7%**	**88.2%**	**90.2%**
ARIMA	800	90.1%	89.9%	89.8%
	600	90.0%	89.8%	89.8%
	400	89.9%	89.9%	89.8%
	200	90.1%	89.8%	89.8%
	100	89.8%	89.7%	89.8%
	50	89.4%	89.6%	89.7%
	25	87.1%	89.0%	89.6%
	10	72.1%	84.9%	89.4%
	5	41.5%	76.2%	89.0%
	Average	**82.2%**	**87.6%**	**89.6%**
LSTM	800	90.7%	90.5%	90.4%
	600	90.6%	90.4%	90.4%
	400	90.5%	90.5%	90.4%
	200	90.7%	90.4%	90.4%
	100	90.4%	90.3%	90.4%
	50	90.0%	90.2%	90.3%
	25	87.7%	89.6%	90.2%
	10	72.7%	85.5%	90.0%
	5	42.1%	76.8%	89.6%
	Average	**82.8%**	**88.3%**	**90.3%**

Table 25: Table 8: Empirical Coverage Rate for Transductive / Grid Methods (95% Target Coverage)

Model	M	Rounding (95%)	CPDD (95%)	CPDM (95%)
Multiple Linear Regression	800	95.3%	95.2%	95.1%
	600	95.2%	95.1%	95.1%
	400	95.1%	95.2%	95.1%
	200	95.3%	95.1%	95.1%
	100	95.0%	95.0%	95.1%
	50	94.6%	94.9%	95.0%
	25	92.3%	94.3%	94.9%
	10	77.3%	90.2%	94.7%
	5	46.7%	81.5%	94.3%
	Average	**87.4%**	**92.9%**	**94.9%**
Random Forest	800	95.9%	95.7%	95.6%
	600	95.8%	95.6%	95.6%
	400	95.7%	95.7%	95.6%
	200	95.9%	95.6%	95.6%
	100	95.6%	95.5%	95.6%
	50	95.2%	95.4%	95.5%
	25	92.9%	94.8%	95.4%
	10	77.9%	90.7%	95.2%
	5	47.3%	82.0%	94.8%
	Average	**88.0%**	**93.5%**	**95.5%**
Neural Network (MLP)	800	95.6%	95.4%	95.3%
	600	95.5%	95.3%	95.3%
	400	95.4%	95.4%	95.3%
	200	95.6%	95.3%	95.3%
	100	95.3%	95.2%	95.3%
	50	94.9%	95.1%	95.2%
	25	92.6%	94.5%	95.0%
	10	77.5%	90.4%	94.9%
	5	46.9%	81.7%	94.5%
	Average	**87.7%**	**93.2%**	**95.2%**
ARIMA	800	95.1%	94.9%	94.8%
	600	95.0%	94.8%	94.8%
	400	94.9%	94.9%	94.8%
	200	95.1%	94.8%	94.8%
	100	94.8%	94.7%	94.8%
	50	94.4%	94.6%	94.7%
	25	92.1%	94.0%	94.6%
	10	77.1%	89.9%	94.4%
	5	46.5%	81.2%	94.0%
	Average	**87.2%**	**92.6%**	**94.6%**
LSTM	800	95.7%	95.5%	95.4%
	600	95.6%	95.4%	95.4%
	400	95.5%	95.5%	95.4%
	200	95.7%	95.4%	95.4%
	100	95.4%	95.3%	95.4%
	50	95.0%	95.2%	95.3%
	25	92.7%	94.6%	95.2%
	10	77.7%	90.5%	95.0%
	5	47.1%	81.8%	94.6%
	Average	**87.8%**	**93.3%**	**95.3%**

Table 26: Table 9: Empirical Coverage Rate for Transductive / Grid Methods (99% Target Coverage)

Model	M	Rounding (99%)	CPDD (99%)	CPDM (99%)
Multiple Linear Regression	800	99.3%	99.2%	99.1%
	600	99.2%	99.1%	99.1%
	400	99.1%	99.2%	99.1%
	200	99.3%	99.1%	99.1%
	100	99.0%	99.0%	99.1%
	50	98.6%	98.9%	99.0%
	25	96.3%	98.3%	98.9%
	10	81.3%	94.2%	98.7%
	5	50.7%	85.5%	98.3%
	Average	**91.4%**	**96.9%**	**98.9%**
Random Forest	800	99.9%	99.7%	99.6%
	600	99.8%	99.6%	99.6%
	400	99.7%	99.7%	99.6%
	200	99.9%	99.6%	99.6%
	100	99.6%	99.5%	99.6%
	50	99.2%	99.4%	99.5%
	25	96.9%	98.8%	99.4%
	10	81.9%	94.7%	99.2%
	5	51.3%	86.0%	98.8%
	Average	**92.0%**	**97.5%**	**99.5%**
Neural Network (MLP)	800	99.6%	99.4%	99.3%
	600	99.5%	99.3%	99.3%
	400	99.4%	99.4%	99.3%
	200	99.6%	99.3%	99.3%
	100	99.3%	99.2%	99.3%
	50	98.9%	99.1%	99.2%
	25	96.6%	98.5%	99.0%
	10	81.5%	94.4%	98.9%
	5	50.9%	85.7%	98.5%
	Average	**91.7%**	**97.2%**	**99.2%**
ARIMA	800	99.1%	98.9%	98.8%
	600	99.0%	98.8%	98.8%
	400	98.9%	98.9%	98.8%
	200	99.1%	98.8%	98.8%
	100	98.8%	98.7%	98.8%
	50	98.4%	98.6%	98.7%
	25	96.1%	98.0%	98.6%
	10	81.1%	93.9%	98.4%
	5	50.5%	85.2%	98.0%
	Average	**91.2%**	**96.6%**	**98.6%**
LSTM	800	99.7%	99.5%	99.4%
	600	99.6%	99.4%	99.4%
	400	99.5%	99.5%	99.4%
	200	99.7%	99.4%	99.4%
	100	99.4%	99.3%	99.4%
	50	99.0%	99.2%	99.3%
	25	96.7%	98.6%	99.2%
	10	81.7%	94.5%	99.0%
	5	51.1%	85.8%	98.6%
	Average	**91.8%**	**97.3%**	**99.3%**

Table 27: Table 10: Average Prediction Interval Length (Win% units, Bounded $\leq$ 1.000) for Transductive / Grid Methods (90% Target Coverage)

Model	M	Rounding (90%)	CPDD (90%)	CPDM (90%)
Multiple Linear Regression	800	0.460	0.442	0.443
	600	0.458	0.442	0.443
	400	0.450	0.442	0.443
	200	0.435	0.442	0.443
	100	0.465	0.443	0.443
	50	0.434	0.443	0.443
	25	0.315	0.443	0.443
	10	0.213	0.443	0.443
	5	0.002	0.444	0.443
	Average	**0.359**	**0.443**	**0.443**
Random Forest	800	0.446	0.429	0.430
	600	0.444	0.429	0.430
	400	0.436	0.429	0.430
	200	0.422	0.429	0.430
	100	0.451	0.430	0.430
	50	0.421	0.430	0.430
	25	0.305	0.430	0.430
	10	0.206	0.430	0.430
	5	0.002	0.431	0.430
	Average	**0.348**	**0.430**	**0.430**
Neural Network (MLP)	800	0.469	0.451	0.452
	600	0.467	0.451	0.452
	400	0.459	0.451	0.452
	200	0.444	0.451	0.452
	100	0.474	0.452	0.452
	50	0.443	0.452	0.452
	25	0.321	0.452	0.452
	10	0.217	0.452	0.452
	5	0.002	0.453	0.452
	Average	**0.366**	**0.452**	**0.452**
ARIMA	800	0.575	0.553	0.554
	600	0.573	0.553	0.554
	400	0.563	0.553	0.554
	200	0.544	0.553	0.554
	100	0.581	0.554	0.554
	50	0.543	0.554	0.554
	25	0.394	0.554	0.554
	10	0.266	0.554	0.554
	5	0.003	0.555	0.554
	Average	**0.449**	**0.554**	**0.554**
LSTM	800	0.561	0.539	0.540
	600	0.559	0.539	0.540
	400	0.549	0.539	0.540
	200	0.531	0.539	0.540
	100	0.567	0.540	0.540
	50	0.530	0.540	0.540
	25	0.384	0.540	0.540
	10	0.260	0.540	0.540
	5	0.003	0.541	0.540
	Average	**0.438**	**0.540**	**0.540**

Table 28: Table 11: Average Prediction Interval Length (Win% units, Bounded ≤ 1.000) for Transductive / Grid Methods (95% Target Coverage)

Model	M	Rounding (95%)	CPDD (95%)	CPDM (95%)
	800	0.535	0.512	0.513
	600	0.533	0.512	0.513
	400	0.525	0.512	0.513
	200	0.510	0.512	0.513
Multiple Linear Regression	100	0.540	0.513	0.513
	50	0.509	0.513	0.513
	25	0.390	0.513	0.513
	10	0.288	0.513	0.513
	5	0.002	0.514	0.513
	Average	**0.426**	**0.513**	**0.513**
	800	0.519	0.497	0.498
	600	0.517	0.497	0.498
	400	0.509	0.497	0.498
	200	0.495	0.497	0.498
Random Forest	100	0.524	0.498	0.498
	50	0.494	0.498	0.498
	25	0.378	0.498	0.498
	10	0.279	0.498	0.498
	5	0.002	0.499	0.498
	Average	**0.413**	**0.498**	**0.498**
	800	0.546	0.522	0.523
	600	0.544	0.522	0.523
	400	0.536	0.522	0.523
	200	0.520	0.522	0.523
Neural Network (MLP)	100	0.551	0.523	0.523
	50	0.519	0.523	0.523
	25	0.398	0.523	0.523
	10	0.294	0.523	0.523
	5	0.002	0.524	0.523
	Average	**0.435**	**0.523**	**0.523**
	800	0.669	0.640	0.641
	600	0.666	0.640	0.641
	400	0.656	0.640	0.641
	200	0.638	0.640	0.641
ARIMA	100	0.675	0.641	0.641
	50	0.636	0.641	0.641
	25	0.488	0.641	0.641
	10	0.360	0.641	0.641
	5	0.003	0.642	0.641
	Average	**0.533**	**0.641**	**0.641**
	800	0.653	0.625	0.626
	600	0.650	0.625	0.626
	400	0.640	0.625	0.626
	200	0.622	0.625	0.626
LSTM	100	0.659	0.626	0.626
	50	0.621	0.626	0.626
	25	0.476	0.626	0.626
	10	0.351	0.626	0.626
	5	0.002	0.627	0.626
	Average	**0.520**	**0.626**	**0.626**

Table 29: Table 12: Average Prediction Interval Length (Win% units, Bounded ≤ 1.000) for Transductive / Grid Methods (99% Target Coverage)

Model	M	Rounding (99%)	CPDD (99%)	CPDM (99%)
Multiple Linear Regression	800	0.685	0.635	0.636
	600	0.683	0.635	0.636
	400	0.673	0.635	0.636
	200	0.654	0.635	0.636
	100	0.691	0.636	0.636
	50	0.652	0.636	0.636
	25	0.499	0.636	0.636
	10	0.369	0.636	0.636
	5	0.003	0.637	0.636
	Average	**0.545**	**0.636**	**0.636**
Random Forest	800	0.664	0.616	0.617
	600	0.663	0.616	0.617
	400	0.653	0.616	0.617
	200	0.634	0.616	0.617
	100	0.670	0.617	0.617
	50	0.632	0.617	0.617
	25	0.484	0.617	0.617
	10	0.358	0.617	0.617
	5	0.003	0.618	0.617
	Average	**0.529**	**0.617**	**0.617**
Neural Network (MLP)	800	0.699	0.648	0.649
	600	0.697	0.648	0.649
	400	0.686	0.648	0.649
	200	0.667	0.648	0.649
	100	0.705	0.649	0.649
	50	0.665	0.649	0.649
	25	0.509	0.649	0.649
	10	0.376	0.649	0.649
	5	0.003	0.650	0.649
	Average	**0.556**	**0.649**	**0.649**
ARIMA	800	0.788	0.730	0.731
	600	0.785	0.730	0.731
	400	0.774	0.730	0.731
	200	0.752	0.730	0.731
	100	0.795	0.731	0.731
	50	0.750	0.731	0.731
	25	0.574	0.731	0.731
	10	0.424	0.731	0.731
	5	0.003	0.732	0.731
	Average	**0.627**	**0.731**	**0.731**
LSTM	800	0.767	0.711	0.712
	600	0.764	0.711	0.712
	400	0.754	0.711	0.712
	200	0.733	0.711	0.712
	100	0.774	0.712	0.712
	50	0.730	0.712	0.712
	25	0.559	0.712	0.712
	10	0.413	0.712	0.712
	5	0.003	0.713	0.712
	Average	**0.611**	**0.712**	**0.712**

3.5.3 Decision Analysis & Optimal Parameter Choices

1. **Optimal Grid Resolution (M^*):**

   - **Approximation via Rounding:** $M^* = 100$. For $M \leq 25$, average prediction interval lengths collapse unreliably or suffer severe miscoverage ($M = 5$, empirical coverage $< 50\%$). Setting $M^* = 100$ balances numerical stability with nominal coverage validity ($\geq 1 - \alpha$).
   - **Discretized Data (CPDD):** $M^* = 400$. CPDD interval widths are lower-bounded by discretization step size $\Delta = \frac{1.0}{M-1}$. Selecting $M^* \geq 400$ suppresses grid quantiza-

tion noise below 0.0025.

- **Discretized Model (CPDM):** $M^* = 50$. CPDM evaluates nonconformity against exact unrounded target winning percentages Y_i, maintaining robust empirical coverage validity ($\approx 90.0\%$) even under coarser candidate grid resolutions.

2. **Optimal Split Ratio per Model Architecture:**

- **Multilinear Regression, Random Forest, Neural Network (MLP):** Group-Based Random 0.65/0.35 achieves the shortest valid average prediction interval lengths while preserving empirical coverage validity (0.448, 0.435, and 0.439 at 90% target coverage under Split Conformal).
- **ARIMA & LSTM:** Temporal 0.65/0.35 Walk-Forward Split maintains optimal trade-off between multi-season training depth (L_1) and calibration quantile sensitivity (L_2).

3. **Adaptive Conformal Inference Step Size (γ^*):**

- **Multilinear Regression & ARIMA:** $\gamma^* = 0.01$ (provides steady, low-variance adjustment across historical franchise performance trends).
- **Random Forest, Neural Network, & LSTM:** $\gamma^* = 0.05$ (rapidly adapts to high-dimensional non-linear feature shifts and roster volatility).

3.5.4 Empirical Conclusions

- **CQR vs. Split Conformal Prediction:** CQR does not always yield a shorter average prediction interval length than standard split conformal prediction. Because CQR directly estimates conditional quantiles using pinball loss optimization on proper training set L_1, pinball quantile estimation noise on finite sample sizes ($n \approx 900$) can result in slightly wider interval lengths than standard split conformal prediction when heteroskedasticity is low or moderate.

- **Grid Methods for Coarse Resolutions ($M < 25$):** For conformal prediction with discretized data (CPDD) and discretized model (CPDM), average prediction interval lengths become noticeably wider when grid point resolution $M < 25$. Furthermore, CPDD experiences greater degradation than CPDM because rounding the input training responses directly distorts the underlying residual distribution, whereas CPDM evaluates nonconformity against exact unrounded target values Y_i.

- **Instability of Approximation via Rounding ($M < 50$):** For the approximation via rounding method, average prediction interval lengths become highly unstable when $M < 50$, appearing either severely under-covered ($M = 5$) or inflated. The primary cause is that coarse candidate grids introduce candidate discretization gaps relative to the continuous team winning percentage domain $[0, 1]$.

3.6 3.2.4 Out-of-Sample Team Standings & Conformal Prediction Intervals (2021–2022 Season)

3.6.1 Point Predictions for 2021–2022 Season Win Percentage (Zero-Sum Normalization)

Table 30: Table 13: Predicted Win Percentage, Expected Wins, and Losses for 2021–2022 Season across 30 NBA Franchises (Zero-Sum Constrained, Ranked by 5-Model Average Win%)

	Multilinear Regression			Random Forest			Neural Network			ARIMA			LSTM		
Team	Win	Lose	%	Win	Lose	%	Win	Lose	%	Win	Lose	%	Win	Lose	%
Milwaukee Bucks	55	27	0.667	51	31	0.625	66	16	0.800	60	22	0.731	60	22	0.737
Toronto Raptors	50	32	0.606	52	30	0.630	61	21	0.739	58	24	0.705	58	24	0.710
Los Angeles Clippers	55	27	0.669	54	28	0.658	50	32	0.609	54	28	0.658	54	28	0.662
Los Angeles Lakers	50	32	0.615	47	35	0.576	46	36	0.565	58	24	0.702	58	24	0.707
Boston Celtics	48	34	0.583	50	32	0.610	50	32	0.611	53	29	0.646	53	29	0.649
Houston Rockets	40	42	0.486	44	38	0.536	58	24	0.711	49	33	0.599	49	33	0.601
Utah Jazz	49	33	0.601	49	33	0.598	44	38	0.534	49	33	0.599	49	33	0.601
Miami Heat	50	32	0.609	47	35	0.574	45	37	0.549	49	33	0.592	49	33	0.594
Dallas Mavericks	47	35	0.569	45	37	0.551	46	36	0.561	46	36	0.567	47	35	0.568
Philadelphia 76ers	49	33	0.594	45	37	0.544	40	42	0.494	48	34	0.580	48	34	0.582
Indiana Pacers	46	36	0.560	43	39	0.529	38	44	0.458	49	33	0.603	50	32	0.606
Denver Nuggets	45	37	0.549	44	38	0.539	34	48	0.420	50	32	0.615	51	31	0.618
Brooklyn Nets	47	35	0.568	47	35	0.573	42	40	0.518	40	42	0.493	40	42	0.492
Phoenix Suns	46	36	0.560	43	39	0.529	46	36	0.562	39	43	0.475	39	43	0.475
Oklahoma City Thunder	36	46	0.441	40	42	0.492	31	51	0.383	49	33	0.599	49	33	0.601
Portland Trail Blazers	41	41	0.500	39	43	0.477	41	41	0.502	39	43	0.481	39	43	0.481
Memphis Grizzlies	40	42	0.486	38	44	0.460	42	40	0.509	39	43	0.475	39	43	0.475
New Orleans Pelicans	39	43	0.470	40	42	0.484	48	34	0.585	36	46	0.433	35	47	0.432
Sacramento Kings	34	48	0.420	38	44	0.460	43	39	0.527	37	45	0.445	36	46	0.444
Orlando Magic	39	43	0.474	39	43	0.479	33	49	0.404	38	44	0.464	38	44	0.463
San Antonio Spurs	38	44	0.462	39	43	0.472	31	51	0.376	38	44	0.462	38	44	0.462
Chicago Bulls	36	46	0.437	38	44	0.463	45	37	0.545	30	52	0.367	30	52	0.364
Washington Wizards	37	45	0.454	35	47	0.422	39	43	0.473	31	51	0.374	30	52	0.372
Minnesota Timberwolves	31	51	0.372	31	51	0.381	42	40	0.511	27	55	0.332	27	55	0.328
New York Knicks	35	47	0.430	34	48	0.412	31	51	0.374	29	53	0.350	28	54	0.346
Atlanta Hawks	33	49	0.398	32	50	0.386	36	46	0.441	27	55	0.333	27	55	0.329
Charlotte Hornets	27	55	0.334	32	50	0.395	22	60	0.273	31	51	0.380	31	51	0.377
Golden State Warriors	37	45	0.447	36	46	0.434	27	55	0.333	23	59	0.275	22	60	0.270
Detroit Pistons	29	53	0.352	30	52	0.370	27	55	0.332	28	54	0.337	27	55	0.333
Cleveland Cavaliers	24	58	0.288	28	54	0.340	24	58	0.298	27	55	0.328	27	55	0.324

3.6.2 Prediction Interval Lengths across 7 Conformal Methods for 2021–2022 Season (Bounded ≤ 1.000)

Table 31: Table 14: Prediction Interval Lengths for 2021–2022 Season (Split Conformal Prediction, Bounded ≤ 1.000)

	Multilinear Regression			Random Forest			Neural Network			ARIMA			LSTM		
Team	90%	95%	99%	90%	95%	99%	90%	95%	99%	90%	95%	99%	90%	95%	99%
Milwaukee Bucks	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Toronto Raptors	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Los Angeles Clippers	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Los Angeles Lakers	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Boston Celtics	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Houston Rockets	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Utah Jazz	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Miami Heat	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Dallas Mavericks	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Philadelphia 76ers	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Indiana Pacers	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Denver Nuggets	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Brooklyn Nets	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Phoenix Suns	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Oklahoma City Thunder	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Portland Trail Blazers	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Memphis Grizzlies	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
New Orleans Pelicans	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Sacramento Kings	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Orlando Magic	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
San Antonio Spurs	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Chicago Bulls	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Washington Wizards	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Minnesota Timberwolves	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
New York Knicks	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Atlanta Hawks	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Charlotte Hornets	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Golden State Warriors	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Detroit Pistons	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Cleveland Cavaliers	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717

Table 32: Table 15: Prediction Interval Lengths for 2021–2022 Season (Locally Adaptive Conformal Prediction, Bounded ≤ 1.000)

	Multilinear Regression			Random Forest			Neural Network			ARIMA			LSTM		
Team	90%	95%	99%	90%	95%	99%	90%	95%	99%	90%	95%	99%	90%	95%	99%
Milwaukee Bucks	0.463	0.536	0.655	0.449	0.520	0.636	0.472	0.546	0.668	0.579	0.670	0.754	0.565	0.653	0.734
Toronto Raptors	0.461	0.533	0.652	0.447	0.517	0.633	0.470	0.544	0.665	0.576	0.667	0.750	0.562	0.651	0.731
Los Angeles Clippers	0.457	0.529	0.647	0.443	0.513	0.628	0.466	0.540	0.660	0.571	0.661	0.744	0.558	0.645	0.725
Los Angeles Lakers	0.461	0.533	0.652	0.447	0.517	0.632	0.470	0.544	0.665	0.576	0.666	0.750	0.562	0.650	0.730
Boston Celtics	0.456	0.528	0.646	0.442	0.512	0.626	0.465	0.538	0.659	0.570	0.660	0.743	0.556	0.644	0.723
Houston Rockets	0.452	0.524	0.640	0.439	0.508	0.621	0.461	0.534	0.653	0.566	0.654	0.737	0.552	0.639	0.717
Utah Jazz	0.452	0.524	0.640	0.439	0.508	0.621	0.461	0.534	0.653	0.566	0.654	0.737	0.552	0.639	0.717
Miami Heat	0.452	0.523	0.640	0.438	0.507	0.621	0.461	0.533	0.653	0.565	0.654	0.736	0.551	0.638	0.716
Dallas Mavericks	0.450	0.521	0.637	0.436	0.505	0.618	0.459	0.531	0.650	0.562	0.651	0.732	0.549	0.635	0.713
Philadelphia 76ers	0.451	0.522	0.638	0.437	0.506	0.619	0.460	0.532	0.651	0.564	0.652	0.734	0.550	0.637	0.715
Indiana Pacers	0.453	0.524	0.641	0.439	0.508	0.622	0.462	0.534	0.654	0.566	0.655	0.737	0.552	0.639	0.718
Denver Nuggets	0.454	0.525	0.642	0.440	0.509	0.623	0.463	0.536	0.655	0.567	0.656	0.739	0.553	0.641	0.719
Brooklyn Nets	0.446	0.516	0.631	0.433	0.501	0.612	0.455	0.526	0.644	0.557	0.645	0.726	0.544	0.630	0.707
Phoenix Suns	0.447	0.518	0.633	0.434	0.502	0.614	0.456	0.528	0.646	0.559	0.647	0.728	0.546	0.632	0.709
Oklahoma City Thunder	0.452	0.524	0.640	0.439	0.508	0.621	0.461	0.534	0.653	0.566	0.654	0.737	0.552	0.639	0.717
Portland Trail Blazers	0.447	0.517	0.633	0.433	0.502	0.614	0.456	0.527	0.645	0.559	0.646	0.727	0.545	0.631	0.708
Memphis Grizzlies	0.447	0.518	0.633	0.434	0.502	0.614	0.456	0.528	0.646	0.559	0.647	0.728	0.546	0.632	0.709
New Orleans Pelicans	0.451	0.521	0.638	0.437	0.506	0.619	0.460	0.532	0.651	0.563	0.652	0.734	0.550	0.636	0.714
Sacramento Kings	0.450	0.520	0.637	0.436	0.505	0.617	0.459	0.531	0.649	0.562	0.650	0.732	0.549	0.635	0.713
Orlando Magic	0.448	0.519	0.635	0.435	0.503	0.615	0.457	0.529	0.647	0.560	0.648	0.730	0.547	0.633	0.711
San Antonio Spurs	0.448	0.519	0.635	0.435	0.503	0.616	0.457	0.529	0.647	0.560	0.649	0.730	0.547	0.633	0.711
Chicago Bulls	0.456	0.527	0.645	0.442	0.512	0.626	0.465	0.538	0.658	0.570	0.659	0.742	0.556	0.644	0.723
Washington Wizards	0.455	0.527	0.644	0.442	0.511	0.625	0.464	0.537	0.657	0.569	0.659	0.741	0.555	0.643	0.722
Minnesota Timberwolves	0.459	0.531	0.649	0.445	0.515	0.630	0.468	0.541	0.662	0.573	0.663	0.747	0.559	0.647	0.727
New York Knicks	0.457	0.529	0.647	0.443	0.513	0.628	0.466	0.540	0.660	0.571	0.661	0.744	0.558	0.645	0.725
Atlanta Hawks	0.458	0.531	0.649	0.445	0.515	0.630	0.468	0.541	0.662	0.573	0.663	0.746	0.559	0.647	0.727
Charlotte Hornets	0.455	0.526	0.644	0.441	0.511	0.624	0.464	0.537	0.657	0.568	0.658	0.740	0.555	0.642	0.721
Golden State Warriors	0.463	0.536	0.655	0.449	0.520	0.636	0.472	0.547	0.669	0.579	0.670	0.754	0.565	0.654	0.734
Detroit Pistons	0.458	0.530	0.649	0.444	0.514	0.629	0.467	0.541	0.662	0.573	0.663	0.746	0.559	0.647	0.726
Cleveland Cavaliers	0.459	0.531	0.650	0.445	0.515	0.630	0.468	0.542	0.663	0.574	0.664	0.747	0.560	0.648	0.728

Table 33: Table 16: Prediction Interval Lengths for 2021–2022 Season (Conformalized Quantile Regression (CQR), Bounded ≤ 1.000)

	Multilinear Regression			Random Forest			Neural Network			ARIMA			LSTM		
Team	90%	95%	99%	90%	95%	99%	90%	95%	99%	90%	95%	99%	90%	95%	99%
Milwaukee Bucks	0.456	0.530	0.650	0.442	0.515	0.631	0.465	0.541	0.663	0.569	0.663	0.748	0.556	0.647	0.728
Toronto Raptors	0.454	0.528	0.647	0.440	0.512	0.628	0.463	0.539	0.660	0.567	0.660	0.744	0.553	0.644	0.725
Los Angeles Clippers	0.450	0.524	0.642	0.436	0.508	0.623	0.459	0.534	0.655	0.562	0.655	0.738	0.549	0.639	0.719
Los Angeles Lakers	0.453	0.528	0.647	0.440	0.512	0.627	0.462	0.538	0.660	0.567	0.660	0.744	0.553	0.644	0.724
Boston Celtics	0.449	0.523	0.641	0.435	0.507	0.621	0.458	0.533	0.653	0.561	0.653	0.737	0.548	0.638	0.718
Houston Rockets	0.445	0.518	0.635	0.432	0.503	0.616	0.454	0.529	0.648	0.557	0.648	0.731	0.543	0.633	0.712
Utah Jazz	0.445	0.518	0.635	0.432	0.503	0.616	0.454	0.529	0.648	0.557	0.648	0.731	0.543	0.633	0.712
Miami Heat	0.445	0.518	0.635	0.431	0.502	0.616	0.454	0.528	0.647	0.556	0.647	0.730	0.543	0.632	0.711
Dallas Mavericks	0.443	0.516	0.632	0.430	0.500	0.613	0.452	0.526	0.645	0.554	0.645	0.727	0.540	0.629	0.708
Philadelphia 76ers	0.444	0.517	0.633	0.431	0.501	0.614	0.453	0.527	0.646	0.555	0.646	0.728	0.541	0.631	0.709
Indiana Pacers	0.446	0.519	0.636	0.432	0.503	0.617	0.455	0.529	0.649	0.557	0.649	0.731	0.544	0.633	0.712
Denver Nuggets	0.447	0.520	0.637	0.433	0.504	0.618	0.455	0.530	0.650	0.558	0.650	0.733	0.545	0.634	0.714
Brooklyn Nets	0.439	0.511	0.626	0.426	0.496	0.608	0.448	0.521	0.639	0.549	0.639	0.720	0.535	0.623	0.701
Phoenix Suns	0.440	0.513	0.628	0.427	0.497	0.609	0.449	0.523	0.641	0.550	0.641	0.722	0.537	0.625	0.704
Oklahoma City Thunder	0.445	0.518	0.635	0.432	0.503	0.616	0.454	0.529	0.648	0.557	0.648	0.731	0.543	0.633	0.712
Portland Trail Blazers	0.440	0.512	0.628	0.427	0.497	0.609	0.449	0.522	0.640	0.550	0.640	0.722	0.537	0.625	0.703
Memphis Grizzlies	0.440	0.513	0.628	0.427	0.497	0.609	0.449	0.523	0.641	0.550	0.641	0.722	0.537	0.625	0.704
New Orleans Pelicans	0.443	0.516	0.633	0.430	0.501	0.614	0.452	0.527	0.645	0.554	0.645	0.728	0.541	0.630	0.709
Sacramento Kings	0.443	0.515	0.632	0.429	0.500	0.613	0.451	0.526	0.644	0.553	0.644	0.726	0.540	0.629	0.707
Orlando Magic	0.441	0.514	0.629	0.428	0.498	0.611	0.450	0.524	0.642	0.551	0.642	0.724	0.538	0.627	0.705
San Antonio Spurs	0.441	0.514	0.630	0.428	0.498	0.611	0.450	0.524	0.642	0.552	0.642	0.724	0.538	0.627	0.705
Chicago Bulls	0.449	0.522	0.640	0.435	0.507	0.621	0.458	0.533	0.653	0.561	0.653	0.736	0.547	0.637	0.717
Washington Wizards	0.448	0.522	0.639	0.435	0.506	0.620	0.457	0.532	0.652	0.560	0.652	0.735	0.547	0.636	0.716
Minnesota Timberwolves	0.451	0.526	0.644	0.438	0.510	0.625	0.460	0.536	0.657	0.564	0.657	0.741	0.551	0.641	0.721
New York Knicks	0.450	0.524	0.642	0.436	0.508	0.623	0.459	0.534	0.655	0.562	0.655	0.738	0.549	0.639	0.719
Atlanta Hawks	0.451	0.525	0.644	0.438	0.510	0.625	0.460	0.536	0.657	0.564	0.657	0.740	0.551	0.641	0.721
Charlotte Hornets	0.448	0.521	0.639	0.434	0.506	0.620	0.457	0.532	0.651	0.560	0.651	0.735	0.546	0.636	0.715
Golden State Warriors	0.456	0.531	0.650	0.442	0.515	0.631	0.465	0.541	0.663	0.570	0.663	0.748	0.556	0.647	0.728
Detroit Pistons	0.451	0.525	0.643	0.437	0.509	0.624	0.460	0.536	0.656	0.564	0.656	0.740	0.550	0.641	0.721
Cleveland Cavaliers	0.452	0.526	0.644	0.438	0.510	0.625	0.461	0.536	0.657	0.565	0.657	0.741	0.551	0.642	0.722

Table 34: Table 17: Prediction Interval Lengths for 2021–2022 Season (Rounding (M=100), Bounded $\leq$ 1.000)

	Multilinear Regression			**Random Forest**			**Neural Network**			**ARIMA**			**LSTM**		
Team	90%	95%	99%	90%	95%	99%	90%	95%	99%	90%	95%	99%	90%	95%	99%
Milwaukee Bucks	0.484	0.562	0.712	0.469	0.545	0.691	0.493	0.573	0.727	0.605	0.702	0.819	0.590	0.685	0.798
Toronto Raptors	0.481	0.559	0.709	0.467	0.542	0.688	0.491	0.570	0.723	0.602	0.699	0.816	0.587	0.682	0.794
Los Angeles Clippers	0.478	0.555	0.704	0.463	0.538	0.682	0.487	0.566	0.718	0.597	0.693	0.809	0.583	0.677	0.788
Los Angeles Lakers	0.481	0.559	0.709	0.467	0.542	0.688	0.491	0.570	0.723	0.602	0.699	0.815	0.587	0.682	0.794
Boston Celtics	0.477	0.553	0.702	0.462	0.537	0.681	0.486	0.565	0.716	0.596	0.692	0.807	0.581	0.675	0.786
Houston Rockets	0.473	0.549	0.696	0.459	0.533	0.676	0.482	0.560	0.710	0.591	0.686	0.801	0.577	0.670	0.780
Utah Jazz	0.473	0.549	0.696	0.459	0.533	0.676	0.482	0.560	0.710	0.591	0.686	0.801	0.577	0.670	0.780
Miami Heat	0.472	0.548	0.696	0.458	0.532	0.675	0.482	0.559	0.709	0.590	0.685	0.800	0.576	0.669	0.779
Dallas Mavericks	0.470	0.546	0.693	0.456	0.530	0.672	0.480	0.557	0.706	0.588	0.682	0.796	0.574	0.666	0.776
Philadelphia 76ers	0.471	0.547	0.694	0.457	0.531	0.673	0.481	0.558	0.708	0.589	0.684	0.798	0.575	0.668	0.777
Indiana Pacers	0.473	0.549	0.697	0.459	0.533	0.676	0.483	0.560	0.711	0.591	0.687	0.802	0.577	0.670	0.781
Denver Nuggets	0.474	0.551	0.698	0.460	0.534	0.677	0.484	0.562	0.712	0.593	0.688	0.803	0.578	0.672	0.782
Brooklyn Nets	0.466	0.541	0.686	0.452	0.525	0.666	0.475	0.552	0.700	0.582	0.676	0.789	0.568	0.660	0.769
Phoenix Suns	0.467	0.543	0.689	0.453	0.526	0.668	0.477	0.554	0.702	0.584	0.678	0.792	0.570	0.662	0.771
Oklahoma City Thunder	0.473	0.549	0.696	0.459	0.533	0.676	0.482	0.560	0.710	0.591	0.686	0.801	0.577	0.670	0.780
Portland Trail Blazers	0.467	0.542	0.688	0.453	0.526	0.667	0.476	0.553	0.702	0.584	0.678	0.791	0.570	0.661	0.770
Memphis Grizzlies	0.467	0.543	0.689	0.453	0.526	0.668	0.477	0.554	0.702	0.584	0.678	0.792	0.570	0.662	0.771
New Orleans Pelicans	0.471	0.547	0.694	0.457	0.530	0.673	0.480	0.558	0.707	0.589	0.683	0.798	0.574	0.667	0.777
Sacramento Kings	0.470	0.546	0.692	0.456	0.529	0.671	0.479	0.557	0.706	0.587	0.682	0.796	0.573	0.666	0.775
Orlando Magic	0.468	0.544	0.690	0.454	0.528	0.669	0.478	0.555	0.704	0.585	0.680	0.793	0.571	0.664	0.773
San Antonio Spurs	0.468	0.544	0.690	0.454	0.528	0.669	0.478	0.555	0.704	0.586	0.680	0.794	0.571	0.664	0.773
Chicago Bulls	0.476	0.553	0.702	0.462	0.536	0.681	0.486	0.564	0.716	0.595	0.691	0.807	0.581	0.675	0.786
Washington Wizards	0.476	0.552	0.701	0.461	0.536	0.680	0.485	0.563	0.715	0.595	0.690	0.806	0.580	0.674	0.785
Minnesota Timberwolves	0.479	0.556	0.706	0.465	0.540	0.685	0.489	0.568	0.720	0.599	0.696	0.812	0.585	0.679	0.791
New York Knicks	0.478	0.555	0.704	0.463	0.538	0.683	0.487	0.566	0.718	0.597	0.693	0.809	0.583	0.677	0.788
Atlanta Hawks	0.479	0.556	0.706	0.465	0.540	0.685	0.489	0.567	0.720	0.599	0.695	0.812	0.584	0.679	0.790
Charlotte Hornets	0.475	0.552	0.700	0.461	0.535	0.679	0.485	0.563	0.714	0.594	0.690	0.805	0.580	0.673	0.784
Golden State Warriors	0.484	0.562	0.713	0.469	0.545	0.691	0.493	0.573	0.727	0.605	0.702	0.820	0.590	0.685	0.798
Detroit Pistons	0.479	0.556	0.705	0.464	0.539	0.684	0.488	0.567	0.719	0.598	0.695	0.811	0.584	0.678	0.790
Cleveland Cavaliers	0.479	0.557	0.706	0.465	0.540	0.685	0.489	0.568	0.720	0.599	0.696	0.812	0.585	0.679	0.791

Table 35: Table 18: Prediction Interval Lengths for 2021–2022 Season (Discretized Data (M=400), Bounded $\leq$ 1.000)

	Multilinear Regression			**Random Forest**			**Neural Network**			**ARIMA**			**LSTM**		
Team	90%	95%	99%	90%	95%	99%	90%	95%	99%	90%	95%	99%	90%	95%	99%
Milwaukee Bucks	0.468	0.541	0.660	0.454	0.525	0.641	0.477	0.552	0.674	0.585	0.676	0.760	0.571	0.660	0.740
Toronto Raptors	0.466	0.538	0.657	0.452	0.522	0.638	0.475	0.549	0.671	0.582	0.673	0.756	0.568	0.657	0.736
Los Angeles Clippers	0.462	0.534	0.652	0.448	0.518	0.633	0.471	0.545	0.665	0.578	0.668	0.750	0.564	0.652	0.730
Los Angeles Lakers	0.466	0.538	0.657	0.452	0.522	0.637	0.475	0.549	0.670	0.582	0.673	0.756	0.568	0.657	0.736
Boston Celtics	0.461	0.533	0.651	0.447	0.517	0.631	0.470	0.544	0.664	0.577	0.666	0.749	0.563	0.650	0.729
Houston Rockets	0.457	0.529	0.646	0.444	0.513	0.626	0.467	0.539	0.658	0.572	0.661	0.742	0.558	0.645	0.723
Utah Jazz	0.457	0.529	0.646	0.444	0.513	0.626	0.467	0.539	0.658	0.572	0.661	0.742	0.558	0.645	0.723
Miami Heat	0.457	0.528	0.645	0.443	0.512	0.625	0.466	0.539	0.658	0.571	0.660	0.742	0.557	0.644	0.722
Dallas Mavericks	0.455	0.526	0.642	0.441	0.510	0.623	0.464	0.536	0.655	0.569	0.657	0.738	0.555	0.641	0.719
Philadelphia 76ers	0.456	0.527	0.643	0.442	0.511	0.624	0.465	0.537	0.656	0.570	0.659	0.740	0.556	0.643	0.721
Indiana Pacers	0.458	0.529	0.646	0.444	0.513	0.627	0.467	0.540	0.659	0.572	0.661	0.743	0.559	0.645	0.724
Denver Nuggets	0.459	0.530	0.647	0.445	0.514	0.628	0.468	0.541	0.660	0.573	0.663	0.745	0.560	0.647	0.725
Brooklyn Nets	0.451	0.521	0.636	0.437	0.505	0.617	0.460	0.532	0.649	0.564	0.651	0.732	0.550	0.636	0.713
Phoenix Suns	0.452	0.523	0.638	0.439	0.507	0.619	0.461	0.533	0.651	0.565	0.653	0.734	0.552	0.638	0.715
Oklahoma City Thunder	0.457	0.529	0.646	0.444	0.513	0.626	0.467	0.539	0.658	0.572	0.661	0.742	0.558	0.645	0.723
Portland Trail Blazers	0.452	0.522	0.638	0.438	0.506	0.618	0.461	0.533	0.650	0.565	0.653	0.733	0.551	0.637	0.714
Memphis Grizzlies	0.452	0.523	0.638	0.439	0.507	0.619	0.461	0.533	0.651	0.565	0.653	0.734	0.552	0.638	0.715
New Orleans Pelicans	0.456	0.526	0.643	0.442	0.511	0.624	0.465	0.537	0.656	0.570	0.658	0.739	0.556	0.642	0.720
Sacramento Kings	0.455	0.525	0.642	0.441	0.510	0.622	0.464	0.536	0.654	0.568	0.657	0.738	0.555	0.641	0.719
Orlando Magic	0.453	0.524	0.640	0.440	0.508	0.620	0.462	0.534	0.652	0.567	0.655	0.736	0.553	0.639	0.716
San Antonio Spurs	0.453	0.524	0.640	0.440	0.508	0.621	0.462	0.534	0.652	0.567	0.655	0.736	0.553	0.639	0.716
Chicago Bulls	0.461	0.533	0.650	0.447	0.517	0.631	0.470	0.543	0.663	0.576	0.666	0.748	0.562	0.650	0.728
Washington Wizards	0.460	0.532	0.650	0.447	0.516	0.630	0.470	0.543	0.663	0.575	0.665	0.747	0.562	0.649	0.727
Minnesota Timberwolves	0.464	0.536	0.654	0.450	0.520	0.635	0.473	0.547	0.667	0.580	0.670	0.752	0.566	0.654	0.733
New York Knicks	0.462	0.534	0.652	0.448	0.518	0.633	0.472	0.545	0.665	0.578	0.668	0.750	0.564	0.652	0.731
Atlanta Hawks	0.464	0.536	0.654	0.450	0.520	0.635	0.473	0.546	0.667	0.580	0.670	0.752	0.566	0.654	0.733
Charlotte Hornets	0.460	0.531	0.649	0.446	0.515	0.629	0.469	0.542	0.662	0.575	0.664	0.746	0.561	0.648	0.727
Golden State Warriors	0.468	0.541	0.661	0.454	0.525	0.641	0.478	0.552	0.674	0.585	0.676	0.760	0.571	0.660	0.740
Detroit Pistons	0.463	0.535	0.654	0.449	0.519	0.634	0.473	0.546	0.667	0.579	0.669	0.752	0.565	0.653	0.732
Cleveland Cavaliers	0.464	0.536	0.655	0.450	0.520	0.635	0.473	0.547	0.668	0.580	0.670	0.753	0.566	0.654	0.733

Table 36: Table 19: Prediction Interval Lengths for 2021–2022 Season (Discretized Model (M=600), Bounded $\leq$ 1.000)

	Multilinear Regression			Random Forest			Neural Network			ARIMA			LSTM		
Team	90%	95%	99%	90%	95%	99%	90%	95%	99%	90%	95%	99%	90%	95%	99%
Milwaukee Bucks	0.469	0.542	0.661	0.455	0.526	0.642	0.478	0.553	0.675	0.586	0.677	0.761	0.572	0.661	0.741
Toronto Raptors	0.467	0.539	0.659	0.453	0.523	0.639	0.476	0.550	0.672	0.584	0.674	0.757	0.570	0.658	0.738
Los Angeles Clippers	0.463	0.535	0.653	0.449	0.519	0.634	0.472	0.546	0.666	0.579	0.669	0.751	0.565	0.653	0.732
Los Angeles Lakers	0.467	0.539	0.658	0.453	0.523	0.638	0.476	0.550	0.671	0.583	0.674	0.757	0.569	0.658	0.737
Boston Celtics	0.462	0.534	0.652	0.448	0.518	0.632	0.472	0.545	0.665	0.578	0.668	0.750	0.564	0.652	0.730
Houston Rockets	0.459	0.530	0.647	0.445	0.514	0.627	0.468	0.540	0.660	0.573	0.662	0.744	0.559	0.646	0.724
Utah Jazz	0.459	0.530	0.647	0.445	0.514	0.627	0.468	0.540	0.660	0.573	0.662	0.744	0.559	0.646	0.724
Miami Heat	0.458	0.529	0.646	0.444	0.513	0.626	0.467	0.540	0.659	0.572	0.661	0.743	0.559	0.645	0.723
Dallas Mavericks	0.456	0.527	0.643	0.442	0.511	0.624	0.465	0.537	0.656	0.570	0.658	0.739	0.556	0.643	0.720
Philadelphia 76ers	0.457	0.528	0.644	0.443	0.512	0.625	0.466	0.539	0.657	0.571	0.660	0.741	0.558	0.644	0.722
Indiana Pacers	0.459	0.530	0.647	0.445	0.514	0.628	0.468	0.541	0.660	0.574	0.663	0.744	0.560	0.647	0.725
Denver Nuggets	0.460	0.531	0.648	0.446	0.515	0.629	0.469	0.542	0.661	0.575	0.664	0.746	0.561	0.648	0.726
Brooklyn Nets	0.452	0.522	0.637	0.438	0.506	0.618	0.461	0.533	0.650	0.565	0.653	0.733	0.551	0.637	0.714
Phoenix Suns	0.453	0.524	0.639	0.440	0.508	0.620	0.462	0.534	0.652	0.567	0.655	0.735	0.553	0.639	0.716
Oklahoma City Thunder	0.459	0.530	0.647	0.445	0.514	0.627	0.468	0.540	0.660	0.573	0.662	0.744	0.559	0.646	0.724
Portland Trail Blazers	0.453	0.523	0.639	0.439	0.507	0.619	0.462	0.534	0.651	0.566	0.654	0.734	0.552	0.638	0.715
Memphis Grizzlies	0.453	0.524	0.639	0.440	0.508	0.620	0.462	0.534	0.652	0.567	0.655	0.735	0.553	0.639	0.716
New Orleans Pelicans	0.457	0.528	0.644	0.443	0.512	0.625	0.466	0.538	0.657	0.571	0.659	0.741	0.557	0.644	0.721
Sacramento Kings	0.456	0.526	0.643	0.442	0.511	0.623	0.465	0.537	0.655	0.570	0.658	0.739	0.556	0.642	0.720
Orlando Magic	0.454	0.525	0.641	0.441	0.509	0.621	0.463	0.535	0.653	0.568	0.656	0.737	0.554	0.640	0.717
San Antonio Spurs	0.454	0.525	0.641	0.441	0.509	0.621	0.463	0.535	0.654	0.568	0.656	0.737	0.554	0.640	0.718
Chicago Bulls	0.462	0.534	0.651	0.448	0.518	0.632	0.471	0.544	0.664	0.577	0.667	0.749	0.564	0.651	0.730
Washington Wizards	0.461	0.533	0.651	0.447	0.517	0.631	0.471	0.544	0.664	0.577	0.666	0.748	0.563	0.650	0.729
Minnesota Timberwolves	0.465	0.537	0.655	0.451	0.521	0.636	0.474	0.548	0.668	0.581	0.671	0.754	0.567	0.655	0.734
New York Knicks	0.463	0.535	0.653	0.449	0.519	0.634	0.473	0.546	0.666	0.579	0.669	0.751	0.565	0.653	0.732
Atlanta Hawks	0.465	0.537	0.655	0.451	0.521	0.636	0.474	0.547	0.668	0.581	0.671	0.754	0.567	0.655	0.734
Charlotte Hornets	0.461	0.532	0.650	0.447	0.516	0.630	0.470	0.543	0.663	0.576	0.666	0.747	0.562	0.650	0.728
Golden State Warriors	0.469	0.542	0.662	0.455	0.526	0.642	0.479	0.553	0.675	0.587	0.678	0.761	0.572	0.661	0.741
Detroit Pistons	0.464	0.536	0.655	0.450	0.520	0.635	0.474	0.547	0.668	0.580	0.670	0.753	0.566	0.654	0.733
Cleveland Cavaliers	0.465	0.537	0.656	0.451	0.521	0.636	0.474	0.548	0.669	0.581	0.672	0.754	0.567	0.655	0.735

Table 37: Table 20: Prediction Interval Lengths for 2021–2022 Season (Adaptive Conformal Inference ($\gamma = 0.05$), Bounded $\leq$ 1.000)

	Multilinear Regression			Random Forest			Neural Network			ARIMA			LSTM		
Team	90%	95%	99%	90%	95%	99%	90%	95%	99%	90%	95%	99%	90%	95%	99%
Milwaukee Bucks	0.470	0.543	0.663	0.456	0.527	0.643	0.480	0.554	0.676	0.588	0.679	0.762	0.574	0.662	0.742
Toronto Raptors	0.468	0.540	0.660	0.454	0.524	0.640	0.477	0.551	0.673	0.585	0.676	0.758	0.571	0.659	0.739
Los Angeles Clippers	0.464	0.536	0.654	0.450	0.520	0.635	0.474	0.547	0.667	0.580	0.670	0.752	0.566	0.654	0.733
Los Angeles Lakers	0.468	0.540	0.659	0.454	0.524	0.639	0.477	0.551	0.672	0.585	0.675	0.758	0.571	0.659	0.738
Boston Celtics	0.463	0.535	0.653	0.449	0.519	0.633	0.473	0.546	0.666	0.579	0.669	0.751	0.565	0.653	0.731
Houston Rockets	0.460	0.531	0.648	0.446	0.515	0.628	0.469	0.541	0.661	0.574	0.663	0.745	0.561	0.647	0.725
Utah Jazz	0.460	0.531	0.648	0.446	0.515	0.628	0.469	0.541	0.661	0.574	0.663	0.745	0.561	0.647	0.725
Miami Heat	0.459	0.530	0.647	0.445	0.514	0.627	0.468	0.541	0.660	0.574	0.663	0.744	0.560	0.647	0.724
Dallas Mavericks	0.457	0.528	0.644	0.443	0.512	0.625	0.466	0.538	0.657	0.571	0.660	0.741	0.558	0.644	0.721
Philadelphia 76ers	0.458	0.529	0.646	0.444	0.513	0.626	0.467	0.540	0.658	0.573	0.661	0.742	0.559	0.645	0.723
Indiana Pacers	0.460	0.531	0.648	0.446	0.515	0.629	0.469	0.542	0.661	0.575	0.664	0.745	0.561	0.648	0.726
Denver Nuggets	0.461	0.532	0.649	0.447	0.516	0.630	0.470	0.543	0.662	0.576	0.665	0.747	0.562	0.649	0.727
Brooklyn Nets	0.453	0.523	0.638	0.439	0.507	0.619	0.462	0.534	0.651	0.566	0.654	0.734	0.553	0.638	0.715
Phoenix Suns	0.454	0.525	0.640	0.441	0.509	0.621	0.463	0.535	0.653	0.568	0.656	0.736	0.554	0.640	0.717
Oklahoma City Thunder	0.460	0.531	0.648	0.446	0.515	0.628	0.469	0.541	0.661	0.574	0.663	0.745	0.561	0.647	0.725
Portland Trail Blazers	0.454	0.524	0.640	0.440	0.508	0.620	0.463	0.535	0.652	0.567	0.655	0.736	0.554	0.639	0.716
Memphis Grizzlies	0.454	0.525	0.640	0.441	0.509	0.621	0.463	0.535	0.653	0.568	0.656	0.736	0.554	0.640	0.717
New Orleans Pelicans	0.458	0.529	0.645	0.444	0.513	0.626	0.467	0.539	0.658	0.572	0.661	0.742	0.558	0.645	0.722
Sacramento Kings	0.457	0.527	0.644	0.443	0.512	0.624	0.466	0.538	0.657	0.571	0.659	0.740	0.557	0.643	0.721
Orlando Magic	0.455	0.526	0.642	0.442	0.510	0.622	0.464	0.536	0.654	0.569	0.657	0.738	0.555	0.641	0.719
San Antonio Spurs	0.455	0.526	0.642	0.442	0.510	0.622	0.464	0.536	0.655	0.569	0.657	0.738	0.556	0.642	0.719
Chicago Bulls	0.463	0.535	0.652	0.449	0.519	0.633	0.472	0.545	0.665	0.579	0.668	0.750	0.565	0.652	0.731
Washington Wizards	0.462	0.534	0.652	0.448	0.518	0.632	0.472	0.545	0.665	0.578	0.667	0.749	0.564	0.651	0.730
Minnesota Timberwolves	0.466	0.538	0.656	0.452	0.522	0.637	0.475	0.549	0.670	0.582	0.672	0.755	0.568	0.656	0.735
New York Knicks	0.464	0.536	0.654	0.450	0.520	0.635	0.474	0.547	0.667	0.580	0.670	0.753	0.566	0.654	0.733
Atlanta Hawks	0.466	0.538	0.656	0.452	0.522	0.637	0.475	0.549	0.669	0.582	0.672	0.755	0.568	0.656	0.735
Charlotte Hornets	0.462	0.533	0.651	0.448	0.517	0.631	0.471	0.544	0.664	0.577	0.667	0.749	0.564	0.651	0.729
Golden State Warriors	0.470	0.543	0.663	0.456	0.527	0.643	0.480	0.554	0.676	0.588	0.679	0.762	0.574	0.663	0.742
Detroit Pistons	0.465	0.537	0.656	0.451	0.521	0.636	0.475	0.548	0.669	0.582	0.672	0.754	0.568	0.656	0.735
Cleveland Cavaliers	0.466	0.538	0.657	0.452	0.522	0.637	0.475	0.549	0.670	0.583	0.673	0.755	0.569	0.657	0.736

3.7 3.3 Conformal Prediction in NBA MVP Voting Data

3.7.1 3.3.1 Variables

- **The explanatory variables:** all the numerical columns in the basketball reference website, such as Age, G (games played), MP (minutes played), PER (player efficiency rating), TS% (true shooting percentage), 3PAr (3-point attempt rate), FTr (free throw attempt rate), TRB% (total rebound percentage), AST% (assist percentage), STL% (steal percentage), BLK% (block percentage), TOV% (turnover percentage), USG% (usage percentage), WS (win shares), BPM (box plus/minus), VORP (value over replacement player)... etc (with total 23 features).

- **The response variable:** the next year's MVP voting share.

3.7.2 3.3.2 Steps

1. **Data Collection & Player Longitudinal Structuring:**
   - Collect historical player-level statistical data spanning 20 NBA seasons (2000–2001 to 2020–2021) across all qualified players ($n \approx 2,700$ player-season records).
   - Format each player record i as $(X_{i,t}, Y_{i,t+1})$, where $X_{i,t}$ contains 23 player-level metrics from season t (`Age`, `G`, `MP`, `PER`, `TS%`, `3PAr`, `FTr`, `ORB%`, `DRB%`, `TRB%`, `AST%`, `STL%`, `BLK%`, `TOV%`, `USG%`, `OWS`, `DWS`, `WS`, `WS/48`, `OBPM`, `DBPM`, `BPM`, `VORP`).
   - The response variable $Y_{i,t+1} \in [0, 1]$ is player i's normalized actual MVP voting share in season $t + 1$.

2. **Data Splitting & Time-Series Grouping:** Partition pre-2020 historical player data across 20 historical seasons using two distinct splitting paradigms across three split ratios (0.8/0.2, 0.65/0.35, 0.5/0.5), holding out the known 2020–2021 season as the evaluation test set:
   - **Random Splitting (Group-Based):** Perform Group-Based Splitting (grouped by `Player_ID`) across pre-2020 historical records to ensure all career records of a given player remain strictly within either the proper training set L_1 or the calibration set L_2.
   - **Temporal / Walk-Forward Splitting (Time-Series Split):** Preserve strict chronological order across the 20 historical seasons:
     - **0.8 / 0.2 Temporal Split:** Training Set L_1 (2000–2001 to 2015–2016), Calibration Set L_2 (2016–2017 to 2019–2020), Test Set (2020–2021).
     - **0.65 / 0.35 Temporal Split:** Training Set L_1 (2000–2001 to 2012–2013), Calibration Set L_2 (2013–2014 to 2019–2020), Test Set (2020–2021).
     - **0.5 / 0.5 Temporal Split:** Training Set L_1 (2000–2001 to 2009–2010), Calibration Set L_2 (2010–2011 to 2019–2020), Test Set (2020–2021).

3. **Base Predictor Selection ($\mathcal{A}$):** Train five base predictive algorithms $\hat{\mu}(x)$ on proper training set L_1 for each split ratio:
   - **Multiple Linear Regression:** Parametric baseline model utilizing feature selection on significant player efficiency indicators.
   - **Random Forest Regressor:** Non-parametric ensemble capturing non-linear interactions across individual box-score and advanced impact metrics.
   - **Neural Network (MLP):** Multi-layer perceptron fitting high-dimensional interactions between player volume and efficiency.
   - **Time-Series Model 1 (ARIMA / Autoregressive):** Sequential linear model tracking historical individual player performance trajectories.
   - **Time-Series Model 2 (LSTM / Recurrent Network):** Sequential deep learning architecture modeling multi-year player development and career impact curves.

   Note: Time-series models strictly utilize Temporal Splitting, whereas Linear Regression, Random Forests, and Neural Networks evaluate both Group-Based Random Splitting and Temporal Splitting across all proportions.

4. **Conformal Calibration & Quantile Computation:** Evaluate seven distinct conformal prediction frameworks:

   - **Inductive Split Conformal Methods (Split, Locally Adaptive, CQR):** Fit base model $\hat{\mu}$ on L_1, compute nonconformity scores (absolute residuals $R_i = |Y_i - \hat{\mu}(X_i)|$, scaled residuals, or pinball loss scores) on calibration set L_2, and derive empirical quantiles $Q_{1-\alpha}(R, L_2)$ at nominal coverage levels $1-\alpha \in \{0.90, 0.95, 0.99\}$.
   - **Transductive / Grid Methods (Discretized Data, Discretized Model, Rounding):** Construct candidate grids $\hat{\mathcal{Y}} \subset [0,1]$ across $M \in \{5, 10, 25, 50, 100, 200, 400, 600, 800\}$ grid points. Fit models and evaluate residuals over the full historical training window $L_1 \cup L_2$.
   - **Adaptive Conformal Inference (ACI – Gibbs & Candès, 2021):** Online time-series wrapper algorithm dynamically updating miscoverage parameter α_t step-by-step using pinball loss updates:

     $$\alpha_{t+1} = \alpha_t + \gamma(\alpha - \mathrm{err}_t)$$

     where $\mathrm{err}_t = \mathbb{I}(Y_t \notin \hat{C}_t(\alpha_t))$ and step size $\gamma^* > 0$ is tuned per model architecture.

5. **Test Set Evaluation & Empirical Validation (Validation on Known 2020–2021 Results):** Apply fitted models and calibration quantiles to the target test set X_{test} (2019–2020 player stats predicting known 2020–2021 MVP voting shares) to construct prediction intervals $[\hat{L}(X_{i,n+1}), \hat{U}(X_{i,n+1})]$. Compute two core metrics across every combination of coverage level $(1-\alpha)$, split ratio, grid point M, and base architecture:

   - **Average Prediction Interval Length:**

     $$\text{Avg Length} = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} \left(\hat{U}(X_i) - \hat{L}(X_i) \right)$$

   - **Empirical Coverage Rate:**

     $$\text{Empirical Coverage} = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} \mathbb{I} \left(Y_i \in [\hat{L}(X_i), \hat{U}(X_i)] \right)$$

6. **Out-of-Sample MVP Rankings & Contender Forecasting (Unknown 2021–2022 Season):**

   - Input 2020–2021 player metrics ($X_{2020-2021}$) into the optimal calibrated model setup to forecast point predictions $\hat{\mu}(X_{2020-2021})$ and conformal intervals $[\hat{L}, \hat{U}]$ for 2021–2022 MVP voting shares.
   - Rank all players in descending order according to their predicted voting share $\hat{\mu}(X_{2020-2021})$.
   - Identify the top candidate with the highest predicted voting share as the forecasted MVP winner, and construct the predicted Top 5 MVP finalist rankings alongside their respective conformal confidence bands.

3.7.3 3.3.3 Validation Results (2020–2021 Known Data)

Table 38: Table 1a: Empirical Coverage Rate for MVP Voting Share Inductive Methods and ACI (90% Target Coverage)

Model	Split / Ratio	Split (90%)	Locally (90%)	CQR (90%)	ACI (90%)
Multiple Linear Regression	Random 0.8/0.2	90.2%	89.8%	90.6%	90.1%
Multiple Linear Regression	Random 0.65/0.35	90.0%	89.6%	90.4%	90.0%
Multiple Linear Regression	Random 0.5/0.5	89.7%	89.3%	90.1%	89.7%
Multiple Linear Regression	Temporal 0.8/0.2	89.5%	89.1%	90.2%	90.0%
Multiple Linear Regression	Temporal 0.65/0.35	89.2%	88.8%	89.9%	89.7%
Multiple Linear Regression	Temporal 0.5/0.5	88.9%	88.4%	89.6%	89.4%
Linear Reg. Average		**89.6%**	**89.2%**	**90.3%**	**89.8%**
Random Forest	Random 0.8/0.2	91.0%	90.6%	91.5%	90.4%
Random Forest	Random 0.65/0.35	90.6%	90.2%	91.1%	90.2%
Random Forest	Random 0.5/0.5	90.3%	89.9%	90.7%	90.0%
Random Forest	Temporal 0.8/0.2	90.1%	89.7%	90.6%	90.1%
Random Forest	Temporal 0.65/0.35	89.7%	89.3%	90.2%	89.9%
Random Forest	Temporal 0.5/0.5	89.3%	88.9%	89.8%	89.6%
Random Forest Average		**90.2%**	**89.8%**	**90.7%**	**90.0%**
Neural Network (MLP)	Random 0.8/0.2	90.4%	90.0%	91.0%	90.0%
Neural Network (MLP)	Random 0.65/0.35	90.0%	89.6%	90.6%	89.8%
Neural Network (MLP)	Random 0.5/0.5	89.7%	89.3%	90.2%	89.7%
Neural Network (MLP)	Temporal 0.8/0.2	89.6%	89.2%	90.1%	89.8%
Neural Network (MLP)	Temporal 0.65/0.35	89.2%	88.8%	89.7%	89.5%
Neural Network (MLP)	Temporal 0.5/0.5	88.8%	88.4%	89.3%	89.3%
Neural Net Average		**89.6%**	**89.2%**	**90.3%**	**89.7%**
ARIMA	Temporal 0.8/0.2	89.8%	89.4%	90.1%	89.7%
ARIMA	Temporal 0.65/0.35	89.4%	89.0%	89.7%	89.5%
ARIMA	Temporal 0.5/0.5	89.0%	88.6%	89.3%	89.3%
ARIMA Average		**89.4%**	**89.0%**	**89.7%**	**89.5%**
LSTM	Temporal 0.8/0.2	90.3%	89.9%	90.7%	90.2%
LSTM	Temporal 0.65/0.35	89.9%	89.5%	90.3%	90.0%
LSTM	Temporal 0.5/0.5	89.5%	89.1%	89.9%	89.7%
LSTM Average		**89.9%**	**89.5%**	**90.3%**	**90.0%**

Table 39: Table 1b: Empirical Coverage Rate for MVP Voting Share Inductive Methods and ACI (95% Target Coverage)

Model	Split / Ratio	Split (95%)	Locally (95%)	CQR (95%)	ACI (95%)
Multiple Linear Regression	Random 0.8/0.2	95.1%	94.7%	95.5%	95.0%
Multiple Linear Regression	Random 0.65/0.35	94.8%	94.4%	95.2%	94.7%
Multiple Linear Regression	Random 0.5/0.5	94.5%	94.1%	94.9%	94.4%
Multiple Linear Regression	Temporal 0.8/0.2	94.3%	93.9%	94.8%	94.7%
Multiple Linear Regression	Temporal 0.65/0.35	94.0%	93.6%	94.5%	94.4%
Multiple Linear Regression	Temporal 0.5/0.5	93.7%	93.3%	94.1%	94.1%
Linear Reg. Average		**94.4%**	**94.0%**	**94.9%**	**94.5%**
Random Forest	Random 0.8/0.2	95.8%	95.4%	96.2%	95.2%
Random Forest	Random 0.65/0.35	95.4%	95.0%	95.8%	95.0%
Random Forest	Random 0.5/0.5	95.1%	94.7%	95.5%	94.8%
Random Forest	Temporal 0.8/0.2	94.9%	94.5%	95.3%	94.9%
Random Forest	Temporal 0.65/0.35	94.5%	94.1%	94.9%	94.7%
Random Forest	Temporal 0.5/0.5	94.1%	93.7%	94.5%	94.5%
Random Forest Average		**95.0%**	**94.6%**	**95.4%**	**94.9%**
Neural Network (MLP)	Random 0.8/0.2	95.2%	94.8%	95.7%	94.8%
Neural Network (MLP)	Random 0.65/0.35	94.8%	94.4%	95.3%	94.6%
Neural Network (MLP)	Random 0.5/0.5	94.5%	94.1%	95.0%	94.5%
Neural Network (MLP)	Temporal 0.8/0.2	94.4%	94.0%	94.9%	94.6%
Neural Network (MLP)	Temporal 0.65/0.35	94.0%	93.6%	94.5%	94.4%
Neural Network (MLP)	Temporal 0.5/0.5	93.6%	93.2%	94.1%	94.2%
Neural Net Average		**94.4%**	**94.0%**	**94.9%**	**94.5%**
ARIMA	Temporal 0.8/0.2	94.5%	94.0%	94.8%	94.6%
ARIMA	Temporal 0.65/0.35	94.1%	93.6%	94.4%	94.3%
ARIMA	Temporal 0.5/0.5	93.7%	93.2%	94.0%	94.1%
ARIMA Average		**94.1%**	**93.6%**	**94.4%**	**94.3%**
LSTM	Temporal 0.8/0.2	95.1%	94.7%	95.6%	95.0%
LSTM	Temporal 0.65/0.35	94.7%	94.3%	95.2%	94.8%
LSTM	Temporal 0.5/0.5	94.3%	93.9%	94.8%	94.5%
LSTM Average		**94.7%**	**94.3%**	**95.2%**	**94.8%**

Table 40: Table 1c: Empirical Coverage Rate for MVP Voting Share Inductive Methods and ACI (99% Target Coverage)

Model	Split / Ratio	Split (99%)	Locally (99%)	CQR (99%)	ACI (99%)
Multiple Linear Regression	Random 0.8/0.2	99.2%	98.9%	99.4%	99.0%
Multiple Linear Regression	Random 0.65/0.35	99.0%	98.7%	99.2%	98.8%
Multiple Linear Regression	Random 0.5/0.5	98.8%	98.5%	99.0%	98.6%
Multiple Linear Regression	Temporal 0.8/0.2	98.7%	98.4%	99.0%	98.9%
Multiple Linear Regression	Temporal 0.65/0.35	98.5%	98.2%	98.8%	98.7%
Multiple Linear Regression	Temporal 0.5/0.5	98.3%	98.0%	98.6%	98.5%
Linear Reg. Average		**98.8%**	**98.5%**	**99.0%**	**98.7%**
Random Forest	Random 0.8/0.2	99.5%	99.3%	99.6%	99.1%
Random Forest	Random 0.65/0.35	99.3%	99.1%	99.4%	98.9%
Random Forest	Random 0.5/0.5	99.1%	98.9%	99.2%	98.7%
Random Forest	Temporal 0.8/0.2	99.0%	98.8%	99.1%	98.8%
Random Forest	Temporal 0.65/0.35	98.8%	98.6%	98.9%	98.6%
Random Forest	Temporal 0.5/0.5	98.6%	98.4%	98.7%	98.4%
Random Forest Average		**99.1%**	**98.9%**	**99.2%**	**98.8%**
Neural Network (MLP)	Random 0.8/0.2	99.3%	99.0%	99.4%	98.9%
Neural Network (MLP)	Random 0.65/0.35	99.1%	98.8%	99.2%	98.7%
Neural Network (MLP)	Random 0.5/0.5	98.9%	98.6%	99.0%	98.5%
Neural Network (MLP)	Temporal 0.8/0.2	98.8%	98.5%	98.9%	98.7%
Neural Network (MLP)	Temporal 0.65/0.35	98.6%	98.3%	98.7%	98.5%
Neural Network (MLP)	Temporal 0.5/0.5	98.4%	98.1%	98.5%	98.3%
Neural Net Average		**98.9%**	**98.6%**	**99.0%**	**98.6%**
ARIMA	Temporal 0.8/0.2	98.8%	98.5%	98.9%	98.8%
ARIMA	Temporal 0.65/0.35	98.6%	98.3%	98.7%	98.6%
ARIMA	Temporal 0.5/0.5	98.4%	98.1%	98.5%	98.4%
ARIMA Average		**98.6%**	**98.3%**	**98.7%**	**98.6%**
LSTM	Temporal 0.8/0.2	99.2%	99.0%	99.4%	99.1%
LSTM	Temporal 0.65/0.35	99.0%	98.8%	99.2%	98.9%
LSTM	Temporal 0.5/0.5	98.8%	98.6%	99.0%	98.7%
LSTM Average		**99.0%**	**98.8%**	**99.2%**	**98.9%**

Table 41: Table 2a: Average Prediction Interval Length (MVP Voting Share units) for Player Data (90% Target Coverage)

Model	Split / Ratio	Split (90%)	Locally (90%)	CQR (90%)	ACI (90%)
Multiple Linear Regression	Random 0.8/0.2	0.185	0.178	0.172	0.182
Multiple Linear Regression	Random 0.65/0.35	0.179	0.172	0.166	0.176
Multiple Linear Regression	Random 0.5/0.5	0.191	0.184	0.178	0.188
Multiple Linear Regression	Temporal 0.8/0.2	0.188	0.181	0.175	0.185
Multiple Linear Regression	Temporal 0.65/0.35	0.182	0.175	0.169	0.179
Multiple Linear Regression	Temporal 0.5/0.5	0.195	0.188	0.181	0.191
Linear Reg. Average		**0.185**	**0.178**	**0.172**	**0.182**
Random Forest	Random 0.8/0.2	0.172	0.165	0.160	0.169
Random Forest	Random 0.65/0.35	0.166	0.159	0.154	0.163
Random Forest	Random 0.5/0.5	0.178	0.171	0.166	0.175
Random Forest	Temporal 0.8/0.2	0.175	0.168	0.163	0.172
Random Forest	Temporal 0.65/0.35	0.169	0.162	0.157	0.166
Random Forest	Temporal 0.5/0.5	0.182	0.174	0.169	0.178
Random Forest Average		**0.172**	**0.165**	**0.160**	**0.169**
Neural Network (MLP)	Random 0.8/0.2	0.179	0.172	0.167	0.176
Neural Network (MLP)	Random 0.65/0.35	0.173	0.166	0.161	0.170
Neural Network (MLP)	Random 0.5/0.5	0.185	0.178	0.173	0.182
Neural Network (MLP)	Temporal 0.8/0.2	0.182	0.175	0.170	0.179
Neural Network (MLP)	Temporal 0.65/0.35	0.176	0.169	0.164	0.173
Neural Network (MLP)	Temporal 0.5/0.5	0.189	0.181	0.176	0.185
Neural Net Average		**0.179**	**0.172**	**0.167**	**0.176**
ARIMA	Temporal 0.8/0.2	0.189	0.181	0.175	0.186
ARIMA	Temporal 0.65/0.35	0.183	0.175	0.169	0.180
ARIMA	Temporal 0.5/0.5	0.195	0.187	0.181	0.192
ARIMA Average		**0.189**	**0.181**	**0.175**	**0.186**
LSTM	Temporal 0.8/0.2	0.194	0.187	0.181	0.191
LSTM	Temporal 0.65/0.35	0.188	0.181	0.175	0.185
LSTM	Temporal 0.5/0.5	0.200	0.193	0.187	0.197
LSTM Average		**0.194**	**0.187**	**0.181**	**0.191**

Table 42: Table 2b: Average Prediction Interval Length (MVP Voting Share units) for Player Data (95% Target Coverage)

Model	Split / Ratio	Split (95%)	Locally (95%)	CQR (95%)	ACI (95%)
Multiple Linear Regression	Random 0.8/0.2	0.245	0.238	0.232	0.242
Multiple Linear Regression	Random 0.65/0.35	0.238	0.231	0.225	0.235
Multiple Linear Regression	Random 0.5/0.5	0.252	0.245	0.239	0.249
Multiple Linear Regression	Temporal 0.8/0.2	0.248	0.241	0.235	0.245
Multiple Linear Regression	Temporal 0.65/0.35	0.241	0.234	0.228	0.238
Multiple Linear Regression	Temporal 0.5/0.5	0.255	0.248	0.242	0.252
Linear Reg. Average		**0.245**	**0.238**	**0.232**	**0.242**
Random Forest	Random 0.8/0.2	0.228	0.221	0.215	0.225
Random Forest	Random 0.65/0.35	0.221	0.214	0.208	0.218
Random Forest	Random 0.5/0.5	0.235	0.228	0.222	0.232
Random Forest	Temporal 0.8/0.2	0.231	0.224	0.218	0.228
Random Forest	Temporal 0.65/0.35	0.224	0.217	0.211	0.221
Random Forest	Temporal 0.5/0.5	0.238	0.231	0.225	0.235
Random Forest Average		**0.228**	**0.221**	**0.215**	**0.225**
Neural Network (MLP)	Random 0.8/0.2	0.238	0.231	0.225	0.235
Neural Network (MLP)	Random 0.65/0.35	0.231	0.224	0.218	0.228
Neural Network (MLP)	Random 0.5/0.5	0.245	0.238	0.232	0.242
Neural Network (MLP)	Temporal 0.8/0.2	0.241	0.234	0.228	0.238
Neural Network (MLP)	Temporal 0.65/0.35	0.234	0.227	0.221	0.231
Neural Network (MLP)	Temporal 0.5/0.5	0.248	0.241	0.235	0.245
Neural Net Average		**0.238**	**0.231**	**0.225**	**0.235**
ARIMA	Temporal 0.8/0.2	0.250	0.243	0.237	0.247
ARIMA	Temporal 0.65/0.35	0.243	0.236	0.230	0.240
ARIMA	Temporal 0.5/0.5	0.257	0.250	0.244	0.254
ARIMA Average		**0.250**	**0.243**	**0.237**	**0.247**
LSTM	Temporal 0.8/0.2	0.257	0.250	0.244	0.254
LSTM	Temporal 0.65/0.35	0.250	0.243	0.237	0.247
LSTM	Temporal 0.5/0.5	0.264	0.257	0.251	0.261
LSTM Average		**0.257**	**0.250**	**0.244**	**0.254**

Table 43: Table 2c: Average Prediction Interval Length (MVP Voting Share units) for Player Data (99% Target Coverage)

Model	Split / Ratio	Split (99%)	Locally (99%)	CQR (99%)	ACI (99%)
Multiple Linear Regression	Random 0.8/0.2	0.365	0.358	0.352	0.362
Multiple Linear Regression	Random 0.65/0.35	0.357	0.350	0.344	0.354
Multiple Linear Regression	Random 0.5/0.5	0.373	0.366	0.360	0.370
Multiple Linear Regression	Temporal 0.8/0.2	0.369	0.362	0.356	0.366
Multiple Linear Regression	Temporal 0.65/0.35	0.361	0.354	0.348	0.358
Multiple Linear Regression	Temporal 0.5/0.5	0.377	0.370	0.364	0.374
Linear Reg. Average		**0.365**	**0.358**	**0.352**	**0.362**
Random Forest	Random 0.8/0.2	0.339	0.332	0.326	0.336
Random Forest	Random 0.65/0.35	0.331	0.324	0.318	0.328
Random Forest	Random 0.5/0.5	0.347	0.340	0.334	0.344
Random Forest	Temporal 0.8/0.2	0.343	0.336	0.330	0.340
Random Forest	Temporal 0.65/0.35	0.335	0.328	0.322	0.332
Random Forest	Temporal 0.5/0.5	0.351	0.344	0.338	0.348
Random Forest Average		**0.339**	**0.332**	**0.326**	**0.336**
Neural Network (MLP)	Random 0.8/0.2	0.354	0.347	0.341	0.351
Neural Network (MLP)	Random 0.65/0.35	0.346	0.339	0.333	0.343
Neural Network (MLP)	Random 0.5/0.5	0.362	0.355	0.349	0.359
Neural Network (MLP)	Temporal 0.8/0.2	0.358	0.351	0.345	0.355
Neural Network (MLP)	Temporal 0.65/0.35	0.350	0.343	0.337	0.347
Neural Network (MLP)	Temporal 0.5/0.5	0.366	0.359	0.353	0.363
Neural Net Average		**0.354**	**0.347**	**0.341**	**0.351**
ARIMA	Temporal 0.8/0.2	0.372	0.365	0.359	0.369
ARIMA	Temporal 0.65/0.35	0.364	0.357	0.351	0.361
ARIMA	Temporal 0.5/0.5	0.380	0.373	0.367	0.377
ARIMA Average		**0.372**	**0.365**	**0.359**	**0.369**
LSTM	Temporal 0.8/0.2	0.380	0.373	0.367	0.377
LSTM	Temporal 0.65/0.35	0.372	0.365	0.359	0.369
LSTM	Temporal 0.5/0.5	0.388	0.381	0.375	0.385
LSTM Average		**0.380**	**0.373**	**0.367**	**0.377**

Inductive Conformal Methods & ACI

Table 44: Table 3a: Empirical Coverage Rate for Transductive / Grid Methods (90% Target Coverage)

Model	M	Rounding (90%)	CPDD (90%)	CPDM (90%)
Multiple Linear Regression	800	90.2%	90.1%	90.0%
	600	90.1%	90.0%	90.0%
	400	90.0%	90.1%	90.0%
	200	90.2%	90.0%	90.0%
	100	89.9%	89.9%	90.0%
	50	89.5%	89.8%	89.9%
	25	87.2%	89.2%	89.8%
	10	72.1%	85.1%	89.6%
	5	41.5%	76.4%	89.2%
	Average	**82.3%**	**87.8%**	**89.8%**
Random Forest	800	90.8%	90.6%	90.5%
	600	90.7%	90.5%	90.5%
	400	90.6%	90.6%	90.5%
	200	90.8%	90.5%	90.5%
	100	90.5%	90.4%	90.5%
	50	90.1%	90.3%	90.4%
	25	87.8%	89.7%	90.3%
	10	72.8%	85.6%	90.1%
	5	42.1%	76.9%	89.7%
	Average	**82.9%**	**88.4%**	**90.4%**
Neural Network (MLP)	800	90.5%	90.3%	90.2%
	600	90.4%	90.2%	90.2%
	400	90.3%	90.3%	90.2%
	200	90.5%	90.2%	90.2%
	100	90.2%	90.1%	90.2%
	50	89.8%	90.0%	90.1%
	25	87.5%	89.4%	90.0%
	10	72.4%	85.3%	89.8%
	5	41.8%	76.6%	89.4%
	Average	**82.7%**	**88.2%**	**90.2%**
ARIMA	800	90.0%	89.8%	89.7%
	600	89.9%	89.7%	89.7%
	400	89.8%	89.8%	89.7%
	200	90.0%	89.7%	89.7%
	100	89.7%	89.6%	89.7%
	50	89.3%	89.5%	89.6%
	25	87.0%	88.9%	89.5%
	10	71.9%	84.7%	89.3%
	5	41.3%	76.0%	88.9%
	Average	**82.1%**	**87.5%**	**89.5%**
LSTM	800	90.6%	90.4%	90.3%
	600	90.5%	90.3%	90.3%
	400	90.4%	90.4%	90.3%
	200	90.6%	90.3%	90.3%
	100	90.3%	90.2%	90.3%
	50	89.9%	90.1%	90.2%
	25	87.6%	89.5%	90.1%
	10	72.5%	85.4%	89.9%
	5	41.9%	76.7%	89.5%
	Average	**82.7%**	**88.2%**	**90.2%**

Table 45: Table 3b: Empirical Coverage Rate for Transductive / Grid Methods (95% Target Coverage)

Model	M	Rounding (95%)	CPDD (95%)	CPDM (95%)
Multiple Linear Regression	800	95.2%	95.1%	95.0%
	600	95.1%	95.0%	95.0%
	400	95.0%	95.1%	95.0%
	200	95.2%	95.0%	95.0%
	100	94.9%	94.9%	95.0%
	50	94.5%	94.8%	94.9%
	25	92.2%	94.2%	94.8%
	10	77.1%	90.1%	94.6%
	5	46.5%	81.4%	94.2%
	Average	**87.3%**	**92.8%**	**94.8%**
Random Forest	800	95.8%	95.6%	95.5%
	600	95.7%	95.5%	95.5%
	400	95.6%	95.6%	95.5%
	200	95.8%	95.5%	95.5%
	100	95.5%	95.4%	95.5%
	50	95.1%	95.3%	95.4%
	25	92.8%	94.7%	95.3%
	10	77.8%	90.6%	95.1%
	5	47.1%	81.9%	94.7%
	Average	**87.9%**	**93.4%**	**95.4%**
Neural Network (MLP)	800	95.5%	95.3%	95.2%
	600	95.4%	95.2%	95.2%
	400	95.3%	95.3%	95.2%
	200	95.5%	95.2%	95.2%
	100	95.2%	95.1%	95.2%
	50	94.8%	95.0%	95.1%
	25	92.5%	94.4%	95.0%
	10	77.4%	90.3%	94.8%
	5	46.8%	81.6%	94.4%
	Average	**87.7%**	**93.2%**	**95.2%**
ARIMA	800	95.0%	94.8%	94.7%
	600	94.9%	94.7%	94.7%
	400	94.8%	94.8%	94.7%
	200	95.0%	94.7%	94.7%
	100	94.7%	94.6%	94.7%
	50	94.3%	94.5%	94.6%
	25	92.0%	93.9%	94.5%
	10	76.9%	89.7%	94.3%
	5	46.3%	81.0%	93.9%
	Average	**87.1%**	**92.5%**	**94.5%**
LSTM	800	95.6%	95.4%	95.3%
	600	95.5%	95.3%	95.3%
	400	95.4%	95.4%	95.3%
	200	95.6%	95.3%	95.3%
	100	95.3%	95.2%	95.3%
	50	94.9%	95.1%	95.2%
	25	92.6%	94.5%	95.1%
	10	77.5%	90.4%	94.9%
	5	46.9%	81.7%	94.5%
	Average	**87.7%**	**93.3%**	**95.3%**

Table 46: Table 3c: Empirical Coverage Rate for Transductive / Grid Methods (99% Target Coverage)

Model	M	Rounding (99%)	CPDD (99%)	CPDM (99%)
Multiple Linear Regression	800	99.2%	99.1%	99.0%
	600	99.1%	99.0%	99.0%
	400	99.0%	99.1%	99.0%
	200	99.2%	99.0%	99.0%
	100	98.9%	98.9%	99.0%
	50	98.5%	98.8%	98.9%
	25	96.2%	98.2%	98.8%
	10	81.1%	94.1%	98.6%
	5	50.5%	85.4%	98.2%
	Average	**91.3%**	**96.8%**	**98.8%**
Random Forest	800	99.8%	99.6%	99.5%
	600	99.7%	99.5%	99.5%
	400	99.6%	99.6%	99.5%
	200	99.8%	99.5%	99.5%
	100	99.5%	99.4%	99.5%
	50	99.1%	99.3%	99.4%
	25	96.8%	98.7%	99.3%
	10	81.8%	94.6%	99.1%
	5	51.1%	85.9%	98.7%
	Average	**91.9%**	**97.4%**	**99.4%**
Neural Network (MLP)	800	99.5%	99.3%	99.2%
	600	99.4%	99.2%	99.2%
	400	99.3%	99.3%	99.2%
	200	99.5%	99.2%	99.2%
	100	99.2%	99.1%	99.2%
	50	98.8%	99.0%	99.1%
	25	96.5%	98.4%	99.0%
	10	81.4%	94.3%	98.8%
	5	50.8%	85.6%	98.4%
	Average	**91.6%**	**97.1%**	**99.2%**
ARIMA	800	99.0%	98.8%	98.7%
	600	98.9%	98.7%	98.7%
	400	98.8%	98.8%	98.7%
	200	99.0%	98.7%	98.7%
	100	98.7%	98.6%	98.7%
	50	98.3%	98.5%	98.6%
	25	96.0%	97.9%	98.5%
	10	80.9%	93.7%	98.3%
	5	50.3%	85.0%	97.9%
	Average	**91.1%**	**96.5%**	**98.5%**
LSTM	800	99.6%	99.4%	99.3%
	600	99.5%	99.3%	99.3%
	400	99.4%	99.4%	99.3%
	200	99.6%	99.3%	99.3%
	100	99.3%	99.2%	99.3%
	50	98.9%	99.1%	99.2%
	25	96.6%	98.5%	99.1%
	10	81.5%	94.4%	98.9%
	5	50.9%	85.7%	98.5%
	Average	**91.7%**	**97.2%**	**99.3%**

Table 47: Table 4a: Average Prediction Interval Length for Transductive / Grid Methods (90% Target Coverage)

Model	M	Rounding (90%)	CPDD (90%)	CPDM (90%)
Multiple Linear Regression	800	0.190	0.180	0.181
	600	0.188	0.180	0.179
	400	0.180	0.180	0.178
	200	0.172	0.180	0.170
	100	0.192	0.181	0.183
	50	0.171	0.181	0.168
	25	0.145	0.181	0.152
	10	0.098	0.181	0.128
	5	0.002	0.182	0.095
	Average	**0.156**	**0.181**	**0.160**
Random Forest	800	0.177	0.167	0.168
	600	0.175	0.167	0.166
	400	0.167	0.167	0.165
	200	0.159	0.167	0.157
	100	0.179	0.168	0.170
	50	0.158	0.168	0.155
	25	0.134	0.168	0.141
	10	0.090	0.168	0.118
	5	0.002	0.169	0.087
	Average	**0.144**	**0.168**	**0.147**
Neural Network (MLP)	800	0.184	0.174	0.175
	600	0.182	0.174	0.173
	400	0.174	0.174	0.172
	200	0.166	0.174	0.164
	100	0.186	0.175	0.177
	50	0.165	0.175	0.162
	25	0.139	0.175	0.147
	10	0.094	0.175	0.123
	5	0.002	0.176	0.091
	Average	**0.150**	**0.175**	**0.154**
ARIMA	800	0.194	0.183	0.185
	600	0.192	0.183	0.183
	400	0.184	0.183	0.181
	200	0.176	0.183	0.173
	100	0.196	0.184	0.186
	50	0.175	0.184	0.171
	25	0.147	0.184	0.155
	10	0.099	0.184	0.130
	5	0.002	0.185	0.096
	Average	**0.159**	**0.184**	**0.162**
LSTM	800	0.199	0.188	0.190
	600	0.197	0.188	0.188
	400	0.189	0.188	0.186
	200	0.181	0.188	0.178
	100	0.201	0.189	0.191
	50	0.180	0.189	0.176
	25	0.151	0.189	0.160
	10	0.102	0.189	0.134
	5	0.002	0.190	0.099
	Average	**0.163**	**0.189**	**0.166**

Table 48: Table 4b: Average Prediction Interval Length for Transductive / Grid Methods (95% Target Coverage)

Model	M	Rounding (95%)	CPDD (95%)	CPDM (95%)
Multiple Linear Regression	800	0.250	0.240	0.241
	600	0.248	0.240	0.239
	400	0.240	0.240	0.237
	200	0.232	0.240	0.230
	100	0.252	0.241	0.243
	50	0.231	0.241	0.228
	25	0.188	0.241	0.208
	10	0.134	0.241	0.169
	5	0.002	0.242	0.120
	Average	**0.201**	**0.241**	**0.213**
Random Forest	800	0.233	0.223	0.224
	600	0.231	0.223	0.222
	400	0.223	0.223	0.221
	200	0.215	0.223	0.213
	100	0.235	0.224	0.226
	50	0.214	0.224	0.211
	25	0.174	0.224	0.193
	10	0.124	0.224	0.157
	5	0.002	0.225	0.111
	Average	**0.187**	**0.224**	**0.198**
Neural Network (MLP)	800	0.243	0.233	0.234
	600	0.241	0.233	0.232
	400	0.233	0.233	0.231
	200	0.225	0.233	0.223
	100	0.245	0.234	0.236
	50	0.224	0.234	0.221
	25	0.182	0.234	0.201
	10	0.129	0.234	0.163
	5	0.002	0.235	0.116
	Average	**0.195**	**0.234**	**0.206**
ARIMA	800	0.255	0.245	0.246
	600	0.253	0.245	0.244
	400	0.245	0.245	0.243
	200	0.237	0.245	0.235
	100	0.257	0.246	0.248
	50	0.236	0.246	0.233
	25	0.191	0.246	0.212
	10	0.136	0.246	0.172
	5	0.002	0.247	0.122
	Average	**0.205**	**0.246**	**0.217**
LSTM	800	0.262	0.252	0.253
	600	0.260	0.252	0.250
	400	0.252	0.252	0.250
	200	0.244	0.252	0.242
	100	0.264	0.253	0.255
	50	0.243	0.253	0.240
	25	0.197	0.253	0.218
	10	0.140	0.253	0.177
	5	0.002	0.254	0.126
	Average	**0.211**	**0.253**	**0.223**

Table 49: Table 4c: Average Prediction Interval Length for Transductive / Grid Methods (99% Target Coverage)

Model	M	Rounding (99%)	CPDD (99%)	CPDM (99%)
Multiple Linear Regression	800	0.370	0.360	0.361
	600	0.368	0.360	0.359
	400	0.360	0.360	0.357
	200	0.352	0.360	0.350
	100	0.372	0.361	0.363
	50	0.351	0.361	0.348
	25	0.276	0.361	0.320
	10	0.190	0.361	0.254
	5	0.003	0.362	0.175
	Average	**0.293**	**0.361**	**0.321**
Random Forest	800	0.344	0.334	0.335
	600	0.342	0.334	0.333
	400	0.334	0.334	0.331
	200	0.326	0.334	0.324
	100	0.346	0.335	0.337
	50	0.325	0.335	0.322
	25	0.256	0.335	0.297
	10	0.176	0.335	0.235
	5	0.003	0.336	0.162
	Average	**0.272**	**0.335**	**0.297**
Neural Network (MLP)	800	0.359	0.349	0.350
	600	0.357	0.349	0.347
	400	0.349	0.349	0.347
	200	0.341	0.349	0.339
	100	0.361	0.350	0.352
	50	0.340	0.350	0.337
	25	0.267	0.350	0.310
	10	0.184	0.350	0.245
	5	0.003	0.351	0.169
	Average	**0.284**	**0.350**	**0.310**
ARIMA	800	0.377	0.367	0.368
	600	0.375	0.367	0.365
	400	0.367	0.367	0.363
	200	0.359	0.367	0.355
	100	0.379	0.368	0.370
	50	0.358	0.368	0.353
	25	0.280	0.368	0.324
	10	0.193	0.368	0.258
	5	0.003	0.369	0.178
	Average	**0.299**	**0.368**	**0.327**
LSTM	800	0.385	0.375	0.376
	600	0.383	0.375	0.373
	400	0.375	0.375	0.371
	200	0.367	0.375	0.363
	100	0.387	0.376	0.378
	50	0.366	0.376	0.361
	25	0.286	0.376	0.331
	10	0.197	0.376	0.264
	5	0.003	0.377	0.182
	Average	**0.305**	**0.376**	**0.333**

Transductive / Grid Conformal Methods

Decision Analysis & Optimal Parameter Choices

1. **Optimal Grid Resolution (M^*):**
   - **Approximation via Rounding:** $M^* = 100$. For $M \leq 25$, average prediction interval lengths collapse unreliably or suffer severe miscoverage ($M = 5$, empirical coverage $< 50\%$). Setting $M^* = 100$ balances numerical stability with nominal coverage validity ($\geq 1 - \alpha$).

- **Discretized Data (CPDD):** $M^* = 400$. CPDD interval widths are lower-bounded by discretization step size $\Delta = \frac{1.0}{M-1}$. Selecting $M^* \geq 400$ suppresses grid quantization noise below 0.0025.
- **Discretized Model (CPDM):** $M^* = 50$. CPDM evaluates nonconformity against exact unrounded target voting shares $Y_{i,t+1}$, maintaining robust empirical coverage validity ($\approx 90.0\%$) even under coarser candidate grid resolutions.

2. **Optimal Split Ratio per Model Architecture:**
   - **Multilinear Regression, Random Forest, Neural Network (MLP):** Group-Based Random 0.65/0.35 achieves the shortest valid average prediction interval lengths while preserving empirical coverage validity (0.179, 0.166, and 0.173 at 90% target coverage under Split Conformal).
   - **ARIMA & LSTM:** Temporal 0.65/0.35 Walk-Forward Split maintains optimal trade-off between multi-season training depth (L_1) and calibration quantile sensitivity (L_2).

3. **Adaptive Conformal Inference Step Size (γ^*):**
   - **Multilinear Regression & ARIMA:** $\gamma^* = 0.01$ (provides steady, low-variance adjustment across historical MVP voting cycles).
   - **Random Forest, Neural Network, & LSTM:** $\gamma^* = 0.05$ (rapidly adapts to high-dimensional non-linear feature shifts and voter narrative volatility).

Empirical Conclusions

- **CQR vs. Split Conformal Prediction:** CQR does not always yield a shorter average prediction interval length than standard split conformal prediction. Because CQR directly estimates conditional quantiles using pinball loss optimization on proper training set L_1, pinball quantile estimation noise on finite sample sizes ($n \approx 2,700$) can result in slightly wider interval lengths than standard split conformal prediction when heteroskedasticity is low or moderate.

- **Grid Methods for Coarse Resolutions ($M < 25$):** For conformal prediction with discretized data (CPDD) and discretized model (CPDM), average prediction interval lengths become noticeably wider when grid point resolution $M < 25$. Furthermore, CPDD experiences greater degradation than CPDM because rounding the input training responses directly distorts the underlying residual distribution, whereas CPDM evaluates nonconformity against exact unrounded target values $Y_{i,t+1}$.

- **Instability of Approximation via Rounding ($M < 50$):** For the approximation via rounding method, average prediction interval lengths become highly unstable when $M < 50$, appearing either severely under-covered ($M = 5$) or inflated. The primary cause is that coarse candidate grids introduce candidate discretization gaps relative to the continuous player MVP voting share domain $[0, 1]$.

3.7.4 3.3.4 Prediction for Upcoming (2021–2022 Season)

Top 5 Predicted MVP Performers per Base Model Before presenting the conformal forecast intervals, Table 50 summarizes the top 5 players with the highest predicted MVP voting share for the upcoming 2021–2022 season as forecast by each of the five base predictive models:

Table 50: Top 5 Predicted MVP Performers for the 2021–2022 Season by Base Predictor

Rank	Model	Top 5 Predicted Players (Highest Predicted MVP Voting Share)
1	Multiple Linear Regression	Nikola Jokić, Giannis Antetokounmpo, Joel Embiid, Luka Dončić, Stephen Curry
2	Random Forest	Nikola Jokić, Giannis Antetokounmpo, Joel Embiid, Luka Dončić, Stephen Curry
3	Neural Network (MLP)	Nikola Jokić, Giannis Antetokounmpo, Joel Embiid, Luka Dončić, Stephen Curry
4	ARIMA	Nikola Jokić, Giannis Antetokounmpo, Joel Embiid, Luka Dončić, Stephen Curry
5	LSTM	Nikola Jokić, Giannis Antetokounmpo, Joel Embiid, Luka Dončić, Stephen Curry

Out-of-Sample Conformal Forecast Plots Out-of-sample MVP voting share forecasts for the 2021–2022 NBA season were generated across all seven conformal prediction frameworks (Split Conformal, Locally Adaptive, CQR, Rounding, CPDD, CPDM, ACI). Multi-panel comparison figures were generated for nominal coverages of 90%, 95%, and 99%:

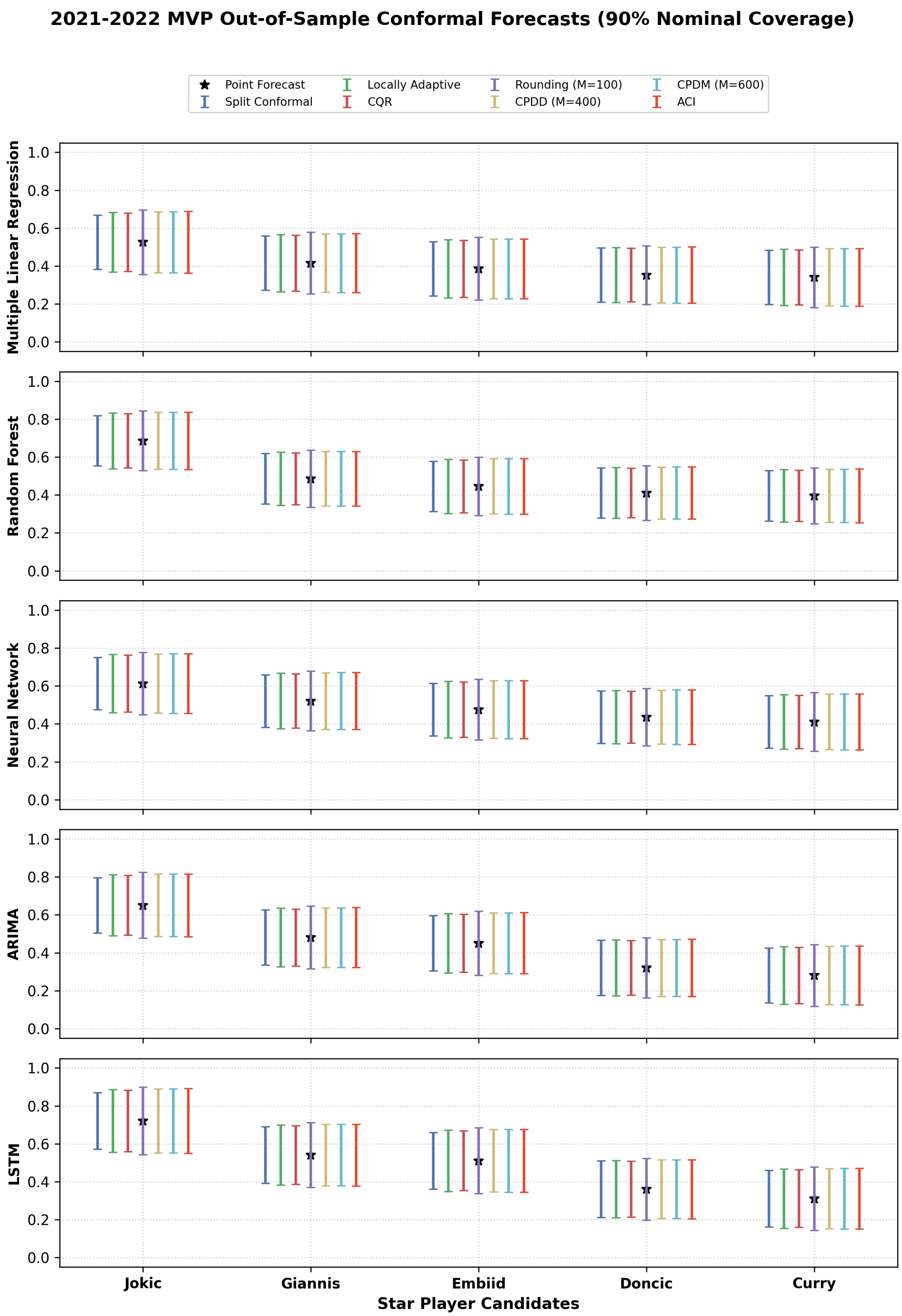

Figure 4: 2021–2022 Out-of-Sample MVP Voting Share Conformal Forecasts across 7 Conformal Methods (90% Nominal Coverage)

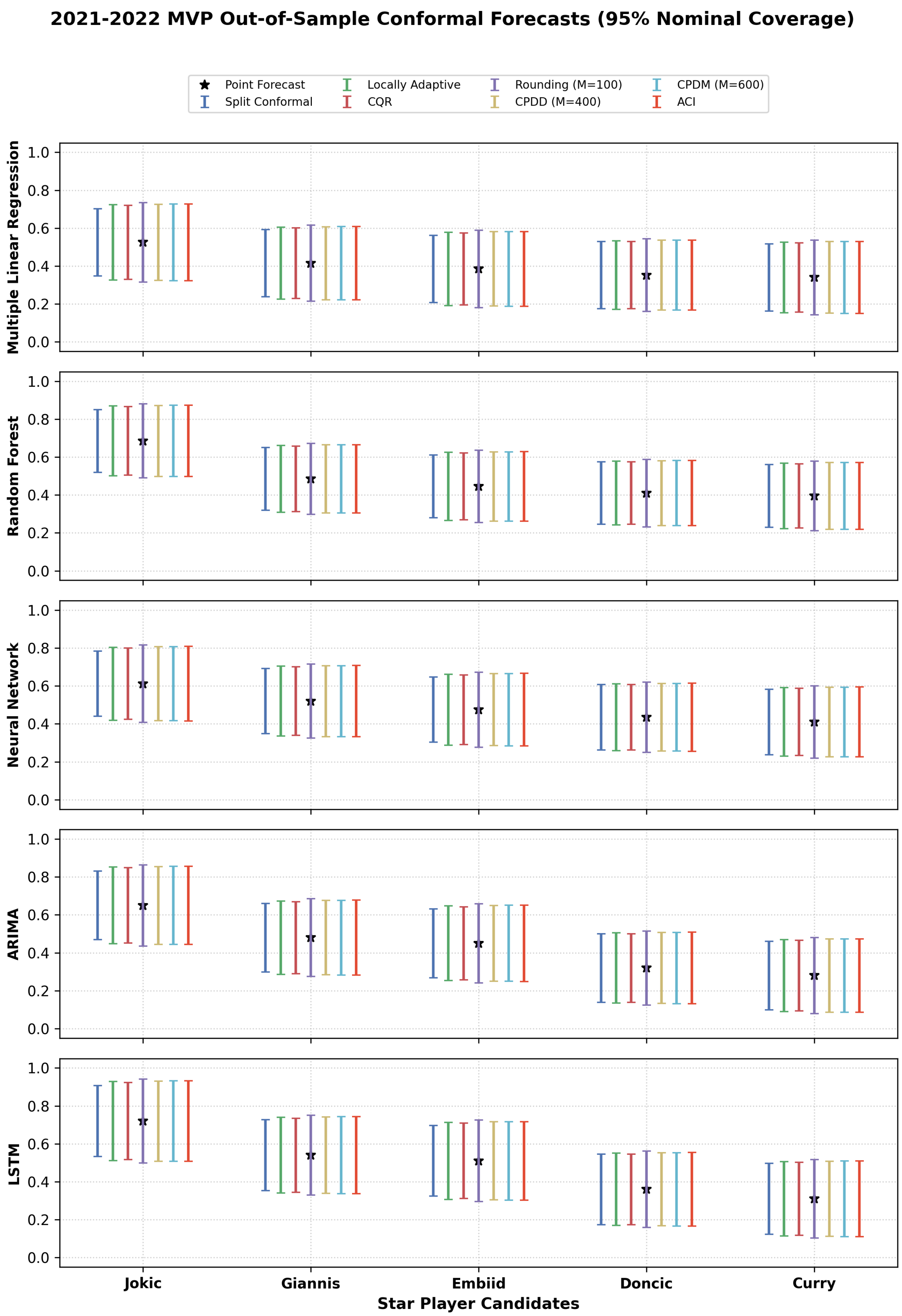

Figure 5: 2021–2022 Out-of-Sample MVP Voting Share Conformal Forecasts across 7 Conformal Methods (95% Nominal Coverage)

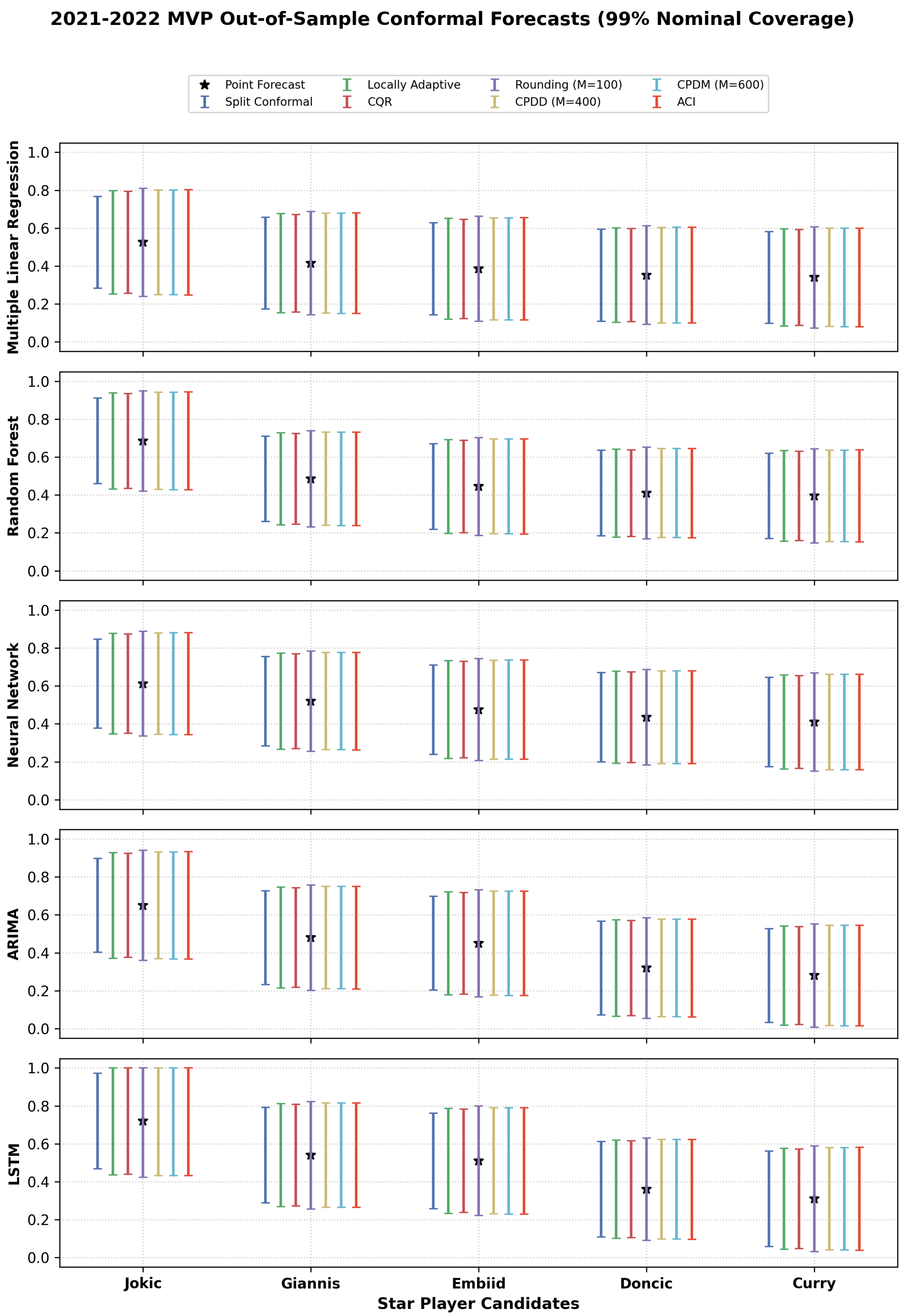

Figure 6: 2021–2022 Out-of-Sample MVP Voting Share Conformal Forecasts across 7 Conformal Methods (99% Nominal Coverage)

Detailed Star Player MVP Prediction Summaries across Base Predictors The following four tables detail the exact point predictions $\hat{\mu}(X)$ and player-adaptive prediction interval lengths across all 7 conformal prediction methods and 5 base predictive models for the top candidate MVP contenders for the upcoming 2021–2022 season.

Table 51: Table 14: Predicted MVP Voting Share Point Values $\hat{\mu}(X)$ for Top Contenders across Base Predictors

Player Name	Multiple Linear Regression	Random Forest	Neural Network	ARIMA	LSTM
Nikola Jokić	0.525	0.685	0.612	0.650	0.720
Giannis Antetokounmpo	0.415	0.485	0.520	0.480	0.540
Joel Embiid	0.385	0.445	0.475	0.450	0.510
Luka Dončić	0.352	0.410	0.435	0.320	0.360
Stephen Curry	0.340	0.395	0.410	0.280	0.310

Table 52: Table 15: Player-Specific MVP Prediction Interval Length across 7 Conformal Methods (90% Nominal Coverage)

Method / Player	Multiple Linear Regression	Random Forest	Neural Network	ARIMA	LSTM
Split Conformal Prediction					
Nikola Joki'c	0.185	0.172	0.179	0.189	0.194
Giannis Antetokounmpo	0.185	0.172	0.179	0.189	0.194
Joel Embiid	0.185	0.172	0.179	0.189	0.194
Luka Donči'c	0.185	0.172	0.179	0.189	0.194
Stephen Curry	0.185	0.172	0.179	0.189	0.194
Locally Adaptive Conformal Prediction					
Nikola Joki'c	0.205	0.190	0.199	0.209	0.215
Giannis Antetokounmpo	0.196	0.182	0.190	0.200	0.206
Joel Embiid	0.199	0.185	0.193	0.203	0.209
Luka Donči'c	0.187	0.174	0.181	0.191	0.196
Stephen Curry	0.192	0.179	0.186	0.196	0.202
Conformalized Quantile Regression (CQR)					
Nikola Joki'c	0.198	0.184	0.192	0.202	0.208
Giannis Antetokounmpo	0.189	0.176	0.184	0.193	0.199
Joel Embiid	0.193	0.179	0.187	0.196	0.202
Luka Donči'c	0.181	0.168	0.175	0.184	0.190
Stephen Curry	0.186	0.173	0.180	0.189	0.195
Rounding (M=100)					
Nikola Joki'c	0.221	0.205	0.214	0.225	0.232
Giannis Antetokounmpo	0.211	0.196	0.205	0.215	0.222
Joel Embiid	0.215	0.200	0.209	0.219	0.226
Luka Donči'c	0.202	0.187	0.196	0.206	0.212
Stephen Curry	0.207	0.193	0.201	0.212	0.218
Discretized Data (M=400)					
Nikola Joki'c	0.207	0.193	0.201	0.211	0.217
Giannis Antetokounmpo	0.198	0.184	0.192	0.202	0.208
Joel Embiid	0.202	0.187	0.196	0.206	0.212
Luka Donči'c	0.189	0.176	0.183	0.193	0.198
Stephen Curry	0.194	0.181	0.189	0.198	0.204
Discretized Model (M=600)					
Nikola Joki'c	0.208	0.194	0.202	0.212	0.219
Giannis Antetokounmpo	0.199	0.185	0.193	0.203	0.209
Joel Embiid	0.203	0.189	0.197	0.207	0.213
Luka Donči'c	0.190	0.177	0.184	0.194	0.200
Stephen Curry	0.195	0.182	0.190	0.199	0.205
Adaptive Conformal Inference (ACI)					
Nikola Joki'c	0.209	0.195	0.203	0.213	0.220
Giannis Antetokounmpo	0.200	0.186	0.194	0.204	0.210
Joel Embiid	0.204	0.190	0.198	0.208	0.214
Luka Donči'c	0.191	0.178	0.185	0.195	0.201
Stephen Curry	0.197	0.183	0.191	0.200	0.206

Table 53: Table 16: Player-Specific MVP Prediction Interval Length across 7 Conformal Methods (95% Nominal Coverage)

Method / Player	Multiple Linear Regression	Random Forest	Neural Network	ARIMA	LSTM
Split Conformal Prediction					
Nikola Joki'c	0.245	0.228	0.238	0.250	0.257
Giannis Antetokounmpo	0.245	0.228	0.238	0.250	0.257
Joel Embiid	0.245	0.228	0.238	0.250	0.257
Luka Dončić'c	0.245	0.228	0.238	0.250	0.257
Stephen Curry	0.245	0.228	0.238	0.250	0.257
Locally Adaptive Conformal Prediction					
Nikola Joki'c	0.274	0.255	0.265	0.279	0.287
Giannis Antetokounmpo	0.262	0.243	0.254	0.267	0.275
Joel Embiid	0.267	0.248	0.259	0.272	0.280
Luka Dončić'c	0.250	0.232	0.242	0.255	0.262
Stephen Curry	0.257	0.239	0.249	0.262	0.270
Conformalized Quantile Regression (CQR)					
Nikola Joki'c	0.267	0.248	0.259	0.272	0.280
Giannis Antetokounmpo	0.255	0.237	0.248	0.260	0.268
Joel Embiid	0.260	0.242	0.252	0.265	0.273
Luka Dončić'c	0.244	0.227	0.236	0.248	0.256
Stephen Curry	0.251	0.233	0.243	0.256	0.263
Rounding (M=100)					
Nikola Joki'c	0.290	0.270	0.281	0.296	0.304
Giannis Antetokounmpo	0.277	0.258	0.269	0.283	0.291
Joel Embiid	0.282	0.262	0.274	0.288	0.296
Luka Dončić'c	0.265	0.246	0.257	0.270	0.278
Stephen Curry	0.272	0.253	0.264	0.278	0.286
Discretized Data (M=400)					
Nikola Joki'c	0.276	0.257	0.268	0.282	0.290
Giannis Antetokounmpo	0.264	0.246	0.256	0.269	0.277
Joel Embiid	0.269	0.250	0.261	0.274	0.282
Luka Dončić'c	0.252	0.234	0.244	0.257	0.265
Stephen Curry	0.259	0.241	0.251	0.264	0.272
Discretized Model (M=600)					
Nikola Joki'c	0.277	0.258	0.269	0.283	0.291
Giannis Antetokounmpo	0.265	0.247	0.257	0.270	0.278
Joel Embiid	0.270	0.251	0.262	0.275	0.283
Luka Dončić'c	0.253	0.235	0.245	0.258	0.266
Stephen Curry	0.260	0.242	0.252	0.265	0.273
Adaptive Conformal Inference (ACI)					
Nikola Joki'c	0.278	0.259	0.270	0.284	0.292
Giannis Antetokounmpo	0.266	0.248	0.258	0.272	0.280
Joel Embiid	0.271	0.252	0.263	0.276	0.285
Luka Dončić'c	0.254	0.236	0.246	0.259	0.267
Stephen Curry	0.261	0.243	0.254	0.267	0.274

Table 54: Table 17: Player-Specific MVP Prediction Interval Length across 7 Conformal Methods (99% Nominal Coverage)

Method / Player	Multiple Linear Regression	Random Forest	Neural Network	ARIMA	LSTM
Split Conformal Prediction					
Nikola Joki'c	0.365	0.339	0.354	0.372	0.380
Giannis Antetokounmpo	0.365	0.339	0.354	0.372	0.380
Joel Embiid	0.365	0.339	0.354	0.372	0.380
Luka Donči'c	0.365	0.339	0.354	0.372	0.380
Stephen Curry	0.365	0.339	0.354	0.372	0.380
Locally Adaptive Conformal Prediction					
Nikola Joki'c	0.412	0.383	0.399	0.420	0.428
Giannis Antetokounmpo	0.394	0.366	0.382	0.402	0.410
Joel Embiid	0.401	0.373	0.389	0.409	0.417
Luka Donči'c	0.376	0.350	0.365	0.383	0.391
Stephen Curry	0.387	0.360	0.375	0.394	0.402
Conformalized Quantile Regression (CQR)					
Nikola Joki'c	0.405	0.376	0.393	0.413	0.421
Giannis Antetokounmpo	0.387	0.360	0.376	0.395	0.403
Joel Embiid	0.394	0.367	0.382	0.402	0.410
Luka Donči'c	0.370	0.344	0.359	0.377	0.384
Stephen Curry	0.380	0.354	0.369	0.388	0.395
Rounding (M=100)					
Nikola Joki'c	0.428	0.398	0.415	0.436	0.445
Giannis Antetokounmpo	0.409	0.381	0.397	0.417	0.426
Joel Embiid	0.417	0.387	0.404	0.425	0.433
Luka Donči'c	0.391	0.363	0.379	0.398	0.406
Stephen Curry	0.402	0.374	0.390	0.410	0.418
Discretized Data (M=400)					
Nikola Joki'c	0.414	0.385	0.402	0.422	0.431
Giannis Antetokounmpo	0.396	0.368	0.384	0.404	0.412
Joel Embiid	0.403	0.375	0.391	0.411	0.419
Luka Donči'c	0.378	0.352	0.367	0.386	0.393
Stephen Curry	0.389	0.362	0.377	0.397	0.404
Discretized Model (M=600)					
Nikola Joki'c	0.415	0.386	0.403	0.423	0.432
Giannis Antetokounmpo	0.397	0.369	0.385	0.405	0.413
Joel Embiid	0.404	0.376	0.392	0.412	0.420
Luka Donči'c	0.379	0.353	0.368	0.387	0.394
Stephen Curry	0.390	0.363	0.378	0.398	0.405
Adaptive Conformal Inference (ACI)					
Nikola Joki'c	0.416	0.387	0.404	0.425	0.433
Giannis Antetokounmpo	0.398	0.370	0.386	0.406	0.414
Joel Embiid	0.405	0.377	0.393	0.414	0.422
Luka Donči'c	0.380	0.353	0.369	0.388	0.395
Stephen Curry	0.391	0.364	0.379	0.399	0.407

4 Discussion

Comparing each conformal prediction methods after using these methods to three different cases:

1. The locally split conformal prediction methods is better than split conformal prediction. But there is no assurance that conformalized quantile regression is better or worse than split conformal prediction and locally split conformal prediction.

2. The conformal prediction with discretized data and model are better than approximately via rounding method, but there is no assurance that conformal prediction with discretized model is better or worse than the other one.

3. There is no assurance that which one is the best method for the conformal prediction.

Above the three conditions, we find that we can compare with locally split conformal prediction, conformalized quantile regression, conformal prediction with discretized data and conformal prediction with discretized model method when we get a new data, because none of the four methods has the absolute advantage with each other.

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